From Seeing to Reporting: A Theory of Reliable Visual Set Generation

Junzhe Shi[*]
Systems Engineering
University of California, Berkeley
Berkeley, CA, USA
junzhe@berkeley.edu

Yuan Gan
Department of AI
Quotr AI
Emeryville, CA, USA
yuan@quotr.ai

Shida Jiang
Systems Engineering
University of California, Berkeley
Berkeley, CA, USA
shida_jiang@berkeley.edu

Abstract

Vision-language models are asked to report what they see: to count objects, box them, parse a screen or pick a click target. We argue that reporting is part of the visual task, not merely its formatting. In a dense scene with 173 shapes, GPT-6.1 lists 173 records that match every shape, yet the counts it states in the same response sum to 188. *From Seeing to Reporting* (FSR) models reporting as a chain from responses to decoded records, constructed sets and task answers, linked by task-compatible matching. The chain gives exact relations for deleting records, averaging their coordinates and switching readouts, and it splits the error of one answer chosen among several calls into missing candidates, lost representatives and wrong choices. These relations guide procedures for counting, detection, document layout, screen parsing and clicks that use the task specification and a shipped configuration, without requiring labels from the target task. On 50 dense synthetic scenes, a one-call reporting-and-decoding protocol cuts per-type count error by 85.2% for GPT-6.1 and 98.8% for Sonnet 5.5. On 600 ScreenSpot-Pro items not used in its design, a three-model click procedure lowers the miss rate from 13.67% for GPT-6.1's two-stage zoom to 9.33%, against 12.67% for three such zooms at about the same cost. Adding a crop-first call as a fourth source shows why more candidates need not bring proportionate gains: seven more items gain a correct candidate, but representative and selection losses leave two fewer misses. We release the protocols, the stored responses and the analysis code.

1 Introduction

Vision-language models are now asked to report a set of objects in an image: to count them, to box them, to parse every element of a screen, or to pick the one element a user means. A typical answer has two parts. The first is a *list*: one entry per object, each giving a class name and a location (a point or a box). We call each entry a *record*. The second, when it is requested, is a *stated count*: the numbers the model writes for how many objects of each class it found. Code can also count the records in the list; we call that the *list count*. The two counts should agree. Often they do not.

Figure 1a shows one response of GPT-6.1 to a dense synthetic scene with 173 shapes of five classes. Its list has 173 records, and each record matches a different shape of the same class within 20 pixels, so the list is exact. In the same response, the model states counts that add up to 188. Across 50 such scenes, GPT-6.1 wrote an exact list in 20 scenes, and in 19 of them its stated counts were wrong. Sonnet 5.5 wrote an exact list in 39 scenes (after mapping its word "diamond" to the class "square"), and its stated counts were wrong in all 39. The response contains a correct enumeration; the model's own summary of that enumeration is not faithful to it.

Visual set reports are becoming the interface between perception models and the systems that use them: inventory counts, document layout, screen parsing for computer-use agents, click targets. Practitioners compensate with folklore: ask again, ask the model to check itself, overlay a grid, tell it not to miss anything, vote across models. Some of these help, some do nothing, and some make things worse. Without a model of where reporting errors come from, it is hard to predict which. Our thesis is that *reporting is part of the visual task, not merely its formatting*: what a model delivers depends on how it is asked to report, how the report is decoded and read, and how several reports are combined, and each of these steps can be analysed and improved.

What we do. We describe reporting as a chain with four objects (Figure 2): a *response*, the raw

[*]Corresponding author.

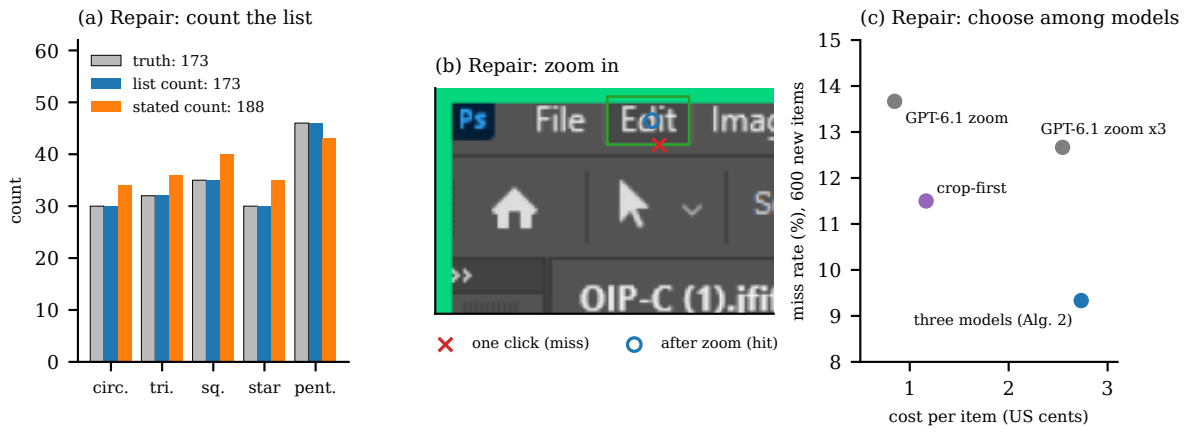


Figure 1: Three lessons in one figure. (a) One response of GPT-6.1 to a dense synthetic scene (SCP-dense image 28, Section 6; 173 shapes of five classes). Grey: true number of shapes per class. Blue: the *list count*, the number of records of each class in the model's list, counted by code; every one of its 173 records matches a different shape. Orange: the *stated count*, the numbers the model wrote for each class in the same response; they sum to 188. Across 50 such scenes, the full list-first reporting and decoding protocol cuts per-type count error by 85.2% (Section 7.2). (b) A single click by Gemini 3.8 misses the "edit menu" (green box); a second call on a zoomed crop (ReGround, Li et al., 2025b) hits it. Two-stage zoom cuts its click error by 65% (Section 7.5). (c) Miss rate against cost per item on 600 ScreenSpot-Pro items not used in designing the method: GPT-6.1's two-stage zoom, three such zooms with a cluster vote, a crop-first baseline in the style of GUI-Lens (Fu et al., 2026), and the three-model tier of Algorithm 2, which keeps every item in the denominator (Section 7.7).

text of one call to a model; the *records* decoded from it; a *set* of records built from one or more responses, for example by keeping only records that two calls agree on; and the *task answer* read from that set, such as a count per class or a click point. Three quantities connect the chain to every loss we study: the maximum number of records that can be paired one-to-one with true objects under the task's matching rule, the number of records that are scored, and the rule that reads the answer. They give exact relations for what each operation does. Deleting unconfirmed records lowers the error exactly when the deletion loses fewer than one match per two scored records removed (with equal costs of the two error types), whichever model or view supplied the confirmation. Averaging the coordinates of agreeing records is a separate effect that can help or hurt. The chain takes reports as they come: from one call on the whole image, or from several views produced by an existing view planner such as tiling or zoom, whose merging and cross-view confirmation are constructions like any other. Alongside these relations we state a behavioural model, kept separate from them: each object has a persistent tendency to be reported or missed, partly shared across models. This analysis of what a model reports complements analyses of what a single call can see (Shi et al., 2026).

Three problems, kept apart. Practice often treats three problems as one. *Detecting* that a report is wrong can be done without labels, from disagreement between reports (Proposition 6). *Judging* which of several reports or configurations is more accurate cannot in general be certified from outputs alone, because two scenes can produce the same outputs (Proposition 13); it needs labels, or a configuration chosen on other data and carried over (Proposition 9). *Repairing* a report works when the needed information is already available: in another field of the same response, in another view of the image, or in another model's report. A method that detects errors well need not judge well, and one that judges well need not repair. The paper is organized around this distinction.

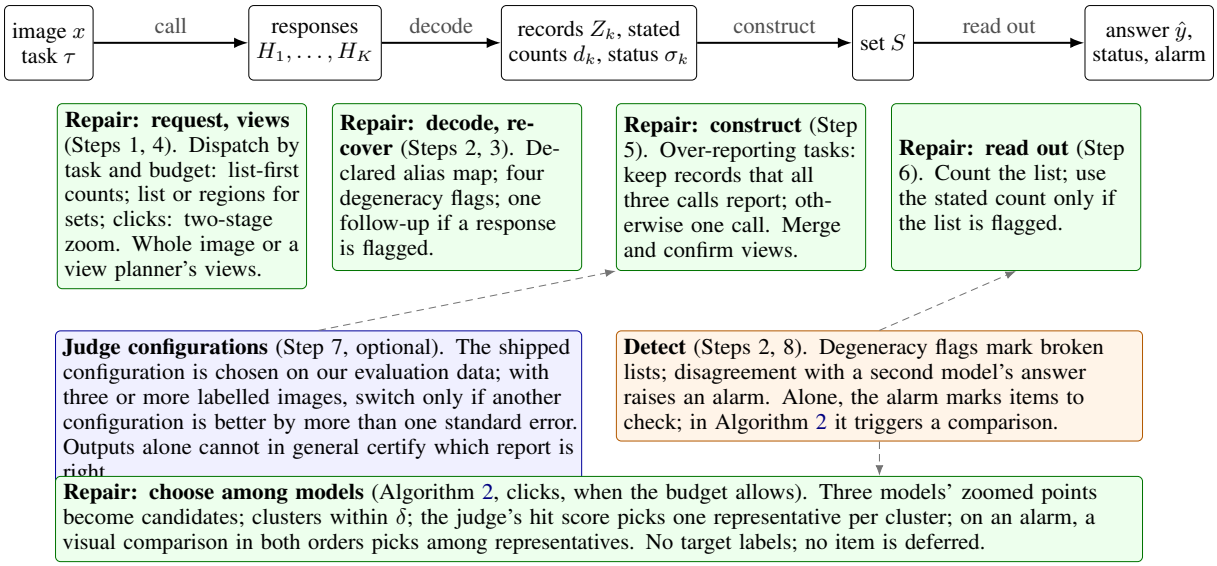


Figure 2: Overview. Top: the reporting chain. A call to a model returns a response; decoding turns it into records, stated counts and a status; a construction combines the records of one or more calls into a set; a readout turns the set into the task answer. Below: where each step of the default pipeline (Algorithm 1, Section 5) acts on the chain, grouped by the three problems the paper keeps apart: repairing a report (green), judging between configurations (blue) and detecting that a report is wrong (orange). Bottom: the several-model tier for clicks, a repair whose run-time comparison of candidates is a model call, not the configuration judge. Each step reads only the task specification, the shipped configuration, the model outputs and, optionally, labels. The exact relations of Section 4 each concern one arrow: the readout bound the last arrow, the deletion and coordinate-averaging identities and the deletion threshold the construction, the impossibility of output-only certification the judging box, and the three-term split of a miss the bottom box.

Three answers. *Do vision-language models report what they see?* Not reliably, and the gap is visible within one response: in dense scenes GPT-6.1 and Sonnet 5.5 often write an exact list while the counts they state in the same response are wrong. *Will better models make the repairs unnecessary?* Some failures are already absent on our tasks: on our synthetic counting scenes, the average coordinate offset is about 1% of the matching tolerance, and Gemini 3.8's default reasoning setting removes most of its broken lists. The gap between list and stated count, and object-level errors that repeat across calls, remain in the newest models we tested. *More calls of the same model, or a different source of evidence?* For counts, clicks and error detection, a different source helped more: another field of the same response, the original pixels of a smaller crop, or a different model. Repeating the same call helps mainly by removing false records that do not repeat.

What we find. Measured against common remedies on the same stored calls (Table 6), the defaults derived from the theory do best where it predicts they should.

- **Repair the readout.** On dense synthetic scenes with five classes, counting the list instead of reading the stated count cuts per-type count error by 71% to 84% in the same responses of GPT-6.1 and Sonnet 5.5; on one-class photographs the difference is not significant. The full list-first protocol (list before counts, row-by-row scan, declared class aliases, and the readout of Section 5) cuts it by 85.2% and 98.8% at the cost of one call; self-consistency, the median of the stated counts of three repeated calls, gains at most 23%. For Gemini 3.8 at its lowest reasoning setting, whose lists often break, the change is $+15.6\%$ and not significant (interval -5.3% to $+39.4\%$); at the same output cap its default setting lowers its error from 52.8 to 24.5 per image (-53.6%). On aerial and document images with several classes the lists themselves miss objects, the stated count stays close to the list, and the protocol changes count error by -20% to $+0.4\%$ for GPT-6.1 and Sonnet 5.5, none significantly, and by -25.8% (DocLayNet) and -28.2% (DOTA, interval reaching zero) for Gemini 3.8 at its default setting. When the label-free gap between the two counts is small, switching between them can change the error by at most that gap (Sections 7.2 and 7.3).
- **Repair the set.** Set error counts false records plus missed objects. When a second call (the *partner*) is used to confirm the records of a first call (the *base*), the change in set

error splits exactly into two terms: deleting the base records the partner does not report, and averaging the coordinates of those it does. Whether deletion helps is decided by one number, the marginal matching rate p_1 of the deleted records, and that number can be estimated from very few labels: on 54 base/partner pairs, five labelled images predict whether deletion helps on the other images in 83% of draws, the task family's default does so without labels in 85%, and the shipped label rule, which starts from that default and switches only when the labelled images favour the other decision by more than one standard error, in 87%, cutting the loss from a wrong decision from 0.43 to 0.17 errors per image. The averaging term is positive in 37 pairs and negative in 13 (Section 7.4).

- **Repair the view.** Two-stage zoom, a regrounding strategy in the style of ReGround (Li et al., 2025b), cuts click error by up to 65% (Gemini 3.8; 45% for Sonnet 5.5, not significant for GPT-6.1); a three-click vote and zooming only on disagreement gain at most 13%. Tiles that keep at least the whole image's tokens per object and contain every object wholly in some tile cannot lower expected recall under the companion paper's monotonicity assumptions (Shi et al., 2026). Reports from four crops, used to confirm the whole-image report, cut set error by up to 45.5%; where they fail, the deletion threshold accounts for it (Section 7.5).

- **Detect.** At a 1.5% distance threshold, a second, cheaper model catches 61% of GPT-6.1's click errors (14 of 23) and a second call of the same model 17% (4 of 23), but the second model flags more items: it leaves 86% of items to act on without review (the *acceptance coverage*) against 96%. At the same acceptance coverage of 93%, it catches 39% against 26%, and only the second model reaches miss rates near 4% on the items acted on. After two-stage zoom the gap remains: the second model catches 68% to 78% of the remaining errors and a second zoomed call of the same model 40% to 50% (Section 7.6).

- **Repair a click by choosing among models.** The miss rate of one answer chosen among several calls is exactly the sum of three terms: no candidate is correct, the correct candidate is not a cluster representative, or the wrong representative is chosen. A tier built on this split (three models' zoomed clicks, a judge's score per candidate, and a comparison only where the models disagree) was run unchanged on 600 ScreenSpot-Pro items whose items and screenshots differ from those it was designed on. It misses 9.33% of targets against 13.67% for GPT-6.1's two-stage zoom (-4.33 points $[-6.33, -2.33]$) and 12.67% for three GPT-6.1 two-stage zooms at about the same cost. Against a crop-first baseline in the style of GUI-Lens (Fu et al., 2026) the difference, -2.17 points, has an interval that reaches zero, and the comparison step adds -0.83 points, not significant. Most of the gain is candidate coverage: on the items GPT-6.1 misses, a second GPT-6.1 call hits 18% of the time, Sonnet 5.5 29% and Gemini 3.8 37%. The split also explains why the crop-first call adds little as a fourth source: it moves 7 items out of the first term and 5 into the other two (Section 7.7).

- **Judge.** Choosing the best set among three calls of one model, with knowledge of the truth, would remove 45% of set error; selectors fitted with labels reach 16%. Constructions, rules that build one set from several calls such as keeping only records that all three calls report, were chosen on five datasets and applied to the sixth; they improve set error significantly in 9 of the 15 cells where models over-report, that is, report more false objects than they miss. A single-call configuration (interface and reasoning setting) chosen on three detection sources raises F1 on the fourth in 5 of 12 cells and lowers it in 1, from a macro F1 of 42.1 to 49.1 (Sections 7.8 and 7.9).

Contributions. Our contribution is a common task-loss model for visual reporting, procedures that act on its stages, and experiments that separate what information is available from the quality of the answer finally delivered. The model covers multi-record outputs and the single-record case of click grounding. Its exact relations do not prescribe one universal interface; they say what each operation can change and what must be measured to decide whether the change is useful.

1. **A common theory of reporting operations**

and task quality. A chain from response to records, set and answer, in which task-compatible matching, scored record counts and the readout express the losses for counts, boxes, screen elements and clicks; exact relations for deletion, coordinate averaging and readout; conditions, stated as inequalities on confirmation rates, under which repetition, a new view or a second model helps, with a behavioural model of persistent errors; a constrained oracle for any declared family of candidate sets; an exact split of the error of one answer chosen among several calls, with the effect of adding a source and of switching an answer; and an impossibility result for certifying a report from outputs alone (Sections 3 and 4).

2. **Executable procedures for recovering, constructing and selecting visual records.** A default pipeline that dispatches by task and budget, and a several-model tier for clicks, in which every input is named (task specification, shipped configuration, computed from the outputs, or optional labels), together with the revisions of the model that led to them (Section 5).

3. **Mechanism-resolving evidence and a fixed-protocol evaluation of complete task outputs.** Measurements on 67,364 calls to five proprietary models and one open-weight model that attribute gains and failures to single operations, and a 600-item click evaluation of the complete tier run unchanged on items not used in its design. All responses, the labels whose licences allow redistribution, a script that rebuilds the others from the official downloads, and the code that regenerates every table and figure are at `https://github.com/shijunzhe/vlm-set-reporting` (Sections 6 and 7).

2 Related Work

Counting and set reporting by VLMs. Counting is a long-standing weak spot of vision-language models (Paiss et al., 2023; Rahmanzadehgervi et al., 2024), and recent work locates part of the failure after perception: count information can be present in internal representations yet misaligned with the emitted number (El-Shangiti et al., 2026), and a visual counting bottleneck has been isolated in the model pipeline (Pang et al., 2026). Training-free fixes divide the image and count per region (Qharabagh et al., 2026), in the spirit of sliced inference for small-object detection (Akyon et al., 2022). Point-and-count supervision (Deitke et al., 2025) and sequence-style detection (Chen et al., 2022; Li et al., 2025a) make the list itself the output. Program-based visual reasoning counts by running code over the items a model finds (Gupta and Kembhavi, 2023; Surís et al., 2023); we show that the same code readout repairs the counts a single model writes about its own list. We work with frozen API models and observe the same gap from the outside: the information is in the model's own list in the same response, so the fix is a readout rule rather than a probe or a training signal. We also extend the analysis from counts to boxes, screen elements and clicks, where the loss is a matching between sets.

Views: zoom, tiles and resolution. Zooming in improves fine-grained perception (Wu and Xie, 2024; Shen et al., 2024; Zhang et al., 2025) and GUI grounding (Jiang et al., 2025; Tang et al., 2026), and region-level consistency can be used at test time (Du et al., 2026). GUI-Lens lets a single general-purpose model crop from coarse to fine, with OCR and detector anchors and a verification step (Fu et al., 2026); Section 7.7 compares our several-model tier with a simplified crop-first baseline in its style and shows that the two err on different items. A companion study analyses when zooming in is safe for small objects (Shi et al., 2026). Uncertainty-driven zoom on open 7B models found that zooming every item can hurt (Tang et al., 2026); on frontier API models we find two-stage zoom on every item, in the style of ReGround (Li et al., 2025b), not significantly different (GPT-6.1) or better (Sonnet 5.5, Gemini 3.8), and cheaper than zooming only on disagreement. The two observations are compatible: whether zoom hurts depends on whether the crop cuts the target and on the relative cost of the second call (Section 7.5).

Visual scaffolds. Overlaid marks and coordinate grids are a popular way to help models localize (Yang et al., 2023; Lei et al., 2025). An overlay adds no Shannon information to the sent image and can only change how the model uses it (Shi et al., 2026); we observe two such changes in current models (fewer omitted class fields for Gemini 3.8, more uses of a class name outside the vocabulary for Sonnet 5.5) and no coordinate correction,

because on synthetic counting scenes their systematic coordinate offset is about 1% of the matching tolerance.

Self-consistency, self-correction and cross-model agreement. Sampling several answers and voting improves reasoning (Wang et al., 2023), but models often fail to correct themselves (Huang et al., 2024; Madaan et al., 2023; Liao et al., 2025), and their errors can be self-consistent and hard to detect (Tan et al., 2025; Bhat and Yamasaki, 2026). Disagreement between different models complements self-consistency for uncertainty estimation (Hamidieh et al., 2026), and multi-VLM agreement has been used to verify OCR (Zhang et al., 2026); we use disagreement among three frontier models' clicks as an alarm and choose among their points with a judge (Section 5.1); test-time scaling can also separate perception from reasoning (Avogaro et al., 2026). In set tasks these effects have a precise form: confirmation changes the set error by an exact identity, and whether a partner helps depends on its confirmation rates for the base model's true and false records, not on its overall quality (Proposition 5).

Ensembles, fusion and selective prediction. Weighted box fusion averages coordinates of matched detections (Solovyev et al., 2021), and detection ensembles combine models by affirmative, consensus or unanimous voting (Casado-García and Heras, 2020); our three-call unanimity is the unanimous strategy applied to repeated calls, and we show that its effect splits into a selection term and an averaging term, the latter of either sign under threshold matching. Error decompositions for detection split error into classification, localization, duplicate and background parts (Bolya et al., 2020; Oksuz et al., 2018); our split is instead by operation, so that each term belongs to one decision. Classical models for aggregating judges assume errors that are independent given the truth (Dawid and Skene, 1979; Grofman et al., 1983), which object-level persistence violates. Work on ensemble diversity measures how correlated the members' errors are (Kuncheva and Whitaker, 2003), and deep ensembles obtain diversity by training members independently (Lakshminarayanan et al., 2017); our members are repeated calls of fixed models, and we measure how strongly their errors on the same object are shared. Our alarms are evaluated as risk-coverage curves, as in selective classification (Geifman and El-Yaniv, 2017). The matroid structure of matching rank (Edmonds and Fulkerson, 1965) and the assignment algorithm (Kuhn, 1955) underlie our identities and oracle, and one-to-one matching between predicted and true sets is also the loss of set-prediction detectors (Carion et al., 2020).

What is known and what is new in the theory. Several of our relations specialise known results to record sets. The deletion threshold is the cost-sensitive decision threshold (Elkan, 2001); its F1 form matches the result that the F1-optimal threshold is half the optimal F1 (Lipton et al., 2014); diminishing coverage is the submodularity of coverage (Nemhauser et al., 1978); the certification bound is a two-point argument (Tsybakov, 2009); and the propensity model is related to labeler-expertise models (Whitehill et al., 2009). The set-reporting setting adds three things. First, a deleted record can lose fewer matches than it held, because a remaining record may take over its target, so the threshold concerns the marginal matching rate p_1 rather than precision, and p_1 estimated on five labelled images, with the task family's default as the starting point, predicts the effect of deletion on other images. Second, a selector and an oracle can be compared fairly only within the same feasible family, which the constrained oracle provides. Third, a view can confirm or reject only the records it contains, which makes cross-view confirmation coverage-aware. The chain itself is the unifying contribution: the same three quantities, matching size, scored record count and readout, carry these results across counts, boxes, screen elements and clicks.

Benchmarks. We use SKU-110K (Goldman et al., 2019), DOTA (Xia et al., 2018), DocLayNet (Pfitzmann et al., 2022), FSC-147 (Ranjan et al., 2021), PixMo-Count (Deitke et al., 2025), ScreenSpot-Pro (Li et al., 2025b), ScreenSpot-v2 (a corrected release of ScreenSpot; Cheng et al., 2024), GroundCUA (Feizi et al., 2026) and ScreenParse (Gurbuz et al., 2026), together with a synthetic counting probe whose labels are exact.

3 Problem Setup

3.1 Terms

Table 1 defines the words used throughout the paper in plain language; the formal objects follow.

Table 1: Terms used in this paper.

Term	Meaning
call, response	One request to a model API, and the raw text it returns. Every attempted call is stored and counted in the cost.
interface	Everything that defines a call except the image: model, reasoning setting (*lowest* or *default*), output cap, prompt, output schema, and the view of the image that is sent.
target	A true object in the image, given by the annotation (a point or box with a class, or the box of the element a click should hit).
record	One object as reported by the model: a class name and a location (a point or a box).
list	The sequence of records in one response. A *list-first* prompt asks for the list before any counts; a *count-first* prompt asks for counts first.
stated count	The numbers the model writes for how many objects of each class it found (for example the field `counts_by_type`).
list count	The number of records of each class in the list, counted by code after the alias map.
readout gap	The difference between the list count and the stated count of the same response. A gap shows that the two disagree; it does not by itself say which is right.
alias map	A declared, task-specific mapping of class names the model uses to the task's classes ("diamond" $\rightarrow$ "square" for rotated squares in our synthetic scenes).
decoding, status	Turning a response into records, stated counts and a status: whether the call returned, whether the text parses as the requested object, which records are valid, and four list-level *degeneracy flags* (Section 5). A *degenerate* list is one with at least one flag.
audit and delivered records	The audit collection holds every record the decoder could recover; the delivered collection holds the records that are valid under the task contract (Table 2). Set scores evaluate the delivered collection.
matching, matched record	Pairing records with targets one-to-one under the task's rule (same class and within a distance or overlap threshold). The *matching size* is the largest number of such pairs. A record in such a pair is *matched*.
set error E, F1	E is the number of unmatched records plus the number of unmatched targets. F1 is the harmonic mean of precision and recall.
readout	The rule that turns a set or a response into the task answer, for example a count per class or one click point.
construction	A rule that builds one set from the records of several calls, for example keeping only records that all three calls agree on (*unanimity*), at least two of three (*majority*), both of two (*intersection*), or any (*union*). *Fused* coordinates average the agreeing records; *deletion only* keeps the base call's own record.
base call, partner	In a confirmation, the call whose records may be deleted, and the call (same model or another model) used to confirm them.
p_1	In a confirmation, the *marginal matching rate* of the base call's unconfirmed records (those the partner does not report): how much the matching size falls per deleted record. It equals their precision unless another record can take over a deleted record's target. Deleting them lowers E when $p_1 < 1/2$ (Proposition 2).
view, tile, two-stage zoom	The part of the image sent in a call. Tiles are overlapping crops sent in separate calls. Two-stage zoom (ReGround, Li et al., 2025b) first asks for a click on the whole screen, then asks again on a crop around that point. The *apparent size* of an object is the number of visual tokens across its shorter side in the image as sent. A *view planner* is any procedure that chooses the views, such as tiling or the safe decomposition of the companion paper.
disagreement alarm	An observable signal that two answers for the same item disagree by more than a threshold. In the risk-coverage analysis of Section 7.6, alarmed items are set aside to trace a risk-coverage curve; in the several-model click tier (Algorithm 2) the alarm triggers a visual comparison, and every item stays in the denominator.
acceptance coverage	In the risk-coverage analysis, the share of items acted on without an alarm. It differs from *candidate coverage*, the share of items with at least one correct candidate, which is measured with the annotations.
candidate, cluster, representative	For a click, a candidate is one point from one call; clusters group candidates within a distance; each cluster's representative is the member the tier would output for it (Section 4.4).
source, cell	A dataset, and one model on one dataset. The six set-task sources are the four *E3 sources* (SKU-110K, DOTA, DocLayNet, FSC-147) and the two *screen sources* (GroundCUA, ScreenParse).
shipped configuration	The fixed values the method uses on any new image (interface and setting per model and loss, flag thresholds, construction per loss, alarm partner, click tier), chosen on our evaluation data and released as a file with the code (Section 5).
over- and under-reporting	Task families where models mostly report false objects (detection, layout, screen parsing) or mostly miss objects (dense counting with points).

3.2 Task

A *visual set-reporting task* τ fixes the following: (i) a class vocabulary $\mathcal{C}_\tau$, with a declared alias map α_τ from strings to $\mathcal{C}_\tau \cup \{\perp\}$ (the map is part of the task, not a global rule); (ii) a record schema (c, g) with class $c \in \mathcal{C}_\tau$ and geometry g (a point, an axis-aligned box, or a click point); (iii) a *compatibility relation* $\rho_\tau(z, y) \in \{0, 1\}$ between a record z and a target y, which decides whether they may be matched; (iv) an *ignore region* $\Omega_\tau(x)$, possibly empty, in which an unmatched record is not scored; (v) a loss ℓ_τ: weighted set error, F1, per-type count error, or click miss. An image x has a true target multiset $Y = Y(x)$ with $n = |Y|$.[1] Table 2 lists, for every task in the paper, the compatibility relation used for scoring, the separate rule used to associate records across calls, and what the decoder does with records that are not valid.

3.3 Reporting chain

One call with interface η consists of a model, a reasoning setting, an output cap, a prompt, an output schema and a view (crop or tile set). The chain (Figure 2) is

$$H_\eta \sim K_\eta(\cdot \mid x, \tau) \xrightarrow{D_{\eta,\tau}} (Z^{\text{all}}, Z, d, \sigma) \xrightarrow{C} S \xrightarrow{r_\tau} \hat{y}.$$

H_η is the raw response text, which is always stored. $D_{\eta,\tau}$ is a deterministic decoder. It returns the audit collection Z^{all} of every recoverable record; the delivered collection $Z \subseteq Z^{\text{all}}$ of records that are valid under the task contract (Table 2), with aliases normalised by α_τ and coordinates in image pixels; the stated count vector $d \in \mathbb{N}^{\mathcal{C}_\tau} \cup \{\varnothing\}$; and a status vector σ that records, among other things, how many records were excluded. Set scores in this paper evaluate the delivered collection; where exclusion is large we also report a sensitivity that keeps the excluded records and scores them as unmatched (Table 9). C is a construction that maps the delivered records of $K \geq 1$ calls, possibly with different interfaces, to a final set S inside a declared feasible family. r_τ is the task readout, for example the per-class count of S, or d when a declared fallback applies. The objective of a procedure π is

$$\mathcal{R}(\pi) = \mathbb{E}\, \ell_\tau(\pi(x), Y) + \lambda\, \mathbb{E}\, \text{cost}(\pi; x),$$

where the cost counts every attempted call, including failed, truncated or retried ones.

3.4 Records, clusters and feasible sets

Calls $1, \ldots, K$ give delivered collections $Z_1, \ldots, Z_K$. A label-free correspondence merges them into clusters $\mathcal{Q} = \{Q_1, \ldots\}$: it first maximizes the number of associated pairs under the association rule of Table 2, then their overlap, against cluster representatives. A cluster has a support set $\text{sup}(Q) \subseteq \{1, \ldots, K\}$ (the calls that reported it), member records, and a fused record $\bar{z}_Q$ (the coordinate mean of its members, with the class of the first member). A construction chooses, for each cluster, either nothing or one representative from a declared candidate set $A(Q)$, so its feasible family is

$$\mathcal{S}_A = \Big\{ \bigcup_Q s_Q : \; s_Q \in \{\varnothing\} \cup \{\{a\} : a \in A(Q)\} \Big\}.$$

Support-k rules use $A(Q) = \{\bar{z}_Q\}$ and keep Q iff $|\text{sup}(Q)| \geq k$. Deletion-only intersection uses the base call's own record. A record gate uses $A(Q) = \{\bar{z}_Q\} \cup \text{members}(Q)$. All oracles and fitted selectors compared with a construction use the same $\mathcal{S}_A$.

3.5 Output contract and status

The decoder reports four status layers for every call. They are distinct, and passing a lower layer never implies a higher one. **L1 transport**: the call returned, and its finish reason (stop, length) is recorded. **L2 outer syntax**: the whole response parses as the requested JSON object after stripping a code fence. **L3 record validity**: which records were recovered (strict parse, or a declared salvage of individual objects when the whole response does not parse), which of them are valid under the task contract, and four list-level degeneracy flags (Section 5). **L4 task completeness**: whether S contains every target. L1 to L3 are checked by code; L4 cannot in general be checked without labels (Section 4.5). The procedures return S, $\hat{y}$ and σ for every input, never drop a call from cost or coverage accounting, and report the strict-JSON failure rate, so the share of salvaged responses is visible.

3.6 Losses

For a scored collection S, let $r(S)$ be the matching size: the largest number of disjoint pairs (z, y) with

[1] Notation differs from the companion paper (Shi et al., 2026): there S is the number of visual tokens across a target side and L the content of a call; here S is a reported set and L1 to L4 are status layers. That paper's S is the apparent size that enters the view effect γ_v of Section 4.2. Our point tolerances (20 or 30 pixels on the synthetic scenes) are about two to three times that paper's tolerance of half the target size.

Table 2: Task contracts. Scoring compatibility decides matches with the truth; association decides which records of different calls are the same object, and uses no labels. Coordinates are mapped from each model's native frame to image pixels before scoring. The detection rows and the multi-class count row use overlapping datasets but different images and different contracts.

Task	Scoring compatibility	Association across calls	Records that are not valid
SCP (synthetic, 100 images)	same class, within 30 px	not used	class unknown after the alias map: kept for status, excluded from the delivered set
SCP-dense (50 images)	same class, within 20 px	not used	as for SCP
Counts, one class (PixMo-Count, FSC-147; Figure 4)	absolute error of the count against the annotated number; no matching	not used	every decoded point is counted
Counts, several classes (30 DOTA windows, 50 DocLayNet pages; Table 8)	per-class count error; DOTA objects marked difficult or less than half inside the window are not targets and there are no ignore regions, so records on them are counted like any other record; list recall (diagnostic only): a record centre inside a true box of the same class, one to one	not used	class unknown after the declared alias map: kept in the response, not counted
Detection boxes (SKU-110K, DOTA, DocLayNet)	same class, IoU ≥ 0.75; unmatched records in ignore regions not scored	same class, IoU ≥ 0.5	label outside the vocabulary or box outside the image: kept for audit, excluded from the delivered set, counted as invalid
FSC-147 points	within half the mean short side of the three exemplar boxes	within the same distance	point outside the image: kept for audit, excluded and counted as invalid
Screen parsing (GroundCUA, ScreenParse)	class-agnostic, IoU ≥ 0.5	IoU ≥ 0.5	coordinates clamped to the image; boxes of zero area dropped
Clicks (ScreenSpot-Pro, ScreenSpot-v2)	point inside the target box	alarm: distance above a share of the longer image side	missing point counted as a miss

$\rho_\tau(z, y) = 1$. Let $c(S)$ be the number of scored records: those matched in the scorer's maximum matching or lying outside Ω_τ. Without an ignore region, $c(S) = |S|$. With one, c can depend on which maximum matching the scorer picks; the relations below hold for any fixed scorer applied consistently, since they are computed from the same c and r. Ignore regions occur in 34 of 240 detection items and in no screen item. The set losses are functions of r, c and n; the count losses also depend on the readout. With costs $c_{\mathrm{FP}}, c_{\mathrm{FN}} > 0$,

$$E_w(S) = c_{\mathrm{FP}}[c(S) - r(S)] + c_{\mathrm{FN}}[n - r(S)],$$
$$\mathrm{F1}(S) = 2\, r(S)/(n + c(S)),$$

with $\mathrm{F1} = 1$ when $n = c(S) = 0$. E denotes E_w with $c_{\mathrm{FP}} = c_{\mathrm{FN}} = 1$; recall is $r(S)/n$. F1 in our tables is the mean of per-image F1. The per-type count error of a readout $\hat{y}$ is $\sum_c |\hat{y}_c - n_c|$, and the click loss is $1 - \mathbb{1}[\hat{p} \in \mathrm{box}(y)]$. Every reported number names the number of calls K, the interface, the setting, and the loss.

4 Theory

The results fall into five groups. Section 4.1 gives exact relations that hold for any outputs and any fixed scorer: what deleting records, averaging their coordinates, or reading a different field does to the loss. Section 4.2 states a behavioural model of how models report objects across repeated calls, views and models, and the conditions it implies; unlike the relations, it is an assumption to be checked. Section 4.3 compares constructions and selectors over a declared family of candidate sets. Section 4.4 splits the error of a single answer chosen among several calls into three parts and gives the exact effect of adding a source and of switching an answer. Section 4.5 states what outputs alone cannot certify. Proofs are in Appendix A.

4.1 Exact relations between operations and losses

Lemma 1 (Matching size)**.** *Fix the targets, a ground set of candidate records and the compatibility graph between them. Then $S \mapsto r(S)$ is*

the rank function of a transversal matroid (Edmonds and Fulkerson, 1965): it never decreases when records are added, increases by 0 or 1 when one record is added, and is submodular (adding a record helps a smaller set at least as much as a larger one).

Proposition 1 (Deletion and addition)**.** *Let $S \subseteq A$, let $\Delta c = c(A) - c(S)$ be the net decrease in scored records and $\Delta r = r(A) - r(S)$ the number of matches lost. Then exactly*

$$E_w(A) - E_w(S) = c_{\mathrm{FP}}\,\Delta c - (c_{\mathrm{FP}} + c_{\mathrm{FN}})\,\Delta r,$$

and in particular $E(A) - E(S) = \Delta c - 2\Delta r$. Addition is the same relation read in reverse.

Proposition 2 (Intersection threshold)**.** *Deleting $A \setminus S$ lowers weighted set error if and only if $c_{\mathrm{FP}}\,\Delta c > (c_{\mathrm{FP}} + c_{\mathrm{FN}})\,\Delta r$, that is, for $\Delta c > 0$, iff $\Delta r/\Delta c < c_{\mathrm{FP}}/(c_{\mathrm{FP}} + c_{\mathrm{FN}})$. Adding a non-empty set lowers weighted set error iff $(c_{\mathrm{FP}} + c_{\mathrm{FN}})$ times the matches gained exceeds c_{FP} times the scored records added.*

In words: deleting records helps when few enough of them were matched. Over a population of images with $\mathbb{E}[\Delta c] > 0$, the expected change has the sign of $c_{\mathrm{FP}}/(c_{\mathrm{FP}} + c_{\mathrm{FN}}) - p_1$, where $p_1 := \mathbb{E}[\Delta r]/\mathbb{E}[\Delta c]$ is the ratio of expected matches lost to expected records removed. We call p_1 the *marginal matching rate* of the deleted records; it is not the average of per-image precisions, and it equals the precision of the deleted records only when no remaining record can take over a deleted record's match. For E the threshold is $1/2$. This is the cost-sensitive decision threshold (Elkan, 2001) applied to the records a partner does not confirm; what is specific to record sets is that Δr can be smaller than the number of matched records deleted, so the relevant rate is p_1 and not their precision. The proposition does not assume that the confirming report comes from the same model, the same view, or any probability model. It concerns deletion only: the retained records keep their coordinates.

Corollary 1 (F1 under deletion)**.** *If $n + c(A) > 0$ and $n + c(S) > 0$, then*

$$\mathrm{F1}(S) - \mathrm{F1}(A) = \frac{2\{r(A)\Delta c - [n + c(A)]\Delta r\}}{[n + c(A)][n + c(S)]},$$

so for $\Delta c > 0$ deletion raises F1 on that image iff $\Delta r/\Delta c < \mathrm{F1}(A)/2$. If $n = 0$, $c(A) > 0$ and every scored record is deleted, F1 rises from 0 to 1.

This is the marginal form of the known result that the F1-optimal threshold of a calibrated classifier is half its optimal F1 (Lipton et al., 2014). Because $\mathrm{F1}(A)/2 \leq 1/2$, a deletion that raises F1 on an image also lowers E there, but not conversely. The threshold is per image; a mean F1 over images changes by the mean of the displayed differences, not by comparing a pooled rate with an average threshold. In 22 of our 54 base/partner pairs, the pooled marginal matching rate lies between half the base call's mean F1 and $1/2$; since the exact condition is per image, this count indicates a direction, consistent with deletion helping E more often than F1 (Section 7.9); which F1 policy to ship is an empirical choice among all constructions, not a consequence of this corollary. In the deletion case, the threshold is the same exchange rate as the price of a false output in the companion analysis of decomposition (Shi et al., 2026).

Proposition 3 (Geometry term)**.** *Let $S_0 \subseteq A$ be the confirmed records with their original coordinates and S_f the same clusters with fused coordinates. Then*

$$E(A) - E(S_f) = \underbrace{[E(A) - E(S_0)]}_{\textit{selection (Prop. 1)}} + \underbrace{[E(S_0) - E(S_f)]}_{\textit{geometry}},$$

and the geometry term equals $[c(S_0) - c(S_f)] - 2[r(S_0) - r(S_f)]$.

By convexity, the mean of a cluster's members is no farther, in mean squared distance, from any fixed target than the members are on average. That does not make every fused record closer than its best member, and matching uses a threshold, so the geometry term can have either sign. On DocLayNet with GPT-6.1 confirmed by Sonnet 5.5 (60 images), $E(A) = 8.567$, $E(S_0) = 7.767$ and $E(S_f) = 8.867$: selection gains 0.800, geometry loses 1.100, and the fused intersection is worse than the single call although deletion alone is better.

Proposition 4 (Two readouts of one response)**.** *Let a decoded response have delivered records Z and stated counts $d \neq \varnothing$, and let $\mathrm{lc}(Z)_c = \#\{z \in Z : \mathrm{class}(z) = c\}$ be the list count. Both readouts come from the same response, so the difference between their errors $\|d - n\|_1$ and $\|\mathrm{lc}(Z) - n\|_1$ isolates the readout and holds generation fixed. The gap $\|d - \mathrm{lc}(Z)\|_1$ is observable without labels,*

and

$$\max\{\|d-n\|_1, \|\mathrm{lc}(Z)-n\|_1\} \;\geq\; \tfrac{1}{2}\|d-\mathrm{lc}(Z)\|_1,$$

$$\big|\,\|d-n\|_1 - \|\mathrm{lc}(Z)-n\|_1\,\big| \;\leq\; \|d-\mathrm{lc}(Z)\|_1.$$

A positive gap shows that at least one readout is wrong; it does not show which. A small gap shows that no choice between the two readouts can change the error by more than the gap, whatever the truth: choosing a readout can never gain more than the gap, and errors that both readouts share, such as objects missing from the list and from the statement alike, are beyond its reach. With a list count of 9 and a stated count of 10, the gap is 1 whether the truth is 9 or 10. Which readout to use is therefore an empirical question per model and task family; the method's rule (list count unless the list is degenerate) is supported by the measurements of Section 7.2, not by this proposition.

4.2 Behaviour under repetition, changed views and changed models

Joint reporting law. Fix a complete interface η: model, prompt, schema, setting, cap and view. A call with interface η reports a true object o with probability $q_{o,\eta}$, and a candidate false object f (a location where false records tend to appear) with probability $q_{f,\eta}$. Given these propensities, repeated calls with the same η are independent. This joint law, with q free to depend on every part of η, is the behavioural model; the consequences below use only it, together with conditional independence given the propensities for calls with different interfaces wherever that is stated. To describe how propensities vary with the model and the view when the rest of the interface is held fixed, that is, for calls that share the prompt, schema, setting and cap, we use the explanatory parametrization

$$\mathrm{logit}\, q_{o,\eta} = a_o + b_{o,m(\eta)} + \gamma_{v(\eta)}(o),$$

$$\textstyle\sum_m b_{o,m} = 0, \qquad \gamma_{v_0}(o) = 0,$$

where v_0 is the whole image, a_o is common to all models (how hard the object is), $b_{o,m}$ is specific to model m, and $\gamma_v(o)$ is the change from the whole image to view v, which depends on the apparent size of o in v. All three terms may differ between prompts, schemas, settings and caps; a comparison that changes one of these compares two such parametrizations, not two terms of one. The form is related to item-response and labeler-expertise models, in which an item's difficulty and a rater's ability combine on the logit scale (Whitehill et al., 2009) (there as a product, here additively, as in Rasch-type models); here the items are objects and the raters are models and views. The two constraints are conventions that make the parts well defined: centering separates the shared part from the model-specific one, and the reference view separates both from the view effect. We do not estimate the parts; the parametrization motivates the predictions below, each of which is tested directly. A reported record contributes to the loss only through the matching: emitting a record near o, producing a match, and adding a match that no other record already provides are different events.

Three consequences are used in the paper. *Persistence*: for fixed η, the probability of missing o in all K calls is $(1-q_{o,\eta})^K$, and averaged over objects it is $\mathbb{E}[(1-q_{o,\eta})^K]$, which tends to the share of objects with $q=0$. The decay is slow when q is small for some objects, so at a fixed budget many misses repeat. *Diminishing coverage*: for calls that are exchangeable given the image, the expected number of targets matched by the union of K calls is concave in K; this follows from the submodularity of coverage (Nemhauser et al., 1978) and of matching rank (Lemma 1). The statement concerns raw coverage, not the loss of a constructed set: repeated calls can still remove false records that do not repeat. For example, with 20 true records that are always reported and 80 candidate false records reported independently with probability 0.2, one call has expected $E=16$ and three-call unanimity has expected $E=0.64$. *Confirmation rates*: if the records of two calls refer to the same candidate objects and the two calls are conditionally independent given the propensities, a partner call B confirms a record of call A with probability

$$h_{\pm} = \frac{\mathbb{E}[q_A q_B \mid \text{true / false}]}{\mathbb{E}[q_A \mid \text{true / false}]},$$

with h_+ for true records and h_- for false ones. For a second call with the same interface $q_B = q_A$, so $h = \mathbb{E}[q_A^2]/\mathbb{E}[q_A]$. A review call that receives the first answer has a different interface and is not covered by this formula.

Proposition 5 (When a different partner is better)**.** *Assume deletion-only confirmation, no match takeover (so Δr equals the number of true records deleted), no ignore region or all of A's false records outside it, and expected counts T (true) and F (false) among A's records, with $h_{\pm}^B$ averaged over*

these records. Then the expected reduction in set error E (equal costs) from replacing A by $A \cap B$ is

$$G(B) = F\,(1 - h_-^B) - T\,(1 - h_+^B),$$

and a partner B beats a same-interface partner A' iff $F\,(h_-^{A'} - h_-^B) > T\,(h_+^{A'} - h_+^B)$.

The partner must reject more of A's false records than it loses of A's true ones, weighted by their counts. Neither similar overall quality nor a larger separation $h_+ - h_-$ is necessary or sufficient: with $T = 100$ and $F = 10$, the pair $(h_+, h_-) = (0.95, 0.85)$ gives $G = -3.5$, while the more separated pair $(0.5, 0)$ gives $G = -40$; and a weaker partner whose errors avoid A's false records can satisfy the inequality. Without the assumptions, the exact relation of Proposition 1 replaces these expectations.

Reports from several views. Nothing in the chain assumes that a report comes from one call on the whole image. A view is part of the interface η, so a view planner (tiling, the safe decomposition of the companion paper, zoom) produces calls like any other, and their records, mapped to image coordinates, enter the same constructions and losses. Merging the reports of several views is itself a construction: the union of their records, with duplicates in the overlaps removed by the association rule. Confirmation across views is covered by Proposition 2 and Proposition 5 with one change. A view can report only objects that it contains, so for a record lying wholly inside no partner view, $h_+^B = h_-^B = 0$: deleting it is a decision about that record's own p_1, on which the partner carries no information. A view should therefore confirm or reject only the records it contains. A different view changes the propensities, but it need not share less of the persistent component: a view can raise how often objects are reported while the misses of the two calls stay as correlated, or become more so. For a true object drawn at random from the image, let A and A' indicate that two calls with the same interface report it and B that a call on another view does, with means $\mu_A = \mu_{A'}$ and μ_B, the calls being conditionally independent given the propensities. Then

$$\begin{aligned} P(A = 0, B = 1) - P(A = 0, A' = 1) \\ = (1 - \mu_A)(\mu_B - \mu_A) \\ + \operatorname{Cov}(A, A') - \operatorname{Cov}(A, B) : \end{aligned}$$

the first term is the gain in how often objects are reported, the second the gain in complementarity. For example, with two equally likely kinds of object reported with probability 0.2 and 0.4 on the whole image, a view that adds $\log 2$ to every logit gives $1/3$ and $4/7$: the mean rises from 0.30 to 0.45 and the covariance with the first call rises from 0.0100 to 0.0119, yet the probability that the first call misses an object and the second reports it is 0.305 for the new view against 0.200 for a repeated call: the gain in how often objects are reported (0.107) more than offsets a small loss of complementarity (0.002). Whether a view helps as a partner is again the inequality of Proposition 5, which also counts its false records.

Proposition 6 (Alarm from disagreement)**.** *Consider a point task in which a record matches a target of the same class within distance θ, with no ignore region, and let E be its set error. Let $U_{2\theta}(A, B) = |A| + |B| - 2\,m_{2\theta}(A, B)$, where $m_{2\theta}$ is the matching size between two reports when records of the same class are paired within 2θ. Then*

$$E(A) + E(B) \;\geq\; U_{2\theta}(A, B),$$

hence $\max\{E(A), E(B)\} \geq \frac{1}{2}\,U_{2\theta}(A, B)$.

The bound needs no labels: when $U > 0$, at least one of the two reports is wrong, although the bound does not say which. Constructed instances attain it, and it can be loose: two identical wrong reports give $U = 0$. For boxes with no ignore region the analogous bound uses IoU $\geq 2t - 1$; at $t = 0.5$ the overlap threshold becomes 0, so the bound keeps differences in the number of records per class but loses all geometric information. The click alarm in our experiments (distance above a share of the longer image side) is an empirical alarm for the hit-in-box loss, evaluated by risk-coverage curves.

4.3 Construction, selection and approximation

Fix a feasible family $\mathcal{S}_A$ and a loss. Define the oracle risk $R_{\text{or}} = \mathbb{E}[\min_{S \in \mathcal{S}_A} \ell(S, Y)]$, which uses the truth to pick the best set; the best risk $R_\Phi^* = \inf_g \mathbb{E}[\ell(g(\Phi), Y)]$ of a rule that maps features Φ computed from outputs alone into $\mathcal{S}_A$; and the risk $R_{\hat{g}}$ of a fitted rule.

Proposition 7 (Risk ordering)**.** $R_{\text{or}} \leq R_\Phi^* \leq R_{\hat{g}}$,

and

$$R_{\hat{g}} - R_{\mathrm{or}} = \underbrace{(R^*_\Phi - R_{\mathrm{or}})}_{\textit{information not in } \Phi} + \underbrace{(R_{\hat{g}} - R^*_\Phi)}_{\textit{selector and estimation}} .$$

An observed gap between a selector and the oracle bounds the sum but does not apportion it. A rule whose fitting uses labels is a *label-fitted output-feature selector*: it reads no labels at inference, but its fitting does. The gap concerns a fixed library of calls; it says nothing about what additional calls or a different library would add.

Proposition 8 (Constrained oracle)**.** *For $\mathcal{S}_A$ with at most one representative per cluster, including the option of choosing none, $\min_{S\in\mathcal{S}_A} E(S) = n - |M|$, where M is a maximum matching in the bipartite graph between clusters and targets that has an edge whenever some $a \in A(Q)$ is compatible with the target. It is attained by selecting one compatible representative for each matched cluster and nothing for the others.*

Unrestricted matching over all records can exceed $|M|$ when two members of one cluster match two targets; that value is not attainable within $\mathcal{S}_A$ (Table 13 reports both).

Proposition 9 (Approximate-model guarantee)**.** *If a decision rule is optimal under a reporting law $\tilde{\Lambda}$, and the true law Λ satisfies $\sup_\pi |\mathcal{R}_\Lambda(\pi) - \mathcal{R}_{\tilde{\Lambda}}(\pi)| \le \varepsilon$, then its regret under Λ is at most 2ε.*

This is the basis for choosing a configuration on source data and carrying it to a new source within a task family. A rule that is only approximately optimal under $\tilde{\Lambda}$, for example one estimated from finite data, adds its own suboptimality to the bound.

4.4 Choosing one answer among several calls

A click has one answer, so the constructions above reduce to choosing one point. Let several calls (models, settings, views) give an item a menu A of candidate points. Group the candidates into *clusters*, the connected components of the graph that joins two candidates within distance δ; fix one *representative* per cluster before any choice is made; and let a rule output one representative. The loss is the miss, $\mathbb{1}[\hat{y} \notin \text{target}]$.

Proposition 10 (Three sources of a miss)**.** *For every rule that outputs a representative, the miss rate is exactly $F + L + W$, where F is the probability that no candidate in A hits, L that some candidate hits but no representative does, and W that some representative hits but the rule outputs one that misses.*

The terms locate where a miss arises; they are not independent effects of interventions. With the candidate menu fixed, changing the representatives leaves F unchanged but can move a miss between L and W. With candidates and representatives fixed, changing only the final choice acts on W. Only changing or enlarging the candidate menu can lower F, and doing so can change all three terms at once, because a new candidate can also change the clusters, the representatives and the comparison. Section 7.7 measures the three terms separately when a source is added. How much an added candidate lowers F needs no independence assumption:

Proposition 11 (Coverage of an added source)**.** *Let F be the probability that every candidate of a menu misses, and $\rho = \Pr(\textit{an added candidate hits} \mid \textit{every earlier candidate misses})$. With the added candidate, the probability becomes $F(1-\rho)$. If the candidates are conditionally independent given per-item hit probabilities $q_1, \dots, q_k$, then $F = \mathbb{E}[\prod_k (1-q_k)]$.*

The relevant quantity is the rescue rate ρ on the items the existing menu misses, not the added source's own accuracy or its correlation alone. For hit indicators A and B with $\Pr(A=0) > 0$,

$$\Pr(B = 1 \mid A = 0) = \mathbb{E}[B] - \frac{\mathrm{Cov}(A, B)}{1 - \mathbb{E}[A]},$$

the binary form of the decomposition into reporting rate and complementarity in Section 4.2. At equal hit rates, lower covariance raises the rescue rate; when hit rates differ, both terms matter. Persistence makes the covariance of a repeated call with the same interface large, which is why a repeat tends to rescue little, but a weak partner can rescue less than a repeat. Section 7.7 therefore measures ρ for each tested source.

Proposition 12 (Net effect of a switch)**.** *Let a base rule output b, and let an action replace b by g on a set T of items decided from outputs. With $e_T = \Pr(b \textit{ misses} \mid T)$, $r = \Pr(g \textit{ hits} \mid b \textit{ misses}, T)$ and $h = \Pr(g \textit{ misses} \mid b \textit{ hits}, T)$, the miss rate falls by exactly*

$$\Pr(T)\,\big[\, e_T\, r - (1 - e_T)\, h \,\big].$$

If, within an event $U \supseteq T$, membership in T is independent of whether b misses, then $e_T = \Pr(b \text{ misses} \mid U)$.

This is the click counterpart of Proposition 2: an action pays when the errors it repairs outweigh the correct answers it breaks, counted on the items it touches. Disagreement alone does not say which candidate is wrong (Proposition 6 bounds the sum of two errors, not the base error), so where to act is an empirical question. On the 600 new screenshots of Section 7.7, the judge's score alone misses 32 of the 101 items whose candidates form two or more clusters (31.7%) and 29 of the 499 whose candidates form one (5.8%); this enrichment, also seen on the 260 design items (10 of 39 against 3 of 221), is consistent with comparing clusters only on items with an alarm. For an action applied to all single-cluster items, or through a trigger whose firing is independent of whether the base misses, $e_T \approx 5.8\%$, and the action improves accuracy iff $h < e_T\, r/(1 - e_T)$: about 3.1% of the correct answers it touches if it repairs half of the misses, and about 6.2% if it repairs all of them. The judge that scores and compares candidates is a design choice of the configuration and not a consequence of the theory: choosing the best of M judges to within Δ in miss rate with probability $1 - \eta$ takes on the order of $\log(M/\eta)/\Delta^2$ labelled items under independent sampling, so the pipeline ships one judge and does not choose it per task.

4.5 What outputs alone can and cannot certify

Proposition 13 (No output-only certificate)**.** *Fix a reported set S, a loss L and a tolerance ϵ. Suppose two admissible worlds induce the same law P of the observable output history H, while $L(S, Y_0) \le \epsilon$ in the first and $L(S, Y_1) > \epsilon$ in the second. Then every output-only acceptance rule $a(H) \in [0, 1]$, possibly randomized, accepts with the same probability in both worlds, so no such rule can accept with probability at least $1 - \beta$ in the first world and at most α in the second if $\alpha + \beta < 1$. More generally, $P_1(\text{accept}) \ge P_0(\text{accept}) - \mathrm{TV}(P_0, P_1)$, where TV is the total variation distance.*

The same argument applies to any property of Y on which two such worlds disagree, for example whether $L(S, Y) < L(S', Y)$ for two reports S and S', so which of two reports is more accurate cannot be certified from outputs alone either. The argument is the two-point method of minimax lower bounds (Tsybakov, 2009); methods that estimate accuracy from the agreement of several classifiers without labels (Platanios et al., 2014) rely on assumptions, such as weakly dependent errors, that rule such pairs out. Such pairs arise from objects that a model rarely reports. If the bad world adds one target that each of K calls reports with probability at most ϑ, given the earlier output, and the target does not change the rest of the output, then $\mathrm{TV}(P_0, P_1) \le K\vartheta$. Any test's advantage in telling the two worlds apart, the difference between its acceptance probabilities in them, is then at most $K\vartheta$ (with equal priors, the best probability of naming the right world is $(1 + \mathrm{TV})/2$), and no test is reliable uniformly over admissible classes in which ϑ can be arbitrarily small. The practical reading: agreement alone does not certify correctness. This does not prevent running a default procedure, ordering interfaces under explicit assumptions tested on other data, as the companion paper does for decomposition (Shi et al., 2026), or operating a configuration chosen on source data without new labels.

Corollary 2 (Sign of a count gap)**.** *Suppose that, for a given report S, some admissible world has $|S| > n$, and that the admissible worlds are closed under adding any number of targets that no interface reports and that do not change the rest of the output. Then the sign of $|S| - n$ is not determined by the outputs: that world has the same output law as one in which S under-counts. (This differs from the sign of a recall gain between two interfaces, which assumptions such as monotonicity in tokens per object can fix.)*

Without further information, neither deleting unconfirmed records nor keeping them dominates for a count objective. Our count procedure keeps them, a conservative choice whose quality we evaluate empirically.

5 Method: A Default Pipeline with Stated Inputs

The pipeline turns an image and a task into a final set, a task answer, a status and an alarm score. Every input it uses comes from one of four named sources: [T] the task specification given by the user (classes, alias map, matching rule, loss); [S] the shipped configuration, a fixed file released with the code (`config/default_pipeline.json`) whose values were chosen on our evaluation data and are carried unchanged to new images; [O] quantities computed from the current outputs; and [L]

Algorithm 1 Default reporting pipeline

Require: image x; task $(\mathcal{C}_\tau, \alpha_\tau, \rho_\tau, \Omega_\tau, \ell_\tau)$ [T]; budget B per item [T]; shipped configuration Θ [S]; optional labels [L]

1: **Dispatch.** From (τ, B, Θ) select the reporting interface, setting and call schedule, and send the selected request: counts, the list-first request (record list first, then counts "for the records you listed"; allowed class names; row-by-row scan; native coordinate frame; no overlays); sets, the list or regions interface that Θ gives the model and loss, K times with K set by the loss; clicks, one model's two-stage zoom, or Algorithm 2 if B covers its worst case; keep every attempt and its cost [T, S]

2: **Decode.** For each response, $H \mapsto (Z^{\text{all}}, Z, d, \sigma)$: audit records, delivered records, stated count, status L1 to L3 with the four degeneracy flags [O]

3: **Recover, at most once per call.** Transport failure (L1): resend the same request. Flagged response (L2 with no recoverable record, or a degeneracy flag): one follow-up that attaches the previous answer; keep it if unflagged, else keep the first response and its flags [S, O]

4: **Views.** Clicks: two-stage zoom on native pixels (ReGround, Li et al., 2025b). Sets and counts: the whole image, or the views of an external view planner if one is configured, one call per view, records mapped to image coordinates [S, O]

5: **Construct.** Merge the reports of all views; map the delivered records of all calls to the final set S with the construction Θ assigns to the loss (Table 3); a view confirms or rejects only records it contains [S]

6: **Read.** $\hat{y} \leftarrow$ list count of S; if the list behind S is flagged and $d \neq \varnothing$, $\hat{y} \leftarrow d$ with status `count_from_statement`; clicks: the final point [O]

7: **Optional labels.** ≥ 3 labelled images: switch to a menu configuration only if its mean paired advantage minus one standard error is positive; 5 to 10 images: may fit the record gate [L]

8: **return** S, $\hat{y}$, readout branch, status σ, alarm score (disagreement with a second report, from a second model if configured), total cost of all attempts

optional labels. No step requires the user to know p_1, a partner's quality or the current F1. Algorithm 1 gives the steps in order. Steps 1 to 6 repair the report, the optional label rule of Step 7 judges between configurations, and the alarm returned in Step 8 detects errors. Section 5.1 adds a second tier for clicks that combines several models, and Section 5.2 describes how the model behind these steps was revised.

Inputs [T, S]. The task fixes the class vocabulary $\mathcal{C}_\tau$, the alias map α_τ (for example {diamond, rhombus, rotated square} $\mapsto$ square), the matching rule ρ_τ of Table 2, the ignore region Ω_τ and the loss ℓ_τ. The loss selects the construction in Step 5. The shipped configuration fixes everything else that the algorithm calls a value: the prompt, the reasoning setting per model, the four flag thresholds, the recovery request, the view settings, the construction per loss, the alarm partner and the alarm thresholds. A model known to write degenerate lists at its lowest setting (Gemini 3.8) is run at its default setting, which at a fixed 16k output cap lowers its count readout error from 52.8 to 24.5 (Table 9).

Steps 1 and 2: dispatch and decode. Step 1 is a dispatcher, not one fixed request: what is shared across tasks is the chain of records, losses and readouts, not one output format. For set tasks scored by F1 the shipped configuration may select the regions interface, in which the model returns one row per region of a fixed coordinate registry (Appendix D). The count request asks for one JSON object in which the list of records comes first and `counts_by_type` follows, declared "for the shapes you listed" (Appendix D gives the prompt). The allowed class names are listed in the format; the model scans row by row from top to bottom and uses its native coordinate frame (0 to 1000 for GPT and Gemini; pixels for Sonnet, with the image pre-resized so that the provider does not resize it again). No grid, ticks or markers are drawn. The decoder keeps every recoverable record in the audit collection Z^{all} and passes the valid ones, after the alias map, to the delivered collection Z (Section 3). It sets four degeneracy flags on the list: more than 10% of records without a valid class; a list length outside $[0.5, 1.5]$ times the stated total; more than 10% repeated coordinates; and the finish reason "length".

Step 3: bounded recovery. The two kinds of failure get different recoveries. A transport failure, in which the call did not return, is a property of the connection and is resent unchanged. A flagged response is a property of how the model wrote this report, and such errors tend to recur, consistent with Section 4.2: in our detection experiments, resending a request whose flag had already repeated cleared the flag in 14.7% of cases, while changing the request cleared it in 58.8%. The change was a follow-up that attached the previous answer and asked for the complete answer under the same contract, or, for reports organised by image region, a new request for the flagged regions only. The pipeline sends the first kind of follow-up, once. If the follow-up still carries a flag, the first response is kept with its flags, and Step 6 decides how to read it. Every attempt enters the cost. The replay at the end of this section reports how often recovery was triggered and what the stored follow-ups changed.

Table 3: Shipped defaults for one model. *Transferred construction*: the best construction on the other five sources, applied to the evaluated source (Table 14). *Final default*: what the pipeline ships. They coincide for set error on over-reporting families and differ elsewhere; the last column gives the reason.

Loss and family	Transferred construction	Final default	Evidence and reason
E, over-reporting families (detection, layout, screen parsing)	three-call unanimity, averaged coordinates (chosen in all six folds)	the same	Significantly better in 9 of 15 cells, significantly worse in none, not significant in 6.
E, under-reporting family (dense points, FSC-147)	three-call unanimity, averaged coordinates	single call, no deletion	The transferred construction is worse in 2 of 3 cells. Deleting unconfirmed records raises E when their marginal matching rate p_1 is above 1/2 (Proposition 2); on FSC-147, p_1 is 0.39 to 0.90 and deletion hurts in 7 of 9 pairs (Table 10).
F1	three-call union (five folds) or majority (one fold)	single call; on detection sources, interface and setting chosen per model (GPT-6.1: regions, lowest; Sonnet 5.5: regions, default; Gemini 3.8: list, default)	The transferred construction is better in 2 of 18 cells, worse in 4 and not significant in 12. A configuration chosen on three detection sources is better on the fourth in 5 of 12 cells and worse in 1 (Table 15).
Count	not part of the transfer	single call; list count, with the stated count as declared fallback	Table 7.

Step 4: views. For clicks we use a re-grounding strategy in the style of the ReGround method that Li et al. (2025b) evaluate. The first call gives a point on the whole screenshot. A crop one third of the screenshot's width and height, centred on that point and shifted to stay inside the image, is sent at native pixels, and the second call gives the point in the crop. If the second call gives no point, the first point is kept; if the first call gives no point, there is nothing to zoom on and the answer is no point, which counts as a miss. Using agreement to decide when to zoom costs more and is not more accurate (Section 7.5). For sets and counts, the pipeline does not plan views itself: it works on whatever reports it is given. By default it sends the whole image. When a *view planner* is configured, for example a tiling scheme or the companion paper's safe decomposition, which places every target wholly inside some view, with neighbouring views overlapping by one object and no view having fewer tokens per object than the whole image, and cannot lower expected recall under that paper's monotonicity assumptions (Shi et al., 2026), each view is one call, its records are mapped to image coordinates, and Step 5 merges them. Leaving the planner and its trigger to the configuration is a measured choice. We computed the apparent size of every annotated object in the whole-image calls on SCP-dense, DocLayNet, DOTA and SKU-110K, using the input geometry measured in the companion paper and its released calibration data (32-pixel tokens for GPT-6.1, 28-pixel tokens for Sonnet 5.5, a fixed grid of about 1,090 tokens for Gemini 3.8). The median object spans 0.62 to 1.43 tokens across its shorter side (the companion paper measures size as the square root of the area, which is never smaller). Below that paper's design value of $S^\star = 1.5$ fall 55% to 79% of objects on the three detection sources and every shape on SCP-dense; yet GPT-6.1 and Sonnet 5.5 write nearly exact lists on SCP-dense. The companion paper likewise finds that these model families reach half recall at well under one token per object side. A fixed threshold would decompose images whose whole-image report is already complete, so when to call a planner is a property of the model and the task that the configuration supplies. These sizes come from the annotations and inform only this design choice; the pipeline reads no annotation at run time. Section 7.5 reports how the pipeline handles reports from four crops, and how our own tiling arm, which did not meet the safe condition, fared on large objects.

Step 5: construction. The construction is chosen per loss (Table 3). Records of different calls are associated without labels (Table 2): boxes of the same class with IoU ≥ 0.5, screen elements at IoU ≥ 0.5 without class, and points of the same class within the task's tolerance. On over-reporting task families and set error, three calls are made and a

record is kept only if all three report it, with its coordinates averaged over the three. Everywhere else a single call is delivered without deletion. Over- or under-reporting is a property of the task family, shipped with the configuration from our evaluation; the user does not supply it. Coordinate averaging is on by default because its contribution is positive in 37 of 54 base/partner pairs and negative in 13 (Table 10); the deletion-only variant is in the menu. Reports of several views are first merged into one report: the union of their records, where a record is dropped if it is associated with a record already kept from another view (a duplicate in an overlap). When reports from different views confirm one another, a view can confirm or reject only records that lie wholly inside it; a record that no partner view contains is kept, because that view carries no evidence about it (Section 4.2). A call that still has no readable answer after recovery counts as an empty report, so three-call unanimity keeps nothing on that image; this is the construction we evaluated, and dropping such a call from the support count is in the menu.

Step 6: readout. The answer is read from the final set. For counts it is the list count of S, the number of records of each class. When the list behind S carries a degeneracy flag and the response stated counts, the answer falls back to the stated count d, and the status says so (`count_from_statement`); a flagged list without stated counts keeps the list count and its flag. With several views, each response states counts for its own crop, and overlapping views would count shared objects twice; stated counts are therefore never combined across views, and a flagged merged list keeps the list count, with the status saying that no stated count was used. The fallback changes only the count: S itself is not repaired, so the count and the set each carry their own status. Proposition 4 explains why the rule is a fallback and not a selector: a large gap between list count and stated count shows that the two readouts of one response are inconsistent, but by itself it does not say which one is closer to the truth. The flags are the evidence that the list is broken.

A second model [S], optional. When a second model is configured, its default role is the alarm of Step 8: its answer is compared with the user's, and items where they disagree by more than a threshold are flagged. The partner is fixed by a rule (GPT-6.1 and Sonnet 5.5: Gemini 3.8; Gemini 3.8: Sonnet 5.5), which picks the cheapest other model per click call (Appendix D), and the threshold δ is a point the user picks on the risk-coverage curve (Figure 6). Using the second model to confirm records is a menu option. The cross-model construction transferred from the other sources improves set error in 12 of 18 transfer cells and F1 in 10 of 18, and is worse in 2 and 3 cells; all of the worse cells are cells in which the user's own model is the strongest single model for that source and loss (Table 14). Because the user may not know whether that is the case, cross-model confirmation is chosen with labels or a shipped family-level ranking, not as a zero-label default.

Step 7: optional labels [L]. Labels are never required. With fewer than three labelled images the pipeline keeps the shipped default and reports the labelled error. With three or more, it switches to another configuration from the menu only if that configuration's mean advantage over the labelled images, paired by image, minus one standard error is positive. For the decision to delete unconfirmed records, this rule predicts the better decision on the other images in 86% to 88% of draws with 3 to 10 labelled images, against 85% for the family default alone (Section 7.4). With 5 to 10 labelled images it may also fit the record gate, a logistic selector on output features (Section 7.8). Certifying the accuracy of a pipeline on a population is a separate service that needs its own labelled sample.

Step 8: delivery. The pipeline returns the final set S, the answer $\hat{y}$, the readout branch, the status σ (L1 to L3, flags, recovery used, views used), the alarm score when a second report exists, and the cost of every attempt.

One entry point, replayed on stored calls. The repository implements Steps 1 to 6 and 8 of Algorithm 1, including merging views and cross-view confirmation, as a single entry point (`fsr_pipeline/`) whose branches read only the task specification, the shipped configuration and the responses a client returns; the label rule of Step 7 is a function in the same module, and the record gate is fitted by the analysis code of Section 7.8. A replay client serves the stored responses of this paper instead of calling a model. A request for which nothing was stored, such as a follow-up that was never sent, returns "unavailable", and the pipeline continues on its documented branch. Run on raw responses, the entry point reproduces the component

analyses: per-type count error of 1.60 for GPT-6.1, 0.42 for Sonnet 5.5 and 24.52 for Gemini 3.8 at its default setting (52.80 at its lowest; Table 9); click hit rates of 90.0%, 87.0% and 87.5% after two-stage zoom on the 200 ScreenSpot-Pro items; and, with recovery switched off, the set error of the shipped construction in all 18 transfer cells to within 0.07 per image, a difference that comes from the 0.1-pixel coordinate rounding of the component analysis. Follow-ups were stored for a sample of flagged detection calls. In the 18 set cells, 171 image runs had at least one flagged call; a stored follow-up cleared the flag in 24 of them, and replaying these changes set error by at most 0.55 per image in any cell. No follow-ups were stored for the count experiments, where 51 runs were flagged, so their results are those of the first call. Synthetic boundary cases go through the same code, and each produces an answer, a readout branch and a status: an empty or unreadable response, truncated JSON, a list without stated counts, a degenerate list with and without stated counts, two transport failures in a row, one unreadable call among three, and a click with no first point or no point in the crop. The replay output is released with the analysis outputs (`pipeline_replay.json`).

5.1 Clicks from several models

Algorithm 1 delivers one model's two-stage zoom point. When the per-item budget covers the worst case of a second tier, the pipeline offers one built on the three sources of a miss (Proposition 10); its mean cost is about 3.2 times that of one zoom. Its run-time inputs are the image, the instruction and the budget; it reads no target label and does not abstain: it returns no point only when no call returned one. The models, their settings, the distance δ, the judge and the number of compared clusters come from the shipped configuration [S].

Algorithm 2 gives the steps, organised by Proposition 10. Generation enlarges the candidate menu, which is the only way to lower F; Section 7.7 measures the rescue rate of each source on the others' misses (Proposition 11). The representative step fixes which member of each cluster can be output, and the comparison chooses among representatives, so these two steps divide the remaining misses between L and W. The comparison runs only on items with an alarm, where the measured miss rate of the judge's score is highest (Proposition 12); items whose candidates form one cluster keep that cluster's representative. The judge's score is used only to rank; we do not treat it as a calibrated probability. The registry makes the point that is scored, drawn, compared and returned the same point: every step refers to a candidate by its identifier, and the code asserts that the coordinates drawn for the judge equal those returned. If a call fails, the item proceeds with the remaining candidates, and it counts as a miss only when no call returned a point. The budget rule uses the worst case, at per-call output caps of 16,384 tokens at the lowest setting and 32,768 at the default, because a mean cost is not a per-item guarantee; technical retries can add to either. On 600 new screenshots the tier costs $0.0273 per item on average (9.45 calls) against $0.0085 (2 calls) for GPT-6.1's two-stage zoom (Section 7.7), and no budget was imposed in our runs. The rules are implemented as one deterministic function over stored or live responses (`gui/code/decision.py` in the release), with request builders for each step; with the

Algorithm 2 Clicks from several models

Require: screenshot x, instruction and budget B [T]; models, settings, per-call output caps, δ, judge, source order, $m_{\max}$ [S]

1: **Tier.** Run this tier only if B covers its worst case at the per-call caps (15 calls besides technical retries: three first clicks, three zooms, three scores, at most six comparisons); otherwise run one model's two-stage zoom [T, S]
2: **Generate.** GPT-6.1 and Sonnet 5.5 at their default setting and Gemini 3.8 at its lowest setting each give a two-stage zoom point (Step 4 of Algorithm 1; a failed zoom keeps the first click, a failed first click gives no candidate); each point is registered as a candidate with an identifier bound to its coordinates on x, its model, setting and parent call [S, O]
3: **Detect.** Clusters are the connected components of the graph that joins candidates within $\delta = 1.5\%$ of the longer side; two or more clusters raise the alarm [O]
4: **Represent.** The judge (Gemini 3.8, lowest setting) sees a native crop with each candidate marked and returns a hit score $p \in [0, 1]$ (a missing or unreadable score counts as -1); each cluster's representative is its member with the highest p, ties broken by the source order GPT-6.1, Sonnet 5.5, Gemini 3.8, fixed before any comparison [S, O]
5: **Compare (alarm only).** Rank clusters by (representative's p, cluster size, earliest source in the cluster); for each pair among the first $m_{\max} = 3$, the judge sees both representatives marked with their identifiers on the downscaled screenshot and on a native crop around each, and names the one that hits the target; each pair is asked in both orders, and a representative wins only if both answers name it; a failed, undecided or split pair gives no win [S, O]
6: **return** the representative of the cluster with most wins, with ties among those clusters broken by the ranking of Step 5; if no cluster has a win, the first cluster of that ranking; if no call gave a point, no answer (a miss); the alarm and the cost of all attempts

evaluation script, run on the stored responses, it regenerates Table 12.

5.2 Revising the model from failed predictions

The theory was built by iteration, and the iteration is part of the method. A model is stated, a consequence is derived and tested, and when a prediction fails we record three things: which assumption failed and how we know; how the revised assumption was checked directly; and what new consequence the revision predicts, and whether it holds. Table 4 lists the five revisions that shaped the final theory. Each row names the kind of evidence: an exact identity, a controlled comparison of interventions on the same images, or an association observed across cells or records.

6 Experimental Setup

Models and settings. The main experiments use three API models: GPT-6.1, Sonnet 5.5 and Gemini 3.8 Flash (written Gemini 3.8). Pilots and the controlled-scene experiments also use GPT-5.6 and Opus 5.5, and an open-weight model, Qwen3-VL-235B, is tested on the readout tasks. Each call uses either the *lowest* reasoning setting the API offers or the *default* setting. Setting labels do not match behaviour, and we report what was measured (Appendix C). Three facts matter for interpretation. First, the default setting also doubled the output cap (16,384 to 32,768 tokens); three runs separate the two: Table 9, a 90-call 16k control on the E3 sources, and a 16k control on controlled scenes. Second, Sonnet 5.5's reasoning share at its default setting varies by task: 0% to 23% of calls in the reasoning experiment (dense scenes, FSC-147, screens) and 0% to 90% across detection cells; we attribute a Sonnet result to reasoning only where most of its calls reasoned. Reasoning tokens are read from the usage fields of every call after the pilots (Qwen3-VL is an instruct model and reports none); in the two pilots, which logged no thinking tokens for the Claude models, adaptive thinking may have run unseen, and no conclusion here depends on those pilots. Third, Gemini 3.8 at its lowest setting still reasoned on click and point outputs (about 99% of GUI click calls), so its click results are labelled "lowest (reasoning observed)". GPT-6.1's lowest setting is `reasoning_effort=low`, and most of its calls carry some reasoning tokens. Images sent to Sonnet 5.5 were resized to at most 1,568 pixels on the long side and 1.15 megapixels (by our code, or in the detection experiment by the provider), and images sent to the other models to at most 2,048 pixels; for Sonnet 5.5 this is below the 2,576-pixel long edge measured for it in the companion paper (Shi et al., 2026).

Data. The *synthetic counting probe* (SCP) has 100 images of 1024×1024 pixels with five shape classes and 15 to 80 shapes each (mean 47.6), drawn over clutter lines; *SCP-dense* uses the same generator with 50 images of 121 to 200 smaller shapes (mean 164.3). Labels are exact. Photographs, aerial images and document pages come from PixMo-Count (100 images, 2 to 10 objects), FSC-147 (100 images in three size strata for the readout and tiling experiments, and 60 further images for the detection experiment), and 60 images each from SKU-110K, DOTA v1.0 and DocLayNet v1.2 (three classes). Screen data are 200 ScreenSpot-Pro and 60 ScreenSpot-v2 click items, 60 GroundCUA screens and 60 ScreenParse screens; the confirmation of Section 7.7 adds 600 further ScreenSpot-Pro click items, sampled with a fixed seed and sharing no item and no screenshot with the 260 click items. The controlled-scene experiments add 96 procedurally drawn scenes with 16 to 384 targets each. A readout experiment on aerial and document images adds 30 DOTA v1.0 validation windows and 50 DocLayNet pages (financial reports, tenders and laws, the categories of the two test shards we use), selected with the annotations to contain at least three (DOTA) or four (DocLayNet) classes and at least 30 or 20 objects, disjoint from the detection images. In total the experiments use 1,746 images with 54,475 annotated targets (each click item counts as one target). We refer to SKU-110K, DOTA, DocLayNet and FSC-147 as the *E3 sources* and to GroundCUA and ScreenParse as the *screen sources*.

Experiments. Table 5 lists the experiments that the results draw on. Every call of these experiments, including smoke tests, pilots and retries, is stored with its raw response; the full registry with costs is in Appendix C. Merges of existing calls (intersections, unanimity, cross-model constructions) are deterministic post-processing of stored responses and count as design evidence: the same items are re-scored under a changed construction. Mechanism tests on existing data, such as the decomposition of a gain into selection and geometry, are labelled as decomposition evidence.

Table 4: Revisions of the model. Each row records the failed assumption and the evidence against it, the revised assumption and its direct check, and the new consequence with its test. The kind of evidence is named in each cell.

Failed assumption	How we know	Revised assumption and direct check	New consequence and test
The intersection threshold, applied to intersections with averaged coordinates, predicts whether confirmation helps.	Wrong sign in 2 of 54 base/partner pairs.	The threshold governs deletion only; coordinate averaging is a separate term (Proposition 2, Proposition 3). Exact identity: both terms add up to the observed change on every image of all 54 pairs.	The averaging term can take either sign and can reverse the result. Observed: positive in 37 pairs, negative in 13; it reverses the deletion-only result in two pairs, which accounts for both former exceptions (Table 10). The deletion term, in turn, can be predicted from p_1 on 5 labelled images in 83% of draws (Section 7.4).
Similar quality is enough for a second model to help as a confirming partner.	On ScreenParse the three models have similar set error (16.2, 13.3, 16.0), yet among constructions with averaged coordinates the best cross-model construction is not significantly different from the best same-model one ($+1.8\%$, $[-3.0, +7.6]$). This refutes sufficiency only.	A partner helps if it rejects more of the base model's false records than it loses of its true ones (Proposition 5). Direct check across pairs: cross-model partners separate the two better than same-model partners in 32 of 36 comparisons; the 4 exceptions are partners that confirm few of the base model's true records.	Conditional consequence: a base with many true and few false records gains only if the partner keeps almost all of its true records ($T(1 - h_+^B)$ dominates G), and a weaker partner usually keeps fewer, so cross-model confirmation is expected to do worst where the base model is strongest. Observed association across transfer cells: all 5 significantly worse cells are cells where the base model is strongest, and 20 of the 24 cells where it is not strongest improve (Table 14).
A count written first, or given in the prompt, pulls the list length like a quota for every model.	Giving a count moves list length with slope at most 0.004 for GPT-6.1 and Sonnet 5.5 on SCP and SCP-dense (controlled comparison).	The pull depends on the model and on how reliable its enumeration is. Direct check: slope 0.17 to 0.20 on PixMo, and 1.63 for Gemini 3.8 on SCP-dense, where its lists degenerate.	Asking for the list before the counts should not hurt and should help where enumeration is reliable. Observed in a controlled comparison under the same mapped readout: list-first lists beat count-first lists for all three models (-39%, -92%, -25%; Table 7). The list-first prompt also adds the row-by-row scan and the allowed class names, so this is a test of the protocol, not of order alone.
Gemini 3.8 has a coordinate spread that a different coordinate convention reduces.	Pixel versus grid convention: $+2\%$, n.s. (controlled comparison).	Current models make no coordinate-frame error on SCP. Direct check: systematic offset at most 0.3 pixels, slope 1.000, spread 0.9 to 1.5 pixels, against a 30-pixel tolerance.	References that correct coordinates (grids, ticks) should not improve localization. Observed in controlled comparisons: localization changes by at most 0.23 pixels in every arm. Any effect of an overlay must come through how the model writes its list (next row).
A grid helps Gemini 3.8 by improving localization.	The grid lowers set error by 3.5 per image at the lowest setting, but localization does not change, and at the default setting the grid makes detection slightly worse (controlled comparisons).	The grid acts on the record format: its gain falls on records that omit their class field. Association within lists: a record omits its class more often when the previous record has the same class (logistic regression controlling for position in the list: $+0.70$ $[0.57, 0.84]$ in log-odds, arm fixed effects, intervals clustered by image). This path explains only a small part of the grid's gain.	An order that interleaves classes should lower omissions, and grouping by class should raise them. Observed in a controlled comparison: a row-by-row scan lowers set error by 11.5 per image against 3.5 for the grid, and grouping by class raises it (Section 7.3).

Table 5: Experiments used in Section 7. Calls include repairs and smoke tests; the registry in Appendix C lists all 67,364 calls.

Experiment	What changes	Calls
Readout	13 prompt arms on SCP, SCP-dense, PixMo	7,900
Readout confirm.	full protocol, SCP-dense; Gemini at 3 settings	250
Multi-class readout	count-first vs. list-first, DOTA and DocLayNet	574
Grid	grids, ticks, cells, scan, type names	1,340
Tiling	count-first, list-only, 2 × 2 tiles (FSC)	1,800
Detection (E3)	list / regions × two settings, 4 datasets	6,523
GUI	click arms incl. zoom; parse arms incl. intersections; pilots	7,232
Zoom, 2nd click	zoom around each model's second click	772
Click choice	scores, default zooms, comparisons, crop-first baselines (260 items)	5,964
Click confirm.	all arms of Table 12, 600 new items	13,126
Fourth source	crop-first point added to the tier, same items	717
Review	blind vs. draft review on crops	1,440
Same-view review	draft review on detection images	663
Comparator	pairwise comparison of disagreeing reports	1,960
Resolution	downscaled / native / upscaled	600
End-to-end	naive / common / proposed arms	1,800
Reasoning	default setting on dense, FSC, GUI	1,110
Open model	Qwen3-VL readout arms	650
Controlled scenes	pilots, subset execution, drift, grids	12,296
Pilots	small runs before the readout experiments	647

Statistics. Intervals are 95% paired bootstrap intervals over images (4,000 resamples unless stated), with relative changes computed on means. "Significant" means the interval excludes 0. Comparisons of two click flows on the same items also report the items repaired and broken and an exact McNemar test, with Holm adjustment where two comparisons share a claim. Fold assignments and random draws use seeds derived from SHA-256 of a fixed string and are persisted, so every table regenerates byte-identically from the saved responses with one script. Costs count every attempted call at list prices.

7 Results

The results follow the three problems of Section 1: a comparison with common practice (Section 7.1), repairing reports (Sections 7.2 to 7.5), detecting errors (Section 7.6), repairing a click by choosing among several models, confirmed on new screenshots (Section 7.7), and judging which report or configuration is more accurate (Sections 7.8 and 7.9). Each table caption states the operation, the comparator, the loss and any label use.

7.1 Against common practice

Figure 1c and Table 12 summarise the result with the widest scope: on 600 ScreenSpot-Pro items not used in designing it, the several-model click tier misses 9.33% of targets against 13.67% for GPT-6.1's two-stage zoom and 12.67% for three such zooms at about the same cost (Section 7.7). The comparisons below use the stored calls of the earlier experiments.

Table 6 compares the defaults of Section 5 with the remedies practitioners use most, on the same stored calls. The pattern follows the three problems. Self-consistency repeats the same source of evidence, so its gains are small: the median of three stated counts helps GPT-6.1 by 23% and the other two models by about 1%, at three times the cost, and a vote of three clicks changes click error by -4.5% to $+2.5\%$. Uncertainty-gated zoom, which zooms only when two clicks of the same model disagree, inherits the same weakness, because persistent errors rarely produce disagreement; it gains 13% for Gemini 3.8 and nothing significant for the other two. The remedies that change the information win: reading the list cuts dense-count error by 85% and 99% for about the cost of one call, and zooming every item cuts click error by 45% and 65% for less than a three-click vote. For set error, three-call unanimity with fused coordinates improves 9 of the 15 over-reporting cells (median -9.6%), against -2.9% for majority vote and -3.7% for box fusion, and it beats box fusion directly in 11 of 18 cells and is not significantly worse in any. For detecting click errors at the same distance threshold, a second model catches 1.9 to 3.5 times the share of errors that a second call of the same model catches, while flagging more items (Section 7.6).

Table 6: Against common practice, from stored calls only. Columns give the common baseline, the usual remedies (self-consistency: median of three stated counts, or a vote of three clicks; uncertainty-gated zoom: zoom only when two clicks disagree; majority vote and equal-weight box fusion over three calls), and our default. Relative changes are against one call (**bold**: paired 95% interval below 0; underlined: above 0). Our count readout costs about as much as one call and our click protocol less than a three-call vote. For Gemini 3.8, "ours" is the list-first protocol at its lowest setting; the method runs Gemini at its default setting, which at the same 16k cap gives a readout error of 24.52 (Table 9). For F1 our default is one call; over the 15 over-reporting cells, majority vote and box fusion change F1 by a median of -0.3% and $+0.2\%$. Per-cell values are in `stageB/analysis/out/common_practice.json`.

	One call	Common remedy	Second common remedy	Ours (Section 5)
Repair the readout: dense counting (SCP-dense, 50 images, per-type count error; cost per image)				
GPT-6.1	10.84 ($0.029)	8.32 ($0.089), **−23.2%**	n/a	1.60 ($0.025), **−85.2%**
Sonnet 5.5	35.02 ($0.031)	34.56 ($0.094), −1.3%	n/a	0.42 ($0.033), **−98.8%**
Gemini 3.8	45.68 ($0.026)	45.22 ($0.076), −1.0%	n/a	52.80 ($0.019), +15.6%
Repair the view: clicks (ScreenSpot-Pro items with all four calls stored, click error; cost per item)				
GPT-6.1 ($n = 199$)	11.1% ($0.0061)	10.5% ($0.0182), −4.5%	11.1% ($0.0123), 0.0%	9.6% ($0.0083), −13.6%
Sonnet 5.5 ($n = 196$)	20.4% ($0.0035)	20.9% ($0.0104), +2.5%	19.4% ($0.0071), −5.0%	11.2% ($0.0058), **−45.0%**
Gemini 3.8 ($n = 200$)	35.5% ($0.0030)	35.0% ($0.0086), −1.4%	31.0% ($0.0063), **−12.7%**	12.5% ($0.0053), **−64.8%**
Repair the set: detection and screen parsing (15 over-reporting cells, set error E, 3 calls each; better / worse vs. one call, median change)				
15 cells vs. box fusion	one call	majority vote: 9 / 0, −2.9%	box fusion: 8 / 0, −3.7%	9 / 0, −9.6% better in 11 of 18 cells, worse in 0
Detect errors: clicks, share of a single call's errors caught by a disagreement alarm ($\delta = 1.5\%$)				
GPT-6.1	n/a	second call, same model: 17%	n/a	second model: 61%
Sonnet 5.5	n/a	second call, same model: 36%	n/a	second model: 71%
Gemini 3.8	n/a	second call, same model: 27%	n/a	second model: 52%

Table 7: Readout effects kept separate (SCP-dense, 50 images, lowest setting, one call; per-type count error; paired image bootstrap 95% intervals). *Pure readout* compares the stated count with the list count of the list in the same response, without the alias map. *Alias map* applies the declared map to the same list. *Protocol* compares count-first and list-first lists under the same mapped readout. *Full method* compares the stated count of a count-first response with the readout of Section 5 on a list-first response. This is the effect of the reporting and decoding protocol.

Effect	Comparison	GPT-6.1	Sonnet 5.5	Gemini 3.8 (lowest)
Pure readout, count-first	same response	10.84 → 2.64 −75.6% [−83.2, −65.8]	35.02 → 7.80 −77.7% [−91.3, −58.4]	45.68 → 129.04 +182.5% [163.3, 202.3]
Pure readout, list-first	same response	10.10 → 1.60 −84.2% [−88.2, −79.6]	18.52 → 5.42 −70.7% [−76.9, −64.8]	46.56 → 95.04 (n=45) +104.2% [72.7, 139.5]
Alias map	same list	1.60 → 1.60 (no change)	5.42 → 0.42 −92.3% [−97.0, −86.2]	96.78 → 96.78 (no change)
Protocol	mapped list count	2.64 → 1.60 −39.4% [−57.6, −13.5]	5.42 → 0.42 −92.3% [−97.7, −44.4]	129.04 → 96.78 −25.0% [−32.7, −17.6]
Full method	count-first stated → readout	10.84 → 1.60 −85.2% [−89.1, −80.3]	35.02 → 0.42 −98.8% [−99.5, −98.1]	45.68 → 52.80 +15.6% [−5.3, 39.4]
Totals, list-first	$\lvert$list length$-n\rvert$ vs. $\lvert$stated total$-n\rvert$	0.04 vs. 7.78	0.02 vs. 10.48	109.38 vs. 46.11

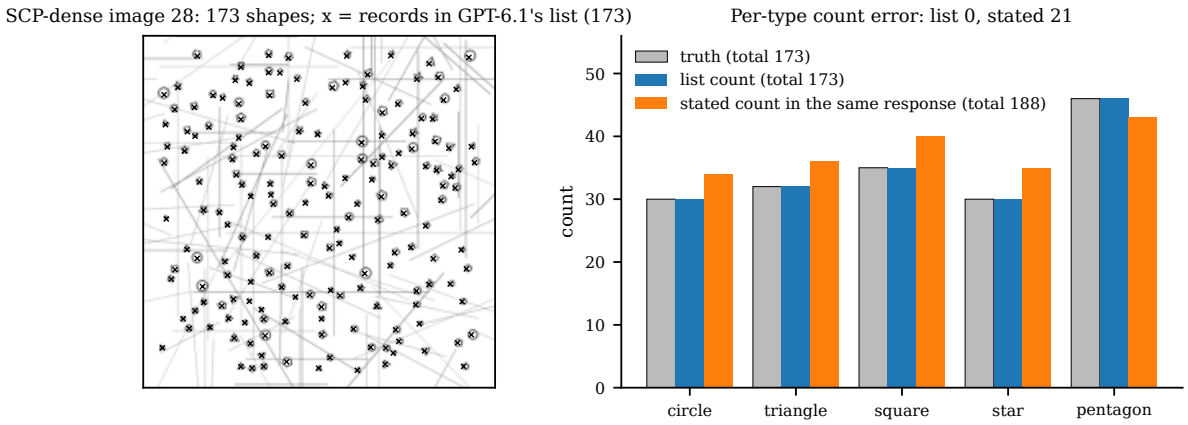


Figure 3: One response, two answers. GPT-6.1 is asked to list every shape in a dense synthetic scene and then give counts by type "for the shapes you listed". The scene has 173 shapes. The list has 173 records, and each matches a different shape of the same class within 20 pixels, so the list count of every class is exact. The counts the model states in the same response sum to 188, and their per-type error is 21. Across 50 such scenes, list length misses the true total by 0.04 objects on average and the stated total misses by 7.78 (Table 7). Image: SCP-dense 28; markers: centres in the model's list.

7.2 Repair the readout: read the list, not the number

Same response, two answers. Figure 3 shows one response in full. When a list-first response has both a list and stated counts, the list length misses the true total by 0.04 objects on average for GPT-6.1 and 0.02 for Sonnet 5.5, while the stated totals miss by 7.78 and 10.48 (Table 7, last row). Reading per-type counts from the list (the list count) instead of the stated counts in the same response cuts per-type count error by 75.6% (GPT-6.1) and 77.7% (Sonnet 5.5) under a count-first prompt, and by 84.2% and 70.7% under the list-first prompt. Generation is held fixed in these comparisons, so the gain is a readout effect: the model's list carries information that its own summary loses. Proposition 4 adds that the size of such a gap shows the two readouts are inconsistent but does not by itself say which one is right; here the labels say it is the list.

The gap grows with the length of a multi-type list. Figure 4 (values in Table 17) repeats the same-response comparison on four datasets. On PixMo (6 objects per image, one class) and FSC-147 (43.8 objects, one class), the stated total tracks the list for GPT-6.1 and Sonnet 5.5 and the difference is not significant. On SCP (47.6 objects, five classes) the list count is better by 44% for GPT-6.1 and by 98.5% for Sonnet 5.5, whose stated per-type counts are 65 times as far from the truth as its list. On SCP-dense (164.3 objects) both models gain 76% to 78%. The gap therefore appears when the stated count must summarize a long list across several classes.

Aliases and protocol. In list-first responses, Sonnet 5.5 wrote 258 "diamond" records across 50 images although the format listed the allowed names. The declared alias map, applied to the same list, takes its error from 5.42 to 0.42 (-92.3%). The prompt alone does not do this: the map is part of the task, and the decoder applies it. Holding the mapped readout fixed, list-first lists are better than count-first lists by 39.4% for GPT-6.1 and 92.3% for Sonnet 5.5. The complete protocol, compared with the stated count of a count-first response, cuts error by 85.2% $[-89.1, -80.3]$ for GPT-6.1 and 98.8% $[-99.5, -98.1]$ for Sonnet 5.5. For Gemini 3.8 at its lowest setting the change is $+15.6\%$ $[-5.3, +39.4]$, not significant: the degeneracy check routes 40 of its 50 responses to the stated count. These are effects of the reporting and decoding protocol. The same-response and same-list comparisons isolate the readout and the alias map; the full protocol also changes how the list is generated, so it does not establish a change in perceptual ability either way.

Aerial and document images with several classes. The readout repair needs a gap to work with. We ran both prompts on 30 DOTA windows and 50 DocLayNet pages chosen to contain several classes and many objects (Table 8; 574 calls). There the lists themselves miss objects: GPT-6.1 and Sonnet

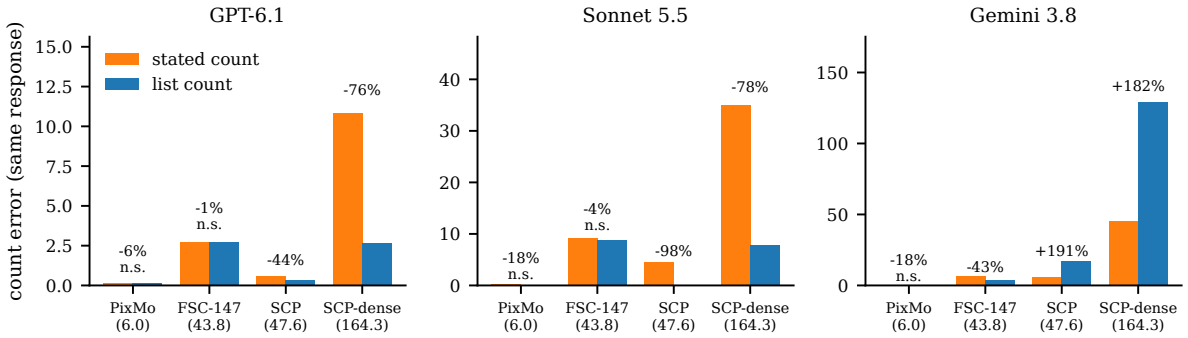


Figure 4: Same-response readout across datasets, count-first prompts, lowest setting. Bars: count error of the stated count and of the list count (without the alias map) in the same response (per-type error on SCP and SCP-dense; total error on PixMo and FSC-147). Labels: relative change from stated count to list count; n.s.: paired 95% interval includes 0. Numbers under dataset names: mean objects per image. Each panel has its own vertical scale. Gemini 3.8's lists degenerate on the synthetic scenes, so reading them is worse (Section 7.3).

5.5 list 68% to 88% of the true objects. The stated count also stays close to the list count of the same response, differing by 0.1 to 7.4 per image against per-type errors of 5 to 46. Switching readouts can change the error by at most this gap (Proposition 4), so the change in the same response is small: none for GPT-6.1, and -13.5% and -19.9% for Sonnet 5.5. The full list-first protocol, here without the recovery step, changes count error by $+0.4\%$ and -2.4% for GPT-6.1 and by -15.7% and -20.1% for Sonnet 5.5, intervals reaching zero. Gemini 3.8 at its lowest setting again writes degenerate lists; the fallback to the stated count holds its change to $+18.0\%$ and -7.9%, both not significant, and at its default setting, the shipped configuration, it changes count error by -28.2% on DOTA (interval reaching zero) and by -25.8% on DocLayNet. The two kinds of scene differ in where the error sits. On dense synthetic scenes the lists are nearly complete and the statement loses information, so the readout repairs most of the error; on these aerial and document images most of the error is already in the list, so switching between the two readouts cannot remove it. The readout gap, which needs no labels, separates the two cases in one direction: a small gap shows that switching between the stated count and the list count can change the error by at most the gap, while a large gap shows only that the switch might matter. The gap says nothing about remedies that produce a new response.

7.3 Repair the readout when the list degenerates

Gemini 3.8 at its lowest setting writes long lists in which most records lack a valid class and some coordinates repeat, so the list count is far worse than its stated count ($+182.5\%$ in the same response; Figure 4). The four-part degeneracy check of Section 5 flags 45 of its 50 dense-scene lists. For 40 of them the readout falls back to the stated count; the other 5 have no stated count and keep the list count with a status flag. On the GPT-6.1 and Sonnet 5.5 list-first responses it flags none.

For Gemini 3.8, degeneration depends mainly on the reasoning setting. At the same 16k output cap, Gemini 3.8's default setting reduces flagged lists from 90% to 12% and the readout error from 52.80 to 24.52 (-53.6% $[-72.4, -29.5]$; Table 9). Doubling the cap to 32k gives a point estimate of 15.44, a further change that is not significant. Even then, Gemini 3.8's dense-count error is about ten times GPT-6.1's, and none of its 50 counts is exact. Strict JSON parsing fails for 70% to 78% of Gemini 3.8 responses at every setting, so the declared record salvage of Section 3.5 is a necessary part of the decoder, and its share is reported.

On the list-first responses of the readout experiment, applying the same check to existing responses recovers 68% (SCP) and 58% (SCP-dense) of Gemini 3.8's error relative to reading its list, and 42% relative to the stated count on FSC-147. For Sonnet 5.5, whose lists are not degenerate, the cost of the check reaches $+23\%$ on SCP and $+13\%$ on SCP-dense (not significant; intervals $[0, 57]$ and $[0, 26]$). On the open model Qwen3-VL, with thresholds unchanged, it lowers SCP-dense error by 7% $[-13, -1]$ relative to the stated count. Gemini 3.8's missing class fields follow a pattern: a record is more likely to omit its type when the previous record has the same type, and a row-by-row scan

Table 8: Readout on aerial and document images with several classes: 30 DOTA windows (71 objects and 3.3 classes per image on average) and 50 DocLayNet pages (25 elements, 5.0 classes); one call per image and arm, lowest setting unless stated, 16k cap. Per-type count error; paired image bootstrap. Difficult DOTA objects and objects less than half inside the window are not targets, so records on them are counted like any other record, for every readout alike. *List recall*: share of true objects matched by a record centre inside a true box of the same class (list-first arm). *Readout gap*: mean per-type difference between the stated count and the list count of the same count-first response, which bounds how much switching readouts can change the error. *Same response*: stated count $\rightarrow$ list count of the count-first response. *Protocol*: readout of the list-first response (list count, or the stated count if the list is flagged) against the stated count of the count-first response; the Gemini default row is the shipped configuration.

Images	Model	List recall	Readout gap	Same response	Change	Protocol readout	Change vs. stated count
DOTA	GPT-6.1	88%	2.2	40.5 $\rightarrow$ 40.8	+0.7% [−5.0, 4.5]	40.7	+0.4% [−17.7, 16.4]
DOTA	Sonnet 5.5	68%	7.4	45.8 $\rightarrow$ 39.7	−13.5% [−34.7, −0.6]	38.6	−15.7% [−38.2, 7.7]
DOTA	Gemini 3.8	26%	44.2	33.2 $\rightarrow$ 63.9	+92.4% [54.2, 145.0]	39.2	+18.0% [−13.4, 61.2]
DOTA	Gemini 3.8, default					23.9	−28.2% [−48.6, 0.1]
DocLayNet	GPT-6.1	87%	0.1	4.9 $\rightarrow$ 4.9	+0.4% [−1.6, 2.2]	4.8	−2.4% [−12.4, 4.5]
DocLayNet	Sonnet 5.5	83%	3.1	8.6 $\rightarrow$ 6.9	−19.9% [−32.5, −10.1]	6.8	−20.1% [−36.7, 3.3]
DocLayNet	Gemini 3.8	68%	5.3	9.4 $\rightarrow$ 11.2	+19.2% [1.6, 41.0]	8.6	−7.9% [−24.7, 10.6]
DocLayNet	Gemini 3.8, default					7.0	−25.8% [−42.5, −8.8]

Table 9: Gemini 3.8 on SCP-dense under the list-first protocol (50 images, one call each). Changing the reasoning setting at a fixed 16k output cap lowers the count readout error by −53.6% [−72.4, −29.5]; raising the cap to 32k changes it by a further −37.0% [−62.4, 7.7] (not significant). Readout error: per-type count error of the Step-6 readout (stated count if the list is flagged, list count otherwise). List-count error: the same for the list count alone. Set error E scores the delivered records, those whose class is valid after the alias map (equal class, 20 px); unknown-class records are excluded from the delivered set and reported in the next column; the column after it is a sensitivity that keeps them and scores them as unmatched. Flagged lists: share with at least one degeneracy flag. JSON fail: share of responses that do not parse as the requested object (records are then recovered by the declared salvage step). The setting comparison is cap-matched, not cost-matched: output tokens and cost rise with reasoning. Cost counts every attempted call.

Setting, cap	Readout error	List-count error	Set error E	Unknown records	E, unknown kept	Flagged	JSON fail	Truncated	Reasoning share	Output tokens	Cost
lowest, 16k	52.80	96.78	103.46	184.76	288.22	90%	72%	4%	0%	4796	$0.95
default, 16k	24.52	25.90	34.58	4.90	39.48	12%	70%	8%	100%	8831	$1.71
default, 32k	15.44	20.14	28.26	8.84	37.10	6%	78%	0%	98%	9487	$1.83

that interleaves types cuts its count error by 60%, while grouping by type makes it worse.

7.4 Repair the set: delete below the threshold, keep geometry separate

Deletion and geometry, measured separately. Table 10 (per-pair values in Table 18) splits the effect of confirming a base call with a partner call into the selection term of Proposition 1 and the geometry term of Proposition 3, for 54 base/partner pairs on six sources. Both identities hold exactly on every image, so the selection term of a pair has the sign of $1 - 2p_1$ by construction (Figure 5a). What the data add is where p_1 lies: 0.00 to 0.39 on SKU-110K and 0.05 to 0.22 on DOTA, where deletion removes 4 to 68 errors per image, against 0.39 to 0.90 on FSC-147. On FSC-147, where models under-report, p_1 is 0.39 to 0.90 and deletion hurts in 7 of 9 pairs, which is why counting families keep a single call. Coordinate averaging is positive in 37 pairs and negative in 13. In two pairs it reverses the sign of the deletion-only result: DocLayNet with GPT-6.1 confirmed by Sonnet 5.5 (selection $+0.80$, geometry -1.10 per image) and Ground-CUA with Sonnet 5.5 confirmed by itself (selection -0.87, geometry $+1.20$). These are the two pairs in which an earlier check of the threshold on fused intersections failed; separating the terms accounts for both.

The threshold from a few labels. The useful question is whether p_1 measured on a handful of labelled images predicts what deletion will do on other images. For each pair we drew 5 labelled images 200 times, estimated p_1 on them, and predicted from Proposition 2 whether deleting unconfirmed records lowers set error on the other 55 images of the pair. The prediction is right in 83% of draws (80% with 3 labelled images, 87% with

Table 10: Confirmation by a partner call, split into selection and geometry (Propositions 1 and 3). Each source has 9 base/partner pairs (3 base models $\times$ same-model or one of two other models), 60 images, one call each; values are per-image reductions of E (positive is better). Both identities hold exactly on every image of every pair. *Sign from 5 labels*: p_1 is estimated on 5 labelled images and Proposition 2 is used to predict whether deletion lowers E on the other 55; the share of correct predictions over 200 draws per pair, averaged over the pairs (Section 7.4). Sign flips: pairs in which the fused total has the opposite sign of the deletion-only selection term. Full per-pair values are in Appendix B.

Source	Pairs	Sign from 5 labels	Deletion helps	p_1 range	Selection range	Geometry + / −	Geometry range	Sign flips
DocLayNet	9	74%	6	0.11 to 0.64	−2.2 to 2.1	6 / 2	−1.1 to 3.1	1
DOTA	9	98%	9	0.05 to 0.22	4.4 to 16.3	7 / 2	−1.5 to 3.9	0
FSC-147	9	81%	2	0.39 to 0.90	−21.8 to 0.3	8 / 0	0.0 to 0.4	0
SKU-110K	9	96%	9	0.00 to 0.39	10.8 to 68.2	7 / 2	−14.1 to 13.8	0
GroundCUA	9	86%	6	0.06 to 0.72	−16.8 to 37.7	4 / 5	−7.7 to 15.4	1
ScreenParse	9	63%	8	0.19 to 0.50	0.0 to 2.1	5 / 2	−0.5 to 0.9	0

10), and the loss against the better decision on the other images is 0.33 errors per image (0.48 and 0.15). The shipped regime alone, which deletes on over-reporting families and keeps on FSC-147 without any label, is right in 85% of draws with a loss of 0.43. The label rule of Step 7 combines the two: it starts from the regime and switches only if the labelled images favour the other decision by more than one standard error. It is right in 86%, 87% and 88% of draws with 3, 5 and 10 labelled images, with losses of 0.22, 0.17 and 0.10 errors per image. In the 8 pairs whose p_1 lies within 0.1 of $1/2$, the gain from deleting is small and its sign is noisy: the plain threshold is right in 47% of draws there, and the regime and the Step 7 rule, which lean towards deletion, in 34% and 37%. Table 10 gives the share by source.

Which partner helps. Proposition 5 says a partner helps if it rejects more of the base call's false records than it loses of the true ones. Cross-model partners separate the two better than same-model partners in 32 of 36 base/partner comparisons on the E3 and screen-parsing sources. The four exceptions are GPT-6.1's two much weaker partners on FSC-147, and Gemini 3.8 confirming Sonnet 5.5 on FSC-147 and SKU-110K, where Gemini 3.8 confirms few of Sonnet 5.5's true records. The misses are consistent with a component shared across models: of the objects that GPT-6.1, the highest-recall model, misses in all three of its calls, a single call of either other model recovers only 0.7% to 9% on SKU-110K, DOTA and FSC-147, and 18% to 26% on DocLayNet. New information flows mainly from higher-recall to lower-recall models.

7.5 Repair the view; fixes that do not help

Zoom and tiles. Two-stage zoom clicks once on the whole screenshot, crops a third of each side around that point at native pixels, and clicks again in the crop; it is a re-grounding strategy in the style of the ReGround method that Li et al. (2025b) evaluate. On ScreenSpot-Pro (200 items) it raises the click hit rate from 64.5% to 87.5% for Gemini 3.8 (+23 points $[16.5, 29.5]$, a 65% cut in click error) and from 78% to 87% for Sonnet 5.5 (+9 $[5, 13.5]$, a 41% cut; 45% on the 196 items that every arm of Table 6 answered); for GPT-6.1, 88.5% to 90.0% is not significant. Zooming only when two different models disagree, a cross-model version of the gated zoom in Table 6, is less accurate than zooming every item for Sonnet 5.5 (2.3 to 2.7 points) and Gemini 3.8 (9.6 to 12.3 points), not significantly different for GPT-6.1, and more expensive. Our tiling arm on FSC-147 used 2×2 crops that overlap by 10% of the side, each enlarged $2\times$ and sent in a separate call. It raises the tokens per object side for every object, but its overlap is smaller than one object for the larger objects (median exemplar side 17.5% of the image side in the largest third), so it does not meet the companion paper's safe-decomposition condition that every target lie inside some view (Shi et al., 2026). The results follow that condition. On the smallest third of images, Sonnet 5.5's count error falls by 70% against whole-image lists; on the largest third, where many objects lie wholly inside no crop, GPT-6.1's rises by 78%; over all images Sonnet 5.5's change (-21%, $[-50, +27]$) is not significant. The small-object gain mixes more tokens per object with less content per call, and the large-object loss is consistent with objects that no crop contains rather than with object size as such. On screens,

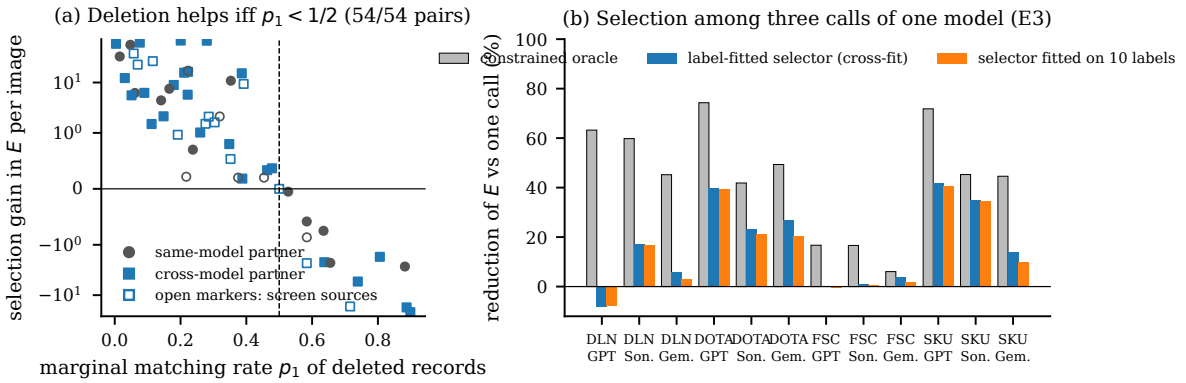


Figure 5: (a) Selection gain of deleting unconfirmed records against their marginal matching rate p_1, for 54 base/partner pairs (symmetric log scale). With equal error costs the gain per image equals $\mathbb{E}[\Delta c](1 - 2p_1)$ exactly (Proposition 2), so this panel checks the implementation of an identity and shows how far each pair lies from the threshold; it is not an independent test. (b) Reduction of set error relative to one call when selecting among three calls of one model: constrained oracle, cross-fitted label-fitted selector, and the selector fitted on 10 labelled images (median of 40 draws).

where tiles overlap by 6%, objects cut by a tile boundary lose 0.16 to 0.42 in report probability in all 6 cells, and separate calls per tile beat one call with several tiles for GPT-6.1 (-15%, $[-23, -5]$). For native-resolution click crops, downscaling lowers Sonnet 5.5's hit rate by 9 points $[3, 16]$, the loss for GPT-6.1 reaches 3 points ($[0, 7]$, not significant), Gemini 3.8 shows no significant response to resizing, consistent with the fixed token grid measured for it in the companion paper, and enlarging these crops does not help.

Reports from a view planner. In the crop calls of the Review experiment (Table 5) the pipeline receives, for each image, a whole-image report and four reports from four native-resolution crops (6% overlap, one call per crop), on 30 GroundCUA screens and 30 SKU-110K images, scored at IoU ≥ 0.5 without class. Merging the four crop reports alone lowers set error relative to one whole-image call for GPT-6.1 on screens (-19.4% $[-27.5, -10.6]$) and for Sonnet 5.5 on SKU-110K (-64.0% $[-70.6, -56.6]$), raises it for Sonnet 5.5 on screens ($+44.9\%$) and for Gemini 3.8 on SKU-110K ($+24.7\%$), and is not significant in the other two cells. Confirming the whole-image report with the merged crops under the shipped rule, in which a crop confirms or rejects only the records it contains wholly, lowers set error in 4 of 6 cells: GPT-6.1 (-22.8%) and Gemini 3.8 (-20.3%) on screens, and GPT-6.1 (-45.5%) and Sonnet 5.5 (-36.6%) on SKU-110K. In these 4 cells it is also better than confirmation by a second whole-image call (-11.8% to -22.6%), and better than the shipped three-call unanimity in 3 of them (-8.0% to -20.2%; Gemini 3.8 on screens -4.7%, not significant), at the price of five calls instead of three. In the other two cells it raises set error: Sonnet 5.5 on screens ($+12.9\%$) and Gemini 3.8 on SKU-110K ($+37.9\%$). The threshold of Proposition 2 accounts for both groups. Among the whole-image records that the crops do not confirm, the share matched to a true object (p_1) is 0.03 to 0.24 in the four cells where confirmation helps, and 0.62 and 0.76 in the two where it hurts; this follows from an identity and is not an independent test. The plain rule, in which every record needs confirmation, differs from the shipped rule by at most 4 percentage points, because with a 6% overlap few records lie wholly inside no crop.

Grids and coordinates. On SCP, the systematic coordinate offset of current models is at most 0.3 pixels, with slope 1.000 and spread 0.9 to 1.5 pixels, against a matching tolerance of 30 pixels. A grid, ticks or cell labels therefore have no coordinate error to correct: in every arm, localization changes by at most 0.23 pixels, and at the lowest setting they do not change detection for GPT-6.1 and Sonnet 5.5. For Gemini 3.8 at its lowest setting, a grid lowers set error by 3.5 per image by reducing records that omit their class, and a row-by-row scan lowers it by 11.5; at its default setting the grid makes detection and localization slightly worse. "No overlay" is therefore the shipped default, chosen from these measurements; the theory does not

forbid an overlay from helping a model whose list format it changes. Markers also make Sonnet 5.5 write "diamond" more often (+0.3 per image). An overlay on the sent image adds no Shannon information and can only change how the model uses its input (Shi et al., 2026). We saw two such changes, fewer omitted class fields for Gemini 3.8 and more uses of the word "diamond" for Sonnet 5.5, and no coordinate correction, because on SCP the offsets are about 1% of the tolerance.

Wording, instructions and review. "Do not miss any" wording, compared with a symmetric instruction not to miss or invent, has no measurable effect on SCP for any of the three models. Asking the model to localize each item relative to structures in the image has no significant effect, because current models have almost no location misses on these scenes. Same-view self-review, in which the model edits its own draft, confirms true and false records at nearly the same rate ($h_+ \approx h_-$) and has no net gain on screen parsing; on FSC-147 it gave significant gains for 2 of 3 models, which we do not explain. Pairwise comparators that ask a model which of two reports is right judge whether something is an object rather than whether the box is precise, and are not used. Reasoning is not uniformly helpful: on 20 GroundCUA screens, GPT-6.1's default setting raises set error by 18.7 per screen $[5.8, 34.3]$, through 16.3 additional false records and no fewer misses.

7.6 Detect errors: they persist, so ask someone else

Persistence. Object-level errors repeat across calls. Intraclass correlations of reporting across repeated calls are 0.60 to 0.98 for true objects and 0.23 to 0.93 for false reports. On the E3 sources, a binomial model with a common reporting probability is rejected in all 12 dataset and model cells, while a beta-binomial model fits. The share of misses that recur in every repeated call is 0.76 to 0.99 on the E3 sources and 0.73 to 0.94 on the screen sources. In clicks, 64% to 83% of a model's errors are repeated by a second call of the same model within the agreement tolerance.

A second model is a better alarm. Persistence means that a second call of the same model rarely disagrees with the first where the first is wrong. A second model shares only part of the propensity. With the partner set by a fixed rule as the cheapest other model, the share of errors in which both calls agree on the same wrong point falls from 83% to 39% for GPT-6.1, from 64% to 29% for Sonnet 5.5, and from 73% to 48% for Gemini 3.8. An alarm defers an item when two points for it are farther apart than δ of the longer image side; *coverage* is the share of items acted on, and the *miss rate when acting* is the share of acted-on items whose point misses the target. At $\delta = 1.5\%$, the same-model alarm catches 17% of GPT-6.1's errors (4 of 23) and the cross-model alarm catches 61% (14 of 23); for Sonnet 5.5 the shares are 36% and 71% (Table 11). The cross-model alarm defers more items, and Figure 6 shows the trade-off: where the two curves overlap, its miss rate at matched coverage is lower by 0.5 to 2.7 points (point estimates on 260 items, without intervals; for GPT-6.1 at 93.1% coverage it catches 9 of the 23 errors against 6 for the same-model alarm), and for GPT-6.1 and Sonnet 5.5 it reaches operating points with a miss rate of 3% to 4%, below the lowest same-model values (7.0% and 10.5%). After two-stage zoom the same pattern holds, so the advantage comes from the second model and not from zoom alone. We zoomed each model's second click as well (772 calls), so that both alarms compare zoomed points. At $\delta = 1.5\%$ a second zoomed call of the same model catches 40% to 50% of the remaining errors, and the partner's zoomed point catches 68% to 78%, with miss rates of 2.6% to 3.6% on the 85% to 88% of items acted on (Table 11). The same-model alarm again acts on more items (93% to 94%); where the two curves overlap, the cross-model miss rate at matched coverage is lower by 0 to 1.5 points.

Re-asking a failed call. Consistent with persistent errors, re-asking also rarely repairs a failed call: in the detection experiments, resending the same request cleared 14.7% of status flags that had already repeated, while changing the request cleared 58.8% (Step 3 of Section 5).

7.7 Repair a click by choosing among models, on new screenshots

The several-model tier of Section 5.1 was designed on the 260 click items of the GUI experiment, where it missed 11 targets against 16 for GPT-6.1's two-stage zoom at its default setting. We then ran it, with its code, prompts, judge and tie rule unchanged and recorded by hash, on 600 further ScreenSpot-Pro items that share no item and no screenshot with those 260, together with the

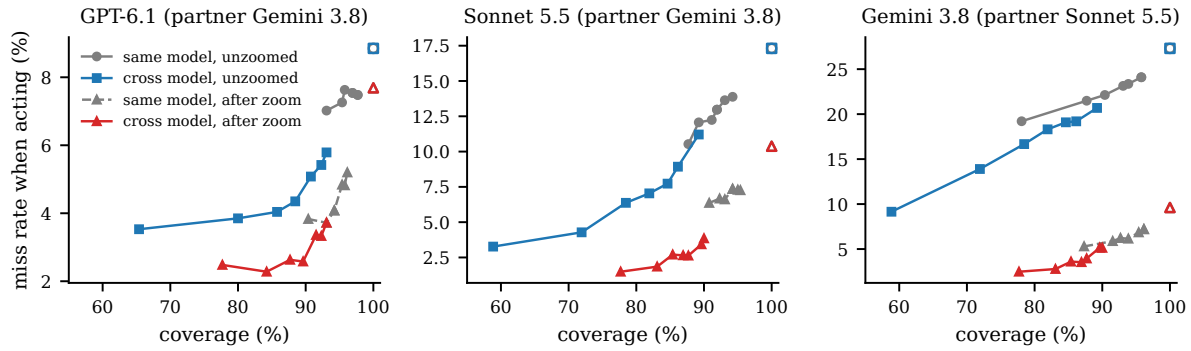


Figure 6: Risk-coverage of click alarms (260 items). An alarm defers an item when the acted-on point and a second point for the same item are farther apart than δ of the longer image side; the system acts on the other items. *Coverage* (x-axis) is the share of items acted on, and the *miss rate when acting* (y-axis) is the share of those items whose point is outside the target. Each curve traces $\delta \in \{0.5, 1, 1.5, 2, 3, 5, 8\}\%$: a smaller δ defers more items, moving left. Hollow markers at 100% coverage are the error rates of acting on every item. Grey: the second point comes from a second call of the same model; blue and red: from the partner, the cheapest other model. Solid grey and blue: first clicks on the whole screenshot; dashed grey and red: both points after two-stage zoom. Lower and further right is better. The cross-model alarm reaches lower miss rates than the same-model alarm both before and after zoom, at the price of deferring more items.

Table 11: Disagreement alarms for clicks (ScreenSpot-Pro and ScreenSpot-v2, 260 items; partner set by a fixed rule as the cheapest other model). The alarm compares the acted-on point with a second point for the same item and fires when the two are farther apart than $\delta = 1.5\%$ of the longer image side; the item is then deferred (sent for review or another step) instead of acted on. Unzoomed alarms compare first clicks; after zoom, the acted-on point is the zoomed first click, and the second point is the same model's zoomed second click or the partner's zoomed first click. Coverage: share of items acted on; miss rate: share of acted-on items whose point is outside the target; errors caught: share of the acted-on prediction's errors (the first click unzoomed, the zoomed first click after zoom) that the alarm defers. These deferrals measure risk and coverage; the automatic tier of Algorithm 2 defers nothing. Other δ values are in Figure 6. Where the same-model and cross-model curves overlap, miss rates at matched coverage (same vs. cross, in %, point estimates without intervals) are, unzoomed: GPT-6.1 (93.1%: 7.0 vs. 5.8); Sonnet 5.5 (87.7%: 10.5 vs. 10.1; 88.5%: 11.3 vs. 10.6; 89.2%: 12.1 vs. 11.2); Gemini 3.8 (78.1%: 19.2 vs. 16.5; 83.7%: 20.5 vs. 18.8; 89.2%: 21.9 vs. 20.7); after zoom: GPT-6.1 (90.4%: 3.8 vs. 2.9; 91.7%: 3.8 vs. 3.4; 93.1%: 3.7 vs. 3.7); Sonnet 5.5 (no overlap); Gemini 3.8 (87.3%: 5.3 vs. 3.8; 88.6%: 5.5 vs. 4.5; 90.0%: 5.7 vs. 5.1).

Model	Partner	Alarm	Error rate, no alarm	Coverage	Miss rate when acting	Errors caught
GPT-6.1	Gemini 3.8	same model, unzoomed	8.9%	95.8%	7.6%	17%
		cross model, unzoomed	8.9%	85.8%	4.0%	61%
		same model, after zoom	7.7%	94.2%	4.1%	50%
		cross model, after zoom	7.7%	87.7%	2.6%	70%
Sonnet 5.5	Gemini 3.8	same model, unzoomed	17.3%	91.1%	12.2%	36%
		cross model, unzoomed	17.3%	78.5%	6.4%	71%
		same model, after zoom	10.4%	93.1%	6.6%	41%
		cross model, after zoom	10.4%	85.4%	2.7%	78%
Gemini 3.8	Sonnet 5.5	same model, unzoomed	27.3%	90.4%	22.1%	27%
		cross model, unzoomed	27.3%	78.5%	16.7%	52%
		same model, after zoom	9.6%	92.7%	6.2%	40%
		cross model, after zoom	9.6%	85.4%	3.6%	68%

Table 12: Choosing one click among several calls, on 600 ScreenSpot-Pro items that share no item and no screenshot with the items used to design the method. Every item is in the denominator; a flow that returns no point counts as a miss, which happened only in single-model flows (column No point). Calls and cost: what the flow needs per item, at list prices. Lower block: paired comparisons, as the difference in miss rate in percentage points with a 95% paired bootstrap interval over screenshots, items repaired and broken, and the exact McNemar p (Holm-adjusted over the two comparisons marked †). D settings: GPT-6.1 and Sonnet 5.5 default, Gemini 3.8 lowest; M: all three lowest.

Flow	Misses	No point	Miss rate	Calls	Cost / item
GPT-6.1, lowest, two-stage zoom	86	4	14.33%	1.99	$0.0080
Sonnet 5.5, lowest, two-stage zoom	104	15	17.33%	1.98	$0.0056
Gemini 3.8, lowest, two-stage zoom	88	0	14.67%	2.00	$0.0054
Sonnet 5.5, default, two-stage zoom	80	0	13.33%	2.00	$0.0066
GPT-6.1, default, two-stage zoom (reference)	82	2	13.67%	2.00	$0.0085
GPT-6.1, three default zooms, cluster vote	76	1	12.67%	5.99	$0.0255
Crop-first baseline, GUI-Lens style (ours)	69	0	11.50%	2.00	$0.0117
Three models, lowest; judge score	66	0	11.00%	8.94	$0.0244
Three models, lowest; score and comparison	61	0	10.17%	9.42	$0.0261
Three models, D settings; judge score	61	0	10.17%	8.99	$0.0257
Three models, D settings; score, comparison (Alg. 2)	56	0	9.33%	9.45	$0.0273

Comparison	Difference	95% interval	Repaired / broken	p
D, score and comparison vs GPT-6.1 default zoom	-4.33	$[-6.33, -2.33]$	34 / 8	7×10^{-5}
D, score and comparison vs GPT-6.1 $\times 3$†	-3.33	$[-5.17, -1.50]$	27 / 7	0.002
D, score and comparison vs crop-first baseline†	-2.17	$[-4.17, 0.00]$	27 / 14	0.060
D, score and comparison vs D, judge score	-0.83	$[-2.00, 0.17]$	8 / 3	0.227
M, score and comparison vs M, judge score	-0.83	$[-2.00, 0.33]$	9 / 4	0.267
GPT-6.1 default zoom vs GPT-6.1 lowest zoom	-0.67	$[-2.67, 1.33]$	19 / 15	0.608
Sonnet 5.5 default zoom vs Sonnet 5.5 lowest zoom	-4.00	$[-6.17, -1.83]$	34 / 10	4×10^{-4}

baselines of Table 12. All 13,120 requests completed (13,126 attempts with retries; $44.93 at list prices for all arms), no flow needed the fallbacks for a failed call, zoom, score or comparison, and every item stays in the denominator. When the run was interrupted once and resumed, at most 8 requests were in flight without a stored response; their cost, at most about $0.05, is not in this total. Single-model flows sometimes returned no point, counted as a miss (Table 12); the three-model flows always returned one.

The tier against single-model practice. On the 600 new items the tier misses 56 targets (9.33%), against 82 (13.67%) for GPT-6.1's two-stage zoom at its default setting: -4.33 points $[-6.33, -2.33]$, with 34 items repaired and 8 broken (exact McNemar $p = 7 \times 10^{-5}$). At about the same cost, three independent default zooms of GPT-6.1 with a cluster vote miss 76, so the tier is better by 3.33 points $[1.50, 5.17]$ (Holm-adjusted $p = 0.002$). Against a crop-first baseline that we implemented in the style of GUI-Lens (Fu et al., 2026), without its OCR and detector anchors or its verification step and with our own prompts, the tier misses 56 against 69: the difference is -2.17 points with an interval that reaches zero ($[-4.17, 0.00]$; Holm-adjusted $p = 0.060$), at 2.3 times that baseline's cost. The data therefore do not show an advantage over crop-first prompting. The comparisons support the tier as a whole; they do not by themselves attribute its gain to the diversity of its models.

Where the gain comes from. Proposition 10 splits the tier's 56 misses into 43 items on which no model's zoomed point hits (F), 3 on which a correct point exists but no cluster representative is correct (L), and 10 on which the comparison picks the wrong representative (W). With the judge's score alone, W is 15 and the other two terms are unchanged. The comparison therefore repairs 8 items and breaks 3: its increment is -0.83 points $[-2.00, 0.17]$, and -0.83 $[-2.00, 0.33]$ with all three models at their lowest setting, neither significant. On the representative pairs in which exactly one point hits, a single comparison call names the right one in 85.9% of calls, and the two orders agree on 71 of 78 pairs, 64 of them correctly; on the design items the single-call rate was 88%. Most of the gain is in F. One default zoom of GPT-6.1 has no correct point on 82 items; three of them, at about the tier's cost, still have none on 62; the tier's three models have none on 43. Proposition 11 names the rate behind this: on the 82 items that GPT-6.1 missed, a second GPT-6.1 zoom hits on 18% and a third, after two misses, on 7.5%, while Sonnet

5.5 hits on 29% and Gemini 3.8 on 37%, and Gemini 3.8 still hits on 26% of the 58 items that both GPT-6.1 and Sonnet 5.5 missed. On these items, repeating GPT-6.1 adds little coverage, consistent with its persistent errors (Section 7.6); the partners rescue more here, which Proposition 11 does not guarantee in general. Counted with the same split, the three-zoom vote's 76 misses are 62 items with no correct candidate and 14 later losses, and the tier's 56 are 43 and 13: of the 20 fewer misses, 19 lie in candidate coverage. This describes the two outputs; it does not separate the roles of the models, settings and judge.

Items without an alarm. On 499 of the 600 items the three points form one cluster and no comparison is made. The tier misses 29 of them (5.8%), which is 52% of its misses. An action applied to all of them, or through a trigger independent of whether the base misses, improves accuracy only if $h < e_T\, r/(1 - e_T)$ with $e_T = 5.8\%$ (Proposition 12): about 3.1% of the correct answers it touches if it repairs half of the misses. The judge's score alone misses 32 of the other 101 items (31.7%), which is where the comparison acts. Nine clusters are chained, with two members farther apart than $\delta = 1.5\%$, and none is wider than 2.13% of the longer side.

Settings. The default setting helps Sonnet 5.5 (misses -4.00 points $[-6.17, -1.83]$ against its lowest setting) and does not change GPT-6.1 significantly (-0.67 $[-2.67, 1.33]$); on the design items the direction was the same. This is why the tier runs Sonnet 5.5 at its default setting.

The crop-first baseline errs elsewhere. The crop-first baseline and the tier miss different items: both miss 42, only the baseline 27, only the tier 14. Of those 14, the baseline supplies a correct point on 7 items where all three of the tier's models miss, so it is a new source in the sense of Proposition 11: added to the tier's menu, it lowers the items with no correct candidate from 43 to 36 ($\rho = 7/43$). On the other 7 the tier had a correct candidate and lost it (L, 2 items) or chose wrongly (W, 5). Conversely, on 14 of the 27 items that only the baseline misses, its wrong point lies within δ of one of the tier's candidates, so adding it to the vote would mostly strengthen wrong clusters (on 12 of the 14 every nearby candidate is wrong).

What a fourth source can add. On the 57 items where the crop-first point and the tier's point are more than δ apart, the crop-first point is right and the tier's wrong on 10, the reverse holds on 25, both are right on 12 and both wrong on 10. By Proposition 12, a rule that switches between the two changes the number of misses by $-(10r - 25h)$, where r and h are the shares of the 10 and of the 25 on which it takes the crop-first point. Within this two-candidate, threshold-triggered menu the largest possible improvement is 10 items (1.67 points). The comparison's accuracy on the representative pairs above does not determine r and h on these different pairs, so we evaluated an enlarged candidate menu directly. We added the crop-first point as a fourth candidate, ranked last among sources, scored by the same judge and handled by the same rules, on the same 600 items whose cross-tabulation above suggested it, so the result describes those items rather than new ones (717 further calls, \$1.44). Items with no correct candidate fall from 43 to 36, as Proposition 11 gives with $\rho = 7/43$. Lost representatives rise from 3 to 7 and wrong choices from 10 to 11: on 3 of the 7 items where only the crop-first point is correct, it joins a cluster of wrong points and loses the representative choice to a wrong member with the same score, and on one further item it displaces a correct member. The four-source tier misses 54 items against 56 (-0.33 points $[-1.17, 0.34]$; 4 repaired, 2 broken) at \$0.041 per item (12.5 calls) instead of \$0.027. Proposition 10 accounts for the result term by term: the added source pays in F and is charged in L and W. The tier of Algorithm 2 therefore stays at three sources, and the crop-first baseline is reported as a baseline.

7.8 Judge: selection has room left, but labels do not close it

Selection among a model's own outputs. Choosing the best feasible set from a model's own three calls would remove 45.2% of set error relative to one call (median over 12 cells; Table 13), and 72% to 74% for GPT-6.1 on SKU-110K and DOTA. The constrained oracle differs from the unrestricted value on 25 of 720 images and has the same median. Record gates on output features, fitted with labels and evaluated by cross-fitting, reach 15.6% (logistic) and 14.6% (boosted trees); fitted on 10 labelled images they reach 13.2%. The more flexible selector does not help under cross-fitting, and its in-sample value (36.5%) is overfit. The empirical gap, median 26.1 points, combines information absent from the output features with selector

Table 13: Selection among three calls of one model (E3, list interface, lowest setting, 60 images per cell; reduction of E relative to the first call, in %). The constrained oracle optimizes over the same feasible family as the selectors (Proposition 8); the unrestricted value lets two members of one cluster match two targets and is not attainable. Selectors are logistic or boosted-tree record gates on output features; they read no labels at inference but are *fitted* with labels (5-fold cross-fitting by image with persisted folds; "10 labels" is the median over 40 draws of 10 labelled images, each evaluated on the images it did not use). The gap (oracle minus cross-fitted logistic, with an evaluation interval) mixes information absent from the features with selector and estimation shortfall (Proposition 7); it is not an information limit.

Source	Model	Oracle	Unrestricted	Differ	Cross-fit logistic	Cross-fit trees	In-sample logistic	10 labels	Gap
DocLayNet	GPT-6.1	63.2	63.2	0	−8.4	−7.4	1.6	−7.7	71.6 [60.1, 87.4]
DocLayNet	Sonnet 5.5	59.8	59.8	0	17.3	14.8	22.6	16.6	42.5 [34.2, 54.0]
DocLayNet	Gemini 3.8	45.2	45.2	0	5.9	1.3	6.9	3.2	39.3 [30.1, 50.3]
DOTA	GPT-6.1	74.3	74.4	1	39.9	39.6	40.4	39.6	34.4 [25.1, 42.6]
DOTA	Sonnet 5.5	41.9	41.9	1	23.3	22.6	24.1	21.3	18.6 [13.7, 25.1]
DOTA	Gemini 3.8	49.3	49.3	0	26.8	24.6	27.5	20.5	22.5 [16.7, 30.1]
FSC-147	GPT-6.1	16.7	19.0	12	0.2	−0.7	1.6	−0.7	16.5 [7.6, 31.7]
FSC-147	Sonnet 5.5	16.6	17.3	9	1.1	1.0	1.2	0.8	15.5 [10.6, 21.4]
FSC-147	Gemini 3.8	6.0	6.2	2	3.9	4.1	4.2	2.0	2.1 [0.8, 4.0]
SKU-110K	GPT-6.1	71.8	71.8	0	42.0	39.9	42.2	40.7	29.8 [27.2, 32.7]
SKU-110K	Sonnet 5.5	45.3	45.3	0	34.9	34.8	35.1	34.6	10.3 [8.4, 12.6]
SKU-110K	Gemini 3.8	44.6	44.6	0	13.9	14.3	14.3	9.8	30.6 [26.4, 35.6]
Median of 12 cells		45.2	45.2	25 (total)	15.6	14.6	18.4	13.2	26.1

and estimation shortfall (Proposition 7); our results bound it but do not apportion it. It measures the distance between the tested selectors and the best set in a fixed library of three calls, so it does not bound what adding calls or changing the library could gain. Within the operations we compared, the larger gains came from changing the information, a different view, model or readout, rather than from selecting among a model's own outputs.

7.9 Judge: defaults that transfer

Table 14 (full values in Table 19) evaluates construction transfer on stored reports: for each source, the construction that does best on the other five sources is applied to the stored calls of the sixth, so it is never chosen on the source it is evaluated on. Table 3 shows where the final defaults follow these transferred constructions and where they differ. For set error, the best same-model construction on the other five sources is three-call unanimity with fused coordinates on every evaluated source. It lowers set error significantly in 9 of the 15 over-reporting cells and raises it significantly in none; on FSC-147, the under-reporting family, it is worse in 2 of 3 cells, consistent with Proposition 2 and Corollary 2. For F1, the transferred same-model construction is better in 2 cells, worse in 4 and not significant in 12, so the default is a single call. Three calls cost three times one call. From logged token usage at list prices, unanimity costs $0.029 to $0.21 per image for GPT-6.1 across the five over-reporting sources, $0.032 to $0.13 for Sonnet 5.5 and $0.011 to $0.058 for Gemini 3.8.

Configurations that transfer. The same logic applies to the configuration of a single call: the reporting interface (one list for the whole image, or one row for each region of a fixed registry) and the reasoning setting. For each model, we chose among the four combinations on three detection sources and applied the choice to the fourth, so the labels of the evaluated source are used only to score it (Table 15). For F1, the chosen configuration is better than the list interface at the lowest setting in 5 of 12 cells and worse in 1 (Sonnet 5.5 on DocLayNet, -6.5 points); macro F1 rises from 42.1 to 49.1, against 51.7 for the best configuration of each cell chosen with its own labels, and the chosen calls cost 1.44 times as much. For set error, the same rule is better in 2 cells, worse in 1 and not significant in 9 (median change 0%), so the set-error defaults stay as in Table 3. One configuration shared by the three models transfers worse (F1: 5 cells better, 3 worse). Proposition 9 is the reason a choice made on related sources carries over when their risks are close, and the one worse cell shows where they are not. These numbers evaluate the selection procedure. The configuration that ships is chosen by the same rule on all four sources (GPT-6.1: regions, lowest; Sonnet 5.5: regions, default; Gemini 3.8: list, default; Table 20); the folds exclude the labels of the source they are evaluated on, so they are not a replay of the shipped configuration on its own data, and the shipped choice coincides with the fold choice in 9 of the 12 cells.

Table 14: Defaults evaluated by leave-one-source-out transfer: for each source, the construction is the one that is best on the other five sources (same model only, or any construction including cross-model ones), and it is evaluated on the evaluated source against a single call of the user's model. Entries are relative changes of the loss (for E, negative is better; for F1, positive is better). **Bold**: paired 95% interval excludes 0 in the favourable direction; underlined: significantly worse. *: the user's model is the strongest single model on that source and loss. Every significantly worse cross-model entry carries a $\star$. Costs, recall and the chosen construction per cell are in Appendix B.

	Same model only						Any construction (incl. cross-model)					
	E			F1			E			F1		
Source	GPT	Son.	Gem.	GPT	Son.	Gem.	GPT	Son.	Gem.	GPT	Son.	Gem.
DocLayNet	$+7\%^\star$	**-10%**	$+3\%$	$-3\%^\star$	-1%	**$+19\%$**	$+4\%^\star$	**-36%**	**-13%**	$-7\%^\star$	**$+15\%$**	**$+75\%$**
DOTA	**-22%**	**$-16\%^\star$**	**-25%**	$-1\%^\star$	-1%	$+2\%$	**-30%**	**$-17\%^\star$**	**-22%**	$-6\%^\star$	$+11\%$	**$+17\%$**
FSC-147	$+3\%^\star$	$+7\%$	$+5\%$	$-2\%^\star$	-6%	$+1\%$	$+58\%^\star$	-1%	-1%	$-35\%^\star$	$+1\%$	**$+68\%$**
SKU-110K	**-25%**	**-32%**	**$-10\%^\star$**	$-8\%^\star$	$+1\%$	**$+13\%$**	**-30%**	**-36%**	**$-8\%^\star$**	$-22\%^\star$	**$+165\%$**	**$+21\%$**
GroundCUA	-4%	$-1\%^\star$	**-21%**	-4%	$+2\%^\star$	$+12\%$	**-26%**	$+52\%^\star$	**-42%**	**$+24\%$**	$-17\%^\star$	**$+61\%$**
ScreenParse	**-3%**	$-2\%^\star$	-3%	0%	$0\%^\star$	$+2\%$	**-14%**	$+4\%^\star$	**-12%**	**$+6\%$**	$-2\%^\star$	**$+11\%$**
Better / worse / n.s.		9 / 2 / 7			2 / 4 / 12			12 / 2 / 4			10 / 3 / 5	

Table 15: Single-call configuration chosen on the other three detection sources and applied to the fourth (F1, primary scoring). Reference: list interface at the lowest setting (low); def.: default setting. Change in F1 points with 95% paired bootstrap interval over images. Macro F1 42.1 (reference) and 49.1 (chosen); the best configuration per cell in hindsight gives 51.7. The table evaluates the selection procedure. The shipped configuration (Table 3) is chosen by the same rule on all four sources and equals the choice shown here in 9 of 12 cells; Table 20 gives the four fixed configurations.

Source	Model	Chosen	Change [95% CI]
DocLayNet	GPT	regions, low	$+1.5$ $[-2.6, +6.0]$
DocLayNet	Sonnet	regions, def.	-6.5 $[-10.4, -3.1]^-$
DocLayNet	Gemini	list, def.	$+14.4$ $[+6.6, +22.5]^+$
DOTA	GPT	regions, low	-1.7 $[-5.3, +1.7]$
DOTA	Sonnet	regions, def.	$+3.1$ $[-1.8, +7.4]$
DOTA	Gemini	regions, def.	$+4.9$ $[-0.5, +10.9]$
FSC-147	GPT	regions, low	$+9.6$ $[+2.2, +17.6]^+$
FSC-147	Sonnet	regions, def.	$+19.0$ $[+12.3, +26.0]^+$
FSC-147	Gemini	regions, def.	$+22.6$ $[+14.3, +31.4]^+$
SKU-110K	GPT	regions, low	$+0.4$ $[-0.5, +1.4]$
SKU-110K	Sonnet	list, def.	$+13.2$ $[+8.9, +17.8]^+$
SKU-110K	Gemini	list, def.	$+2.9$ $[-2.2, +8.1]$

When cross-model constructions are allowed, the transferred default improves set error in 12 of 18 cells and F1 in 10 of 18. It is better in 20 of the 24 cells in which the user's model is not the strongest single model for that source and loss, and not significantly different in the other 4. All 5 cells in which it is significantly worse are cells where the user's model is the strongest; for example, a three-model majority raises Sonnet 5.5's set error on GroundCUA by 52% relative to one call, and a two-model intersection raises GPT-6.1's on FSC-147 by 58%. Among constructions with averaged coordinates, against the best same-model construction chosen with all labels (a reference, not a deployable rule), the best cross-model construction is significantly better only on DOTA (-5.8%) and significantly worse on GroundCUA ($+26.0\%$). When the user does not know which model is strongest, three-model unanimity is significantly better than three-call unanimity of a randomly chosen model on 4 of 5 over-reporting sources (-4% to -11%). Voting a strong model together with weaker ones is not a safe default: a three-model majority is 52% worse than GPT-6.1's own three-call majority on FSC-147 and 59% worse than Sonnet 5.5's on GroundCUA, three-model click voting is 3.5 points below GPT-6.1 alone, and the per-type median of three models' dense counts is 0.60 worse than Sonnet 5.5 alone. On aerial and shelf detection the same vote helps instead: it is 9.5% better than Sonnet 5.5's three-call majority on DOTA and 4.7% better than Gemini 3.8's on SKU-110K; on DocLayNet and ScreenParse it is not significantly different. Cross-model confirmation is therefore a menu option chosen with a few labels or a family-level ranking, while the cross-model alarm is a default whenever a second model is available.

8 Discussion

What a practitioner can take away. The results reduce to a short list of diagnostics, each computable from outputs alone, and a fix with its failure condition (Table 16). Three quantities do most of the work. The readout gap, the difference between the list count and the stated count of one response, shows that the two disagree; the degeneracy flags

then decide which to read, and by default the list count is read unless the list is degenerate. Agreement between two calls measures how stable the output is, not whether it is right; under the repeated-call model, high agreement means that another call with the same interface is expected to add little new coverage, while whether an additional call or source of evidence is worth paying for is decided by its measured benefit and cost in the evaluated configuration. The marginal matching rate p_1 of unconfirmed records says whether to delete or keep them; the shipped configuration encodes it as the task family's regime, and with three or more labelled images the label rule of Step 7 can switch to a construction that deletes or keeps.

Why the gains come from changing information. The theory and the measurements point the same way. Object-level persistence means that repeated calls of one model share most of their errors. In our comparisons, votes over the same outputs gained at most 23% for counts and 4.5% for clicks, and even a perfect selector over three calls can only choose among the records those calls contain. The largest gains we measured came from operations that change what is available: reading a different field of the same response, looking at a different view, or asking a model with a different model-specific component. Where those operations fail, the failures have the predicted form: deletion hurts when $p_1 > 1/2$, tiling can lose objects that no tile contains whole, and a partner hurts when its additional loss of true matches outweighs its additional rejection of false records, with the effect of averaged coordinates counted separately.

Where click accuracy can still come from. The split of Proposition 10 locates the tier's remaining misses on the 600 new screenshots: 43 items with no correct candidate, 10 wrong choices and 3 lost representatives, and 29 of the 56 misses lie on items where the three models agree. A rule that chooses among existing candidates acts on at most the 13 misses of the last two terms. The other 43 need a source with a high rescue rate ρ on exactly those items; the crop-first call has $\rho = 7/43$, but on 4 of the 7 items where only its point is correct the tier still misses (on 3 its point loses the representative choice inside a cluster of wrong points), and on one further item it displaces a correct answer, so most of its gain moves into the other two terms. On items where the models agree, Proposition 12 gives the condition for any further action, $h < e_T\, r/(1 - e_T)$ on the items it touches: if it acts on all of them, or through a trigger independent of whether the base misses, $e_T = 5.8\%$, and an action that repairs half of its misses must break fewer than about 3.1% of its correct answers. These two measurable quantities, the rescue rate on the items no candidate covers and the break rate on agreeing items, decide whether a further source or action raises accuracy, and both are measured from stored calls on labelled items.

What future models should change. The measurements point to properties that model builders can target directly. A model's stated summary should be computed from its own enumeration: GPT-6.1 and Sonnet 5.5 already list almost exactly the right number of objects, and their stated counts lose that information. Long lists should not degrade with position; Gemini 3.8 at its lowest setting drops class fields more often when consecutive records share a class; an order that interleaves classes lowers its count error and grouping by class raises it (Section 7.3). Class vocabularies given in the prompt should be respected, since Sonnet 5.5 still wrote "diamond" 258 times when the format named the allowed classes. Finally, errors that persist across calls limit what a fixed budget of resampling can repair; diversity across calls, or calibrated uncertainty at the object level, would make the cheaper same-model alarm as useful as a second model.

Scope of the results. The theory's guarantees hold under their stated conditions. The identities of Section 4.1 hold for any outputs and any fixed scorer. Proposition 5 assumes deletion-only confirmation, no match takeover, and no ignore region or none of the base report's false records inside one; outside these conditions the exact identity of Proposition 1 replaces it. The disagreement floor is a bound for point tasks and for boxes at IoU thresholds above 0.5; the click alarm is empirical and is evaluated by risk and coverage. Proposition 13 concerns model classes that admit scenes with identical or nearly identical output laws; when an object is reported with probability at most ϑ per call, K calls separate the two scenes by at most $K\vartheta$ in total variation. The readout repair applies where a readout gap exists: on our dense synthetic scenes the gap is most of the error, while on aerial and document images it is a small part of it for GPT-6.1 and Sonnet 5.5. In all but three runs, changing the reasoning setting also changed the output cap, and

Table 16: Symptoms, label-free diagnostics, fixes and what not to do. Each row is backed by a result in Section 7.

Symptom	Diagnostic (no labels at run time)	Fix	Do not
Stated count disagrees with the list	readout gap in the same response	read the list count unless the list is flagged as degenerate (§7.2)	read the stated count when the list is intact
Class names outside the vocabulary	share of records whose class fails α_τ	list allowed names; map declared aliases in code (§7.2)	rely on the prompt alone
List degrades toward its end	four degeneracy flags	fall back to the stated count; scan row by row; use the default setting (§7.3)	group the list by type
Small objects missed, clicks off target	apparent size of objects; the model's measured recall curve	views from a view planner that keeps every object inside some view (one-object overlap, one call per view), merged; confirm with them only the records each view contains, and only where unconfirmed records are mostly false ($p_1 < 1/2$); two-stage zoom on native pixels (§7.5)	tile without a one-object overlap; pack tiles into one call; let a view reject records it cannot see
Misses that may repeat across calls	agreement between two calls of the same model: high agreement means stable outputs and little new coverage from another call of the same interface; it is not proof of correctness or of zero remaining value	confirm with another view or model where it passes Proposition 5; cross-model disagreement as an alarm (§7.6)	buy more calls of the same model when they already agree; same-view self-review on screens
Unsure whether to delete	the shipped family regime; with 3 or more labelled images, the label rule of Step 7	with equal error costs, delete unconfirmed records iff $p_1 < 1/2$ (§7.4)	intersect for count objectives
Unsure whether to trust the output	disagreement floor; cross-model disagreement	alarm and status; labels to certify (§4.5)	read agreement as correctness
Click must be right; budget covers the several-model tier's worst case	number of clusters among the models' zoomed points (alarm when two or more)	three models' zoomed points, a judge's score per candidate, a two-order comparison only on alarmed items (§7.7)	spend the budget on repeated calls of one model without measuring their rescue rate; read agreement among models as proof that an answer is right
Several models available	family-level ranking or a few labels	alarm by default; cross-model confirmation from the menu (§7.9)	an equal-weight vote of all models on points, screens, clicks or counts when one model is strongest (worse or not significantly different there; better on DOTA and SKU-110K, §7.9)

Sonnet 5.5's reasoning share at its default setting ranged from 0% to 90% by task, so Sonnet 5.5 results are attributed to reasoning only where most calls reasoned. All results are for the model versions listed in Appendix C, at a total cost of about $758; we release every response so that the identities and tables can be recomputed for new versions without repeating our calls.

Broader impact. Better-calibrated set reports can make inventory, document and screen-automation systems more reliable, and an explicit status and alarm make failures visible instead of silent. The same techniques could make automated agents act with more confidence; our pipeline always returns a status and never claims certification without labels, which we consider the appropriate default.

9 Conclusion

Reporting is part of the visual task, not merely its formatting. When vision-language models report sets of objects, part of their error is in the report rather than in the seeing, and treating reporting as a chain from responses to records, sets and readouts makes that error analysable. The chain gives exact identities for what each operation does, a behavioural model of persistent errors, and a proof that outputs alone cannot certify correctness when two scenes produce the same outputs. It also separates three problems. Repair works where the information is already present: read the count from the model's list of records, not from the numbers it states, unless the list has degenerated; delete unconfirmed records only below the cost threshold; change the report structure, its decoding (declared aliases applied in code), the view or the model, rather than restating the same request more insistently. Detection benefits most from a different source of evidence; on our click tasks a second model provided it better than a second call. For one click, the error splits exactly into missing candidates, lost representatives and wrong choices, which tells where to spend calls: a tier that adds two other models' candidates and compares them only where they disagree lowered the miss rate on 600 new screenshots by about a third relative to one model's zoom. Judging between reports needs labels or a configuration transferred from other data; three-call unanimity, chosen on five sources, improved set error on most over-reporting cells of the sixth and did not significantly worsen it in any of them. Every number in this paper can be regenerated from the released responses, with the DOTA, SKU-110K and FSC-147 labels rebuilt from their official downloads.

References

Fatih Cagatay Akyon, Sinan Onur Altinuc, and Alptekin Temizel. 2022. Slicing aided hyper inference and fine-tuning for small object detection. In *IEEE International Conference on Image Processing (ICIP)*.

Niccolo Avogaro, Nayanika Debnath, Li Mi, Thomas Frick, Junling Wang, Zexue He, Hang Hua, Konrad Schindler, and Mattia Rigotti. 2026. SPARC: Separating perception and reasoning circuits for test-time scaling of VLMs. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*.

S. Divakar Bhat and Toshihiko Yamasaki. 2026. Consistent yet wrong: Evidence insensitivity in spatial vision-language models. *Preprint*, arXiv:2606.02742.

Daniel Bolya, Sean Foley, James Hays, and Judy Hoffman. 2020. TIDE: A general toolbox for identifying object detection errors. In *European Conference on Computer Vision (ECCV)*.

Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. 2020. End-to-end object detection with transformers. In *European Conference on Computer Vision (ECCV)*.

Ángela Casado-García and Jónathan Heras. 2020. Ensemble methods for object detection. In *European Conference on Artificial Intelligence (ECAI)*.

Ting Chen, Saurabh Saxena, Lala Li, David J. Fleet, and Geoffrey Hinton. 2022. Pix2seq: A language modeling framework for object detection. In *International Conference on Learning Representations (ICLR)*.

Kanzhi Cheng, Qiushi Sun, Yougang Chu, Fangzhi Xu, Yantao Li, Jianbing Zhang, and Zhiyong Wu. 2024. SeeClick: Harnessing GUI grounding for advanced visual GUI agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9313–9332.

A. P. Dawid and A. M. Skene. 1979. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Journal of the Royal Statistical Society. Series C (Applied Statistics)*, 28(1):20–28.

Matt Deitke, Christopher Clark, et al. 2025. Molmo and PixMo: Open weights and open data for state-of-the-art vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Yong Du, Yuchen Yan, Fei Tang, Zhengxi Lu, Chang Zong, Weiming Lu, Shengpei Jiang, and Yongliang Shen. 2026. Test-time reinforcement learning for GUI grounding via region consistency. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*.

Jack Edmonds and D. R. Fulkerson. 1965. Transversals and matroid partition. *Journal of Research of the National Bureau of Standards, Section B: Mathematics and Mathematical Physics*, 69B(3):147–153.

Ahmed Oumar El-Shangiti, Abzal Nurgazy, Hilal AlQuabeh, Nikolai Rozanov, and Kentaro Inui. 2026. The count is there, but misaligned: Understanding and correcting counting failures in VLMs. *Preprint*, arXiv:2607.09544.

Charles Elkan. 2001. The foundations of cost-sensitive learning. In *Proceedings of the Seventeenth International Joint Conference on Artificial Intelligence (IJCAI)*, pages 973–978.

Aarash Feizi, Shravan Nayak, Xiangru Jian, Kevin Qinghong Lin, Kaixin Li, Rabiul Awal, Xing Han Lù, Johan Obando-Ceron, Juan A. Rodriguez, Nicolas Chapados, David Vazquez, Adriana Romero-Soriano, Reihaneh Rabbany, Perouz Taslakian, Christopher Pal, Spandana Gella, and Sai Rajeswar. 2026. Grounding computer use agents on human demonstrations. In *International Conference on Learning Representations (ICLR)*.

Zichuan Fu, Shirong Wang, Wenlin Zhang, Guojing Li, Yimin Deng, Jingtong Gao, Junjia Qi, Hanyu Yan, Yefeng Zheng, Xiaopeng Li, Wanyu Wang, Xian Wu, and Xiangyu Zhao. 2026. GUI-Lens: Coarse-to-fine cropping for GUI grounding with general-purpose VLMs. *arXiv preprint arXiv:2608.03270*.

Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30.

Eran Goldman, Roei Herzig, Aviv Eisenschtat, Oria Ratzon, Itsik Levi, Jacob Goldberger, and Tal Hassner. 2019. Precise detection in densely packed scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5222–5231.

Bernard Grofman, Guillermo Owen, and Scott L. Feld. 1983. Thirteen theorems in search of the truth. *Theory and Decision*, 15(3):261–278.

Tanmay Gupta and Aniruddha Kembhavi. 2023. Visual programming: Compositional visual reasoning without training. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

A. Said Gurbuz, Sunghwan Hong, Ahmed Nassar, Marc Pollefeys, and Peter Staar. 2026. ScreenParse: Moving beyond sparse grounding with complete screen parsing supervision. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*.

Kimia Hamidieh, Veronika Thost, Walter Gerych, Mikhail Yurochkin, and Marzyeh Ghassemi. 2026. Complementing self-consistency with cross-model disagreement for uncertainty quantification. In *International Conference on Learning Representations (ICLR)*.

Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. 2024. Large language models cannot self-correct reasoning yet. In *International Conference on Learning Representations (ICLR)*.

Zhiyuan Jiang, Shenghao Xie, Wenyi Li, Wenqiang Zu, Peihang Li, Jiahao Qiu, Siqi Pei, Lei Ma, Tiejun Huang, Mengdi Wang, and Shilong Liu. 2025. Zoom in, click out: Unlocking and evaluating the potential of zooming for GUI grounding. *Preprint*, arXiv:2512.05941.

Harold W. Kuhn. 1955. The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2):83–97.

Ludmila I. Kuncheva and Christopher J. Whitaker. 2003. Measures of diversity in classifier ensembles and their relationship with the ensemble accuracy. *Machine Learning*, 51(2):181–207.

Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. 2017. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30.

Xuanyu Lei, Zonghan Yang, Xinrui Chen, Peng Li, and Yang Liu. 2025. Scaffolding coordinates to promote vision-language coordination in large multi-modal models. In *Proceedings of the 31st International Conference on Computational Linguistics (COLING 2025)*, pages 2886–2903.

Jincheng Li, Chunyu Xie, Ji Ao, Dawei Leng, and Yuhui Yin. 2025a. LMM-Det: Make large multimodal models excel in object detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*.

Kaixin Li, Ziyang Meng, Hongzhan Lin, Ziyang Luo, Yuchen Tian, Jing Ma, Zhiyong Huang, and Tat-Seng Chua. 2025b. ScreenSpot-Pro: GUI grounding for professional high-resolution computer use. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM)*, pages 8778–8786.

Yuan-Hong Liao, Rafid Mahmood, Sanja Fidler, and David Acuna. 2025. Can large vision-language models correct semantic grounding errors by themselves? In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Zachary C. Lipton, Charles Elkan, and Balakrishnan Narayanaswamy. 2014. Optimal thresholding of classifiers to maximize F1 measure. In *Machine Learning and Knowledge Discovery in Databases (ECML PKDD)*, pages 225–239.

Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. Self-Refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36.

George L. Nemhauser, Laurence A. Wolsey, and Marshall L. Fisher. 1978. An analysis of approximations for maximizing submodular set functions, I. *Mathematical Programming*, 14(1):265–294.

Kemal Oksuz, Baris Can Cam, Emre Akbas, and Sinan Kalkan. 2018. Localization recall precision (LRP): A new performance metric for object detection. In *European Conference on Computer Vision (ECCV)*.

Roni Paiss, Ariel Ephrat, Omer Tov, Shiran Zada, Inbar Mosseri, Michal Irani, and Tali Dekel. 2023. Teaching CLIP to count to ten. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3170–3180.

Xingzhou Pang, Yifan Hou, Junling Wang, and Mrinmaya Sachan. 2026. Unveiling the visual counting bottleneck in vision-language models. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*.

Birgit Pfitzmann, Christoph Auer, Michele Dolfi, Ahmed S. Nassar, and Peter Staar. 2022. DocLayNet: A large human-annotated dataset for document-layout segmentation. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, pages 3743–3751.

Emmanouil Antonios Platanios, Avrim Blum, and Tom Mitchell. 2014. Estimating accuracy from unlabeled data. In *Proceedings of the Thirtieth Conference on Uncertainty in Artificial Intelligence (UAI)*.

Muhammad Fetrat Qharabagh, Mohammadreza Ghofrani, and Kimon Fountoulakis. 2026. LVLM-Count: Enhancing the counting ability of large vision-language models. *Transactions on Machine Learning Research*.

Pooyan Rahmanzadehgervi, Logan Bolton, Mohammad Reza Taesiri, and Anh Totti Nguyen. 2024. Vision language models are blind. In *Proceedings of the Asian Conference on Computer Vision (ACCV)*.

Viresh Ranjan, Udbhav Sharma, Thu Nguyen, and Minh Hoai. 2021. Learning to count everything. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3394–3403.

Haozhan Shen, Kangjia Zhao, Tiancheng Zhao, Ruochen Xu, Zilun Zhang, Mingwei Zhu, and Jianwei Yin. 2024. ZoomEye: Enhancing multimodal LLMs with human-like zooming capabilities through tree-based image exploration. *arXiv preprint arXiv:2411.16044*.

Junzhe Shi, Yuan Gan, and Shida Jiang. 2026. Why VLMs miss small objects, and when zooming in is safe. *Preprint*, arXiv:2610.09313.

Roman Solovyev, Weimin Wang, and Tatiana Gabruseva. 2021. Weighted boxes fusion: Ensembling boxes from different object detection models. *Image and Vision Computing*, 107:104117.

Dídac Surís, Sachit Menon, and Carl Vondrick. 2023. ViperGPT: Visual inference via Python execution for reasoning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*.

Hexiang Tan, Fei Sun, Sha Liu, Du Su, Qi Cao, Xin Chen, Jingang Wang, Xunliang Cai, Yuanzhuo Wang, Huawei Shen, and Xueqi Cheng. 2025. Too consistent to detect: A study of self-consistent errors in LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Fei Tang, Bofan Chen, Zhengxi Lu, Tongbo Chen, Songqin Nong, Tao Jiang, Wenhao Xu, Weiming Lu, Jun Xiao, Yueting Zhuang, and Yongliang Shen. 2026. UI-Zoomer: Uncertainty-driven adaptive zoom-in for GUI grounding. *Preprint*, arXiv:2604.14113.

Alexandre B. Tsybakov. 2009. *Introduction to Nonparametric Estimation*. Springer.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*.

Jacob Whitehill, Paul Ruvolo, Tingfan Wu, Jacob Bergsma, and Javier Movellan. 2009. Whose vote should count more: Optimal integration of labels from labelers of unknown expertise. In *Advances in Neural Information Processing Systems (NIPS)*.

Penghao Wu and Saining Xie. 2024. V*: Guided visual search as a core mechanism in multimodal LLMs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Gui-Song Xia, Xiang Bai, Jian Ding, Zhen Zhu, Serge Belongie, Jiebo Luo, Mihai Datcu, Marcello Pelillo, and Liangpei Zhang. 2018. DOTA: A large-scale dataset for object detection in aerial images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3974–3983.

Jianwei Yang, Hao Zhang, Feng Li, Xueyan Zou, Chunyuan Li, and Jianfeng Gao. 2023. Set-of-mark prompting unleashes extraordinary visual grounding in GPT-4V. *arXiv preprint arXiv:2310.11441*.

Jiarui Zhang, Mahyar Khayatkhoei, Prateek Chhikara, and Filip Ilievski. 2025. MLLMs know where to look: Training-free perception of small visual details

with multimodal LLMs. In *International Conference on Learning Representations (ICLR)*.

Yulong Zhang, Tianyi Liang, Xinyue Huang, Erfei Cui, Guoqing Wang, Xu Guo, Chenhui Li, and Gongshen Liu. 2026. Consensus entropy: Harnessing multi-VLM agreement for self-verifying and self-improving OCR. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

A Proofs

Throughout, $E(S) = c(S) + n - 2r(S)$ and $E_w(S) = c_{\mathrm{FP}}[c(S) - r(S)] + c_{\mathrm{FN}}[n - r(S)]$, with c and r computed by one fixed scorer.

Lemma 1. Consider the bipartite graph between records and targets with edges given by ρ_τ. The subsets of records that can be matched simultaneously form the independent sets of a transversal matroid (Edmonds and Fulkerson, 1965), and the maximum matching size of S is the rank of S in that matroid. Matroid rank functions are monotone, satisfy $r(S \cup \{z\}) - r(S) \in \{0, 1\}$, and are submodular. □

Proposition 1. Substituting the definition, $E_w(A) - E_w(S) = c_{\mathrm{FP}}[(c(A) - c(S)) - (r(A) - r(S))] - c_{\mathrm{FN}}[r(A) - r(S)] = c_{\mathrm{FP}}\Delta c - (c_{\mathrm{FP}} + c_{\mathrm{FN}})\Delta r$. Without an ignore region every record is scored, so $\Delta c = |A| - |S|$. □

Proposition 2. Deleting $A \setminus S$ lowers E_w iff $E_w(A) - E_w(S) > 0$, which by Proposition 1 is $c_{\mathrm{FP}}\Delta c > (c_{\mathrm{FP}} + c_{\mathrm{FN}})\Delta r$. If $\Delta c > 0$, divide by $(c_{\mathrm{FP}} + c_{\mathrm{FN}})\Delta c$. If $\Delta c = 0$, deletion lowers E_w iff $\Delta r < 0$, which cannot happen by monotonicity (Lemma 1), so deletion of no scored record never helps. Addition of B' to A is the same identity with the roles of $A \cup B'$ and A: it lowers E_w iff $(c_{\mathrm{FP}} + c_{\mathrm{FN}})[r(A \cup B') - r(A)] > c_{\mathrm{FP}}[c(A \cup B') - c(A)]$. For the population rule, the expected change is $c_{\mathrm{FP}}\mathbb{E}[\Delta c] - (c_{\mathrm{FP}} + c_{\mathrm{FN}})\mathbb{E}[\Delta r]$, whose sign is that of $c_{\mathrm{FP}}/(c_{\mathrm{FP}} + c_{\mathrm{FN}}) - p_1$ when $\mathbb{E}[\Delta c] > 0$. □

Corollary 1. Write $r = r(A)$, $c = c(A)$, so that $n + c(S) = n + c - \Delta c$. Then $\mathrm{F1}(S) - \mathrm{F1}(A) = \frac{2(r-\Delta r)}{n+c-\Delta c} - \frac{2r}{n+c} = \frac{2[(r-\Delta r)(n+c) - r(n+c-\Delta c)]}{(n+c)(n+c-\Delta c)} = \frac{2[r\,\Delta c - (n+c)\Delta r]}{(n+c)(n+c-\Delta c)}$. Both denominators are positive, so for $\Delta c > 0$ the sign is that of $r/(n+c) - \Delta r/\Delta c = \mathrm{F1}(A)/2 - \Delta r/\Delta c$. If $n = 0$, then $r = 0$ and $\mathrm{F1}(A) = 0$ whenever $c(A) > 0$; deleting every scored record gives $n = c = 0$ and $\mathrm{F1} = 1$ by convention. □

Proposition 3. The decomposition is a telescoping sum. Applying $E = c + n - 2r$ to S_0 and S_f gives the geometry term $[c(S_0) - c(S_f)] - 2[r(S_0) - r(S_f)]$. Convexity of the squared distance gives $\|\bar{z} - y\|^2 \le \frac{1}{m}\sum_i \|z_i - y\|^2$ for any target y, so averaging does not increase mean squared error to any fixed target. A threshold matching is not monotone in squared error, however. If one member is within tolerance of y and the other is far from every target, their mean can leave the tolerance ball of y and a match is lost; conversely, two members just outside the tolerance on opposite sides of y have a mean inside it, and a match is gained. Hence the geometry term can have either sign. □

Proposition 4. Both errors are evaluated against the same n for the same response, so their difference depends only on which field is read. By the triangle inequality, $\|d - \mathrm{lc}(Z)\|_1 \le \|d - n\|_1 + \|n - \mathrm{lc}(Z)\|_1 \le 2\max\{\|d - n\|_1, \|\mathrm{lc}(Z) - n\|_1\}$. For the second inequality, the reverse triangle inequality gives $\big||d_c - n_c| - |\mathrm{lc}(Z)_c - n_c|\big| \le |d_c - \mathrm{lc}(Z)_c|$ for every class c; summing over classes and using $|\sum_c x_c| \le \sum_c |x_c|$ gives the bound. □

Diminishing coverage. Let $U_K = Z_1 \cup \cdots \cup Z_K$ and $\delta_K = \mathbb{E}[r(U_K) - r(U_{K-1})]$. By submodularity, $r(U_K \cup Z_{K+1}) - r(U_K) \le r(U_{K-1} \cup Z_{K+1}) - r(U_{K-1})$. Taking expectations and using exchangeability of Z_K and Z_{K+1} given the image, the right side has expectation δ_K, so $\delta_{K+1} \le \delta_K$. □

Proposition 5. Under deletion-only confirmation, the records of A not confirmed by B are deleted. In expectation, $T(1 - h_+^B)$ true and $F(1 - h_-^B)$ false records are deleted. With no takeover, Δr is the number of deleted true records, and with all false records scored, Δc is the number of deleted records. By Proposition 1, $\mathbb{E}[E(A) - E(A \cap B)] = T(1 - h_+^B) + F(1 - h_-^B) - 2T(1 - h_+^B) = F(1 - h_-^B) - T(1 - h_+^B)$. The comparison follows from $G(B) - G(A') = F(h_-^{A'} - h_-^B) - T(h_+^{A'} - h_+^B)$. For the examples, $10(0.15) - 100(0.05) = -3.5$ and $10(1) - 100(0.5) = -40$. □

Proposition 6. Let M_A and M_B be maximum matchings of A and B to Y at tolerance θ. At least $r(A) + r(B) - n$ targets are matched in both. For each such target y, its partners $a \in A$ and $b \in B$ satisfy $\|a - b\| \le \|a - y\| + \|y - b\| \le 2\theta$, and distinct targets give disjoint pairs. Hence $m_{2\theta}(A, B) \ge r(A) + r(B) - n$. With no ignore region, $E(A) + E(B) = |A| + |B| + 2n - 2r(A) - 2r(B) \ge |A| + |B| - 2m_{2\theta}(A, B) = U_{2\theta}(A, B)$. The bound on the maximum follows. For boxes, if $\mathrm{IoU}(a, y) \ge t$

and $\mathrm{IoU}(b, y) \geq t$, then $1 - \mathrm{IoU}$ is a metric (the Jaccard distance), so $\mathrm{IoU}(a, b) \geq 2t - 1$, and the same argument applies. □

Proposition 7. Every rule $g(\Phi)$ takes values in $\mathcal{S}_A$, so pointwise $\ell(g(\Phi), Y) \geq \min_{S \in \mathcal{S}_A} \ell(S, Y)$; taking expectations and the infimum over g gives $R_{\mathrm{or}} \leq R^*_\Phi$. A fitted rule is one such g, so $R^*_\Phi \leq R_{\hat{g}}$. The decomposition is an identity. □

Proposition 8. Any $S \in \mathcal{S}_A$ contains at most one record per cluster, so a maximum matching of S to targets induces a cluster-target matching of size $r(S)$ in the stated graph; hence $r(S) \leq |M|$. Records unmatched in the scorer's matching and lying outside Ω add to c, so $c(S) \geq r(S)$ with or without ignore regions, and $E(S) = c(S) + n - 2r(S) \geq n - r(S) \geq n - |M|$. The construction that selects, for each cluster matched in M, a representative compatible with its target, and nothing else, has $r = c = |M|$ and attains the bound. □

Proposition 9. Let $\tilde{\pi}$ be optimal under $\tilde{\Lambda}$ and π^* optimal under Λ. Then $\mathcal{R}_\Lambda(\tilde{\pi}) - \mathcal{R}_\Lambda(\pi^*) = [\mathcal{R}_\Lambda(\tilde{\pi}) - \mathcal{R}_{\tilde{\Lambda}}(\tilde{\pi})] + [\mathcal{R}_{\tilde{\Lambda}}(\tilde{\pi}) - \mathcal{R}_{\tilde{\Lambda}}(\pi^*)] + [\mathcal{R}_{\tilde{\Lambda}}(\pi^*) - \mathcal{R}_\Lambda(\pi^*)] \leq \varepsilon + 0 + \varepsilon$. □

Proposition 10. The output is a representative. If no candidate hits, the output misses (event of F). Otherwise, if no representative hits, the output misses (event of L). Otherwise some representative hits, and the output misses exactly in the event of W. The three events are disjoint and their union is the event that the output misses.

Proposition 11. Every candidate of the enlarged menu misses iff every earlier candidate misses and the added one misses, so the probability is F $\Pr(\text{added misses} \mid \text{all earlier miss}) = F(1 - \rho)$ when $F > 0$, and 0 when $F = 0$. Under conditional independence given q, $\Pr(\text{all miss} \mid q) = \prod_k (1 - q_k)$; taking expectations gives F.

Proposition 12. The change in the miss indicator is $\nVdash_T (\nVdash[b \text{ misses}] - \nVdash[g \text{ misses}])$, which is $+1$ when b misses and g hits, -1 when b hits and g misses, and 0 otherwise. Its expectation is $\Pr(T)[\Pr(b \text{ misses}, g \text{ hits} \mid T) - \Pr(b \text{ hits}, g \text{ misses} \mid T)] = \Pr(T)[e_T r - (1 - e_T)h]$. If $\nVdash_T$ and $\nVdash[b \text{ misses}]$ are independent given U and $T \subseteq U$, then $\Pr(b \text{ misses} \mid T) = \Pr(b \text{ misses} \mid U, T) = \Pr(b \text{ misses} \mid U)$. The judge sample size follows from Hoeffding's inequality for each judge's empirical miss rate with a union bound over the M judges.

Proposition 13. For any acceptance rule $a(H) \in [0, 1]$, the acceptance probability in world j is $\int a \, dP_j$. If $P_0 = P_1$ the two are equal, so acceptance at least $1 - \beta$ in world 0 forces acceptance at least $1 - \beta > \alpha$ in world 1 when $\alpha + \beta < 1$. In general, $|\int a \, dP_0 - \int a \, dP_1| \leq \mathrm{TV}(P_0, P_1)$ for $0 \leq a \leq 1$. For the hidden-target construction, couple the two worlds so that the output histories coincide unless the added target is reported in at least one of the K calls, an event of probability at most $K\vartheta$ by the union bound, since the per-call bound ϑ holds given the earlier history; non-interference means that, conditional on the target not being reported, the rest of the output has the same law in both worlds. The coupling inequality gives $\mathrm{TV}(P_0, P_1) \leq K\vartheta$. □

Corollary 2. By assumption there is an admissible world in which the report over-counts, $|S| > n_0$. Add $m > |S| - n_0$ targets that no interface reports and that do not change the rest of the output ($\vartheta = 0$); by closure the result is admissible. The output law is unchanged and $|S| < n_0 + m$, so the sign of the count gap differs between two worlds with the same output law. □

B Full Tables

All tables in this appendix are written by `paper/make_tables.py` from the outputs of `stageB/analysis/run_all.sh`, which recomputes them from saved responses and labels without model calls. The same numbers are available in machine-readable form in the JSON files of `stageB/analysis/out/` and in `stageB/analysis/out/main_tables.md`. The click and configuration tables (Tables 12, 15 and 20) are written by `stageB/analysis/accfirst/make_accuracy_tables.py`.

Table 17: Same-response readout across datasets (count-first prompts, lowest setting, one call; data of Figure 4). List count without and with the declared alias map; paired image bootstrap (a separate run from Table 7, so shared rows can differ in the last digit of an interval).

Dataset	Objects	Error	Model	n	Stated	List count (unmapped)	List count (alias map)
PixMo-Count	6.0	total	GPT-6.1	100	0.16	0.15, −6.3% [−23.1, 0.0]	same
			Sonnet 5.5	100	0.17	0.14, −17.6% [−38.5, 0.0]	same
			Gemini 3.8	100	0.33	0.27, −18.2% [−37.5, 0.0]	same
FSC-147	43.8	total	GPT-6.1	100	2.73	2.70, −1.1% [−8.5, 8.8]	same
			Sonnet 5.5	100	9.24	8.85, −4.2% [−11.9, 2.1]	same
			Gemini 3.8	100	6.87	3.93, −42.8% [−65.7, −17.4]	same
SCP	47.6	per type	GPT-6.1	100	0.59	0.33, −44.1% [−60.8, −18.6]	0.33, −44.1% [−61.1, −19.0]
			Sonnet 5.5	100	4.52	0.07, −98.5% [−99.6, −96.9]	0.07, −98.5% [−99.6, −96.9]
			Gemini 3.8	100	5.92	17.25, +191.4% [145.8, 239.3]	17.25, +191.4% [146.5, 234.5]
SCP-dense	164.3	per type	GPT-6.1	50	10.84	2.64, −75.6% [−83.0, −65.9]	2.64, −75.6% [−83.0, −65.7]
			Sonnet 5.5	50	35.02	7.80, −77.7% [−91.5, −58.4]	5.42, −84.5% [−97.6, −65.9]
			Gemini 3.8	50	45.68	129.04, +182.5% [163.4, 202.0]	129.04, +182.5% [163.2, 202.4]

Table 18: Selection and geometry terms for every base/partner pair (per image, reduction of E; positive is better). Δc: deleted scored records; Δr: matches lost.

Source	Base	Partner	Δc	Δr	p_1	Selection	Geometry	Total
DocLayNet	GPT-6.1	same model	0.92	0.48	0.527	−0.05	−0.30	−0.35
DocLayNet	GPT-6.1	Sonnet 5.5	2.63	0.92	0.348	0.80	−1.10	−0.30
DocLayNet	GPT-6.1	Gemini 3.8	8.08	5.15	0.637	−2.22	0.43	−1.78
DocLayNet	Sonnet 5.5	same model	1.33	0.32	0.237	0.70	0.10	0.80
DocLayNet	Sonnet 5.5	GPT-6.1	3.03	0.45	0.148	2.13	2.77	4.90
DocLayNet	Sonnet 5.5	Gemini 3.8	8.30	3.97	0.478	0.37	3.07	3.43
DocLayNet	Gemini 3.8	same model	3.48	2.03	0.584	−0.58	0.30	−0.28
DocLayNet	Gemini 3.8	GPT-6.1	2.12	0.55	0.260	1.02	0.47	1.48
DocLayNet	Gemini 3.8	Sonnet 5.5	1.93	0.22	0.112	1.50	0.00	1.50
DOTA	GPT-6.1	same model	7.07	0.43	0.061	6.20	0.50	6.70
DOTA	GPT-6.1	Sonnet 5.5	29.33	6.53	0.223	16.27	−0.65	15.62
DOTA	GPT-6.1	Gemini 3.8	27.08	5.72	0.211	15.65	−1.50	14.15
DOTA	Sonnet 5.5	same model	6.15	0.87	0.141	4.42	0.05	4.47
DOTA	Sonnet 5.5	GPT-6.1	6.20	0.32	0.051	5.57	2.92	8.48
DOTA	Sonnet 5.5	Gemini 3.8	10.30	2.28	0.222	5.73	1.25	6.98
DOTA	Gemini 3.8	same model	11.25	1.87	0.166	7.52	0.82	8.33
DOTA	Gemini 3.8	GPT-6.1	7.60	0.68	0.090	6.23	3.90	10.13
DOTA	Gemini 3.8	Sonnet 5.5	13.93	2.50	0.179	8.93	1.17	10.10
FSC-147	GPT-6.1	same model	2.78	1.77	0.635	−0.75	0.27	−0.48
FSC-147	GPT-6.1	Sonnet 5.5	22.60	20.05	0.887	−17.50	0.00	−17.50
FSC-147	GPT-6.1	Gemini 3.8	27.28	24.52	0.899	−21.75	0.10	−21.65
FSC-147	Sonnet 5.5	same model	7.35	4.82	0.655	−2.28	0.10	−2.18
FSC-147	Sonnet 5.5	GPT-6.1	4.53	2.10	0.463	0.33	0.37	0.70
FSC-147	Sonnet 5.5	Gemini 3.8	11.23	8.30	0.739	−5.37	0.03	−5.33
FSC-147	Gemini 3.8	same model	3.53	3.12	0.882	−2.70	0.07	−2.63
FSC-147	Gemini 3.8	GPT-6.1	0.82	0.32	0.388	0.18	0.10	0.28
FSC-147	Gemini 3.8	Sonnet 5.5	2.83	2.28	0.806	−1.73	0.13	−1.60
SKU-110K	GPT-6.1	same model	33.92	0.52	0.015	32.88	5.83	38.72
SKU-110K	GPT-6.1	Sonnet 5.5	153.82	43.05	0.280	67.72	−14.07	53.65
SKU-110K	GPT-6.1	Gemini 3.8	113.82	22.83	0.201	68.15	1.17	69.32
SKU-110K	Sonnet 5.5	same model	61.95	2.95	0.048	56.05	0.87	56.92
SKU-110K	Sonnet 5.5	GPT-6.1	59.23	0.23	0.004	58.77	13.83	72.60
SKU-110K	Sonnet 5.5	Gemini 3.8	73.15	5.57	0.076	62.02	9.47	71.48
SKU-110K	Gemini 3.8	same model	36.88	13.02	0.353	10.85	1.43	12.28
SKU-110K	Gemini 3.8	GPT-6.1	13.07	0.40	0.031	12.27	9.67	21.93
SKU-110K	Gemini 3.8	Sonnet 5.5	66.98	25.87	0.386	15.25	−10.10	5.15
GroundCUA	GPT-6.1	same model	5.85	1.87	0.319	2.12	−0.67	1.45
GroundCUA	GPT-6.1	Sonnet 5.5	34.58	4.00	0.116	26.58	13.03	39.62
GroundCUA	GPT-6.1	Gemini 3.8	43.42	17.02	0.392	9.38	−6.30	3.08
GroundCUA	Sonnet 5.5	same model	5.17	3.02	0.584	−0.87	1.20	0.33
GroundCUA	Sonnet 5.5	GPT-6.1	13.82	8.07	0.584	−2.32	−6.50	−8.82

Source	Base	Partner	Δc	Δr	p_1	Selection	Geometry	Total
GroundCUA	Sonnet 5.5	Gemini 3.8	38.90	27.83	0.716	−16.77	−7.73	−24.50
GroundCUA	Gemini 3.8	same model	30.98	6.92	0.223	17.15	−0.93	16.22
GroundCUA	Gemini 3.8	GPT-6.1	26.32	1.83	0.070	22.65	9.60	32.25
GroundCUA	Gemini 3.8	Sonnet 5.5	42.57	2.45	0.058	37.67	15.43	53.10
ScreenParse	GPT-6.1	same model	0.38	0.08	0.217	0.22	0.07	0.28
ScreenParse	GPT-6.1	Sonnet 5.5	4.90	1.40	0.286	2.10	0.93	3.03
ScreenParse	GPT-6.1	Gemini 3.8	4.12	1.25	0.304	1.62	0.03	1.65
ScreenParse	Sonnet 5.5	same model	0.80	0.30	0.375	0.20	0.00	0.20
ScreenParse	Sonnet 5.5	GPT-6.1	1.80	0.63	0.352	0.53	−0.47	0.07
ScreenParse	Sonnet 5.5	Gemini 3.8	2.83	1.42	0.500	0.00	−0.37	−0.37
ScreenParse	Gemini 3.8	same model	2.17	0.98	0.454	0.20	0.00	0.20
ScreenParse	Gemini 3.8	GPT-6.1	1.57	0.30	0.192	0.97	0.47	1.43
ScreenParse	Gemini 3.8	Sonnet 5.5	3.38	0.93	0.276	1.52	0.87	2.38

Table 19: Leave-one-source-out transfer, every cell. Construction names: unan3 / maj3 / union3 = three same-model calls kept by unanimity / majority / union; x2_inter, x2_union = two-model intersection or union with the named partner; x3_maj = three-model majority; _fus = fused coordinates, _del = deletion only. Interval: paired 95% bootstrap of the change. Cost per image counts every call.

Pool	Loss	Source	Model	Default applied	Single → default	95% CI	Recall	Cost/img
same	E	DocLayNet	GPT-6.1	unan3_fus	8.57 → 9.15	[−0.35, 1.65]	0.677	$0.0323
same	E	DocLayNet	Sonnet 5.5	unan3_fus	13.77 → 12.33	[−2.25, −0.70]	0.571	$0.0384
same	E	DocLayNet	Gemini 3.8	unan3_fus	11.83 → 12.18	[−0.63, 1.42]	0.230	$0.0132
same	E	DOTA	GPT-6.1	unan3_fus	45.00 → 34.98	[−16.73, −4.97]	0.464	$0.0785
same	E	DOTA	Sonnet 5.5	unan3_fus	37.87 → 31.95	[−8.70, −3.77]	0.321	$0.0472
same	E	DOTA	Gemini 3.8	unan3_fus	40.98 → 30.80	[−16.38, −5.38]	0.258	$0.0178
same	E	FSC-147	GPT-6.1	unan3_fus	30.12 → 30.90	[−0.10, 1.78]	0.637	$0.0334
same	E	FSC-147	Sonnet 5.5	unan3_fus	48.32 → 51.73	[2.07, 4.83]	0.292	$0.0187
same	E	FSC-147	Gemini 3.8	unan3_fus	52.05 → 54.63	[0.42, 5.42]	0.189	$0.0097
same	E	SKU-110K	GPT-6.1	unan3_fus	198.35 → 148.32	[−58.67, −41.80]	0.529	$0.2121
same	E	SKU-110K	Sonnet 5.5	unan3_fus	217.30 → 148.67	[−80.43, −57.82]	0.084	$0.1258
same	E	SKU-110K	Gemini 3.8	unan3_fus	150.97 → 136.48	[−23.90, −6.18]	0.192	$0.0584
same	E	GroundCUA	GPT-6.1	unan3_fus	92.58 → 89.25	[−7.40, 0.50]	0.492	$0.0920
same	E	GroundCUA	Sonnet 5.5	unan3_fus	44.15 → 43.67	[−3.13, 2.43]	0.680	$0.0999
same	E	GroundCUA	Gemini 3.8	unan3_fus	121.75 → 95.58	[−34.90, −17.95]	0.144	$0.0438
same	E	ScreenParse	GPT-6.1	unan3_fus	16.23 → 15.78	[−0.83, −0.10]	0.597	$0.0292
same	E	ScreenParse	Sonnet 5.5	unan3_fus	13.27 → 12.97	[−0.75, 0.17]	0.597	$0.0318
same	E	ScreenParse	Gemini 3.8	unan3_fus	16.02 → 15.53	[−1.28, 0.20]	0.477	$0.0109
same	F1	DocLayNet	GPT-6.1	union3	0.710 → 0.686	[−0.049, −0.003]	0.708	$0.0323
same	F1	DocLayNet	Sonnet 5.5	union3	0.573 → 0.569	[−0.021, 0.013]	0.629	$0.0384
same	F1	DocLayNet	Gemini 3.8	union3	0.384 → 0.459	[0.028, 0.129]	0.484	$0.0132
same	F1	DOTA	GPT-6.1	union3	0.395 → 0.389	[−0.024, 0.012]	0.509	$0.0785
same	F1	DOTA	Sonnet 5.5	union3	0.334 → 0.329	[−0.022, 0.014]	0.376	$0.0472
same	F1	DOTA	Gemini 3.8	union3	0.312 → 0.319	[−0.025, 0.044]	0.361	$0.0178
same	F1	FSC-147	GPT-6.1	maj3_fus	0.707 → 0.695	[−0.044, 0.008]	0.670	$0.0334
same	F1	FSC-147	Sonnet 5.5	maj3_fus	0.452 → 0.425	[−0.046, −0.008]	0.355	$0.0187
same	F1	FSC-147	Gemini 3.8	maj3_fus	0.273 → 0.277	[−0.034, 0.042]	0.245	$0.0097
same	F1	SKU-110K	GPT-6.1	union3	0.437 → 0.402	[−0.046, −0.024]	0.541	$0.2121
same	F1	SKU-110K	Sonnet 5.5	union3	0.128 → 0.130	[−0.007, 0.011]	0.157	$0.1258
same	F1	SKU-110K	Gemini 3.8	union3	0.354 → 0.402	[0.022, 0.080]	0.390	$0.0584
same	F1	GroundCUA	GPT-6.1	union3	0.483 → 0.462	[−0.032, −0.010]	0.527	$0.0920
same	F1	GroundCUA	Sonnet 5.5	union3	0.717 → 0.730	[−0.002, 0.029]	0.745	$0.0999
same	F1	GroundCUA	Gemini 3.8	union3	0.256 → 0.286	[−0.004, 0.067]	0.374	$0.0438
same	F1	ScreenParse	GPT-6.1	union3	0.564 → 0.566	[−0.004, 0.009]	0.618	$0.0292
same	F1	ScreenParse	Sonnet 5.5	union3	0.615 → 0.615	[−0.009, 0.010]	0.629	$0.0318
same	F1	ScreenParse	Gemini 3.8	union3	0.536 → 0.547	[−0.007, 0.031]	0.588	$0.0109
any	E	DocLayNet	GPT-6.1	x2_inter_fus w/ Sonnet	8.57 → 8.87	[−1.23, 1.77]	0.651	$0.0236
any	E	DocLayNet	Sonnet 5.5	x2_inter_fus w/ GPT	13.77 → 8.87	[−6.63, −3.37]	0.651	$0.0236
any	E	DocLayNet	Gemini 3.8	x2_inter_fus w/ GPT	11.83 → 10.35	[−2.67, −0.35]	0.357	$0.0151
any	E	DOTA	GPT-6.1	x3_maj_fus	45.00 → 31.43	[−21.90, −7.27]	0.410	$0.0481
any	E	DOTA	Sonnet 5.5	x3_maj_fus	37.87 → 31.47	[−10.40, −3.15]	0.409	$0.0481
any	E	DOTA	Gemini 3.8	x3_maj_fus	40.98 → 31.85	[−13.83, −5.02]	0.401	$0.0481
any	E	FSC-147	GPT-6.1	x2_inter_fus w/ Sonnet	30.12 → 47.62	[11.28, 24.48]	0.356	$0.0175

Pool	Loss	Source	Model	Default applied	Single → default	95% CI	Recall	Cost/img
any	E	FSC-147	Sonnet 5.5	x2_inter_fus w/ GPT	48.32 → 47.62	[−2.28, 0.75]	0.356	$0.0175
any	E	FSC-147	Gemini 3.8	x2_inter_fus w/ GPT	52.05 → 51.77	[−0.80, 0.08]	0.234	$0.0142
any	E	SKU-110K	GPT-6.1	x3_maj_fus	198.35 → 138.55	[−72.30, −47.70]	0.382	$0.1353
any	E	SKU-110K	Sonnet 5.5	x3_maj_fus	217.30 → 139.43	[−92.02, −64.67]	0.379	$0.1353
any	E	SKU-110K	Gemini 3.8	x3_maj_fus	150.97 → 139.13	[−17.20, −6.78]	0.379	$0.1353
any	E	GroundCUA	GPT-6.1	x3_maj_fus	92.58 → 68.75	[−29.70, −18.27]	0.557	$0.0787
any	E	GroundCUA	Sonnet 5.5	x3_maj_fus	44.15 → 67.02	[17.92, 27.90]	0.569	$0.0787
any	E	GroundCUA	Gemini 3.8	x3_maj_fus	121.75 → 71.03	[−59.05, −42.70]	0.543	$0.0787
any	E	ScreenParse	GPT-6.1	x3_maj_fus	16.23 → 14.02	[−3.18, −1.28]	0.613	$0.0241
any	E	ScreenParse	Sonnet 5.5	x3_maj_fus	13.27 → 13.82	[−0.58, 1.70]	0.617	$0.0241
any	E	ScreenParse	Gemini 3.8	x3_maj_fus	16.02 → 14.02	[−2.85, −1.28]	0.613	$0.0241
any	F1	DocLayNet	GPT-6.1	x2_union w/ Sonnet	0.710 → 0.660	[−0.101, 0.001]	0.728	$0.0236
any	F1	DocLayNet	Sonnet 5.5	x2_union w/ GPT	0.573 → 0.660	[0.056, 0.120]	0.728	$0.0236
any	F1	DocLayNet	Gemini 3.8	x2_union w/ GPT	0.384 → 0.674	[0.215, 0.368]	0.736	$0.0151
any	F1	DOTA	GPT-6.1	x2_union w/ Sonnet	0.395 → 0.369	[−0.052, 0.003]	0.491	$0.0419
any	F1	DOTA	Sonnet 5.5	x2_union w/ GPT	0.334 → 0.369	[−0.025, 0.095]	0.491	$0.0419
any	F1	DOTA	Gemini 3.8	x2_union w/ GPT	0.312 → 0.366	[0.001, 0.107]	0.488	$0.0324
any	F1	FSC-147	GPT-6.1	x3_maj_fus	0.707 → 0.458	[−0.328, −0.173]	0.390	$0.0207
any	F1	FSC-147	Sonnet 5.5	x3_maj_fus	0.452 → 0.457	[−0.023, 0.034]	0.388	$0.0207
any	F1	FSC-147	Gemini 3.8	x3_maj_fus	0.273 → 0.458	[0.121, 0.252]	0.390	$0.0207
any	F1	SKU-110K	GPT-6.1	x2_union w/ Sonnet	0.437 → 0.339	[−0.116, −0.079]	0.461	$0.1155
any	F1	SKU-110K	Sonnet 5.5	x2_union w/ GPT	0.128 → 0.339	[0.181, 0.244]	0.461	$0.1155
any	F1	SKU-110K	Gemini 3.8	x2_union w/ GPT	0.354 → 0.429	[0.035, 0.121]	0.518	$0.0898
any	F1	GroundCUA	GPT-6.1	x2_union w/ Sonnet	0.483 → 0.597	[0.091, 0.137]	0.701	$0.0640
any	F1	GroundCUA	Sonnet 5.5	x2_union w/ GPT	0.717 → 0.597	[−0.143, −0.098]	0.701	$0.0640
any	F1	GroundCUA	Gemini 3.8	x2_union w/ GPT	0.256 → 0.412	[0.112, 0.205]	0.503	$0.0454
any	F1	ScreenParse	GPT-6.1	x3_maj_fus	0.564 → 0.596	[0.011, 0.055]	0.613	$0.0241
any	F1	ScreenParse	Sonnet 5.5	x3_maj_fus	0.615 → 0.601	[−0.041, 0.013]	0.617	$0.0241
any	F1	ScreenParse	Gemini 3.8	x3_maj_fus	0.536 → 0.596	[0.040, 0.084]	0.613	$0.0241

Table 20: The four fixed single-call configurations on the four detection sources (first repetition; mean over sources of per-source mean F1 at the primary scoring, and mean cost per image at list prices). low: lowest setting; def.: default setting. The shipped F1 configuration per model is the highest F1 of its column (bold); for Gemini 3.8 the two default-setting configurations differ by 0.2 points.

	GPT-6.1		Sonnet 5.5		Gemini 3.8	
Configuration	F1	Cost	F1	Cost	F1	Cost
list, low	56.2	$0.0295	37.1	$0.0201	33.1	$0.0084
regions, low	**58.6**	$0.0327	31.6	$0.0291	39.4	$0.0101
list, def.	54.0	$0.0436	44.7	$0.0237	**45.0**	$0.0192
regions, def.	56.7	$0.0475	**48.1**	$0.0477	44.8	$0.0200

C Experiment Registry, Costs and Settings

Registry. Table 21 lists every experiment with model calls. Counts include smoke tests, pilots, repairs and retries; costs are list prices for every attempted call. The open model was served through a routing provider and is priced at 0 here (0.82M input and 1.75M output tokens in the open-model experiment). The registry covers the models evaluated here; calls to an earlier generation of models, used only for background, are not part of it.

Table 21: All experiments with model calls (67,364 calls, about $758).

Experiment	Data and arms	Models	Calls	Cost
Pilot v1	12 procedural scenes, 768 px; responsibility-split arms; cap 8,192	G5.6, G6.1, S5.5, O5.5, Q	3,540	$29.30
Pilot v2	24 scenes, 1024 px, 16 to 384 targets; dose response	same	4,080	$47.50

Experiment	Data and arms	Models	Calls	Cost
Subset execution	12 scenes, 96 or 192 targets; representation and drift arms	G5.6, G6.1, O5.5, S5.5	1,924	$53.84
Coverage and grids	48 scenes; coverage responsibility, grid on/off, granularity; repairs, smoke, supplement	above + Gem	2,752	$113.80
Detection (E3)	SKU-110K, DOTA, DocLayNet, FSC-147, 60 images each; list / regions × lowest / default; 16k control	G6.1, S5.5, Gem	6,523	$171.13
GUI	ScreenSpot-Pro 200, ScreenSpot-v2 60, GroundCUA 60, ScreenParse 60; click and parse arms; pilots 1,050	same	7,232	$67.77
Same-view review	SKU, DOTA, FSC, 20 images each; draft review (same view, crop)	same	663	$7.40
Readout	SCP 100, SCP-dense 50, PixMo 100; 13 prompt arms	same	7,900	$88.48
Tiling	FSC-147 100 (three size strata); count-first, list-only, 2 × 2 tiles	same	1,800	$8.02
Review	GroundCUA 30 + SKU 30, four crops each; blind vs. draft (2 deleted, 2 injected, 1 moved)	same	1,440	$15.41
Comparator	80 disagreeing clicks; SKU and DOTA crops, 150 per model; both orders	same	1,960	$4.76
Resolution	ScreenSpot-Pro 100; down / native / up	same	600	$1.52
End-to-end	SCP-dense 50, FSC 100, clicks 200; naive / common / proposed	same	1,800	$22.65
Reasoning	SCP-dense 30, FSC 30, GroundCUA 20; default setting	same	1,110	$26.37
Open model	SCP 100, SCP-dense 50; readout arms	Q	650	not priced
Grid	SCP 100, SCP-dense 40; grid, ticks, cells, scan, type names; 100 default calls	G6.1, S5.5, Gem	1,340	$15.40
Pilots	small runs before the readout experiments	same	647	$5.54
Readout confirmation	SCP-dense 50; full list-first protocol, lowest setting, 16k cap	same	150	$3.87
Gemini, default 16k	same images; default setting, 16k cap	Gem	50	$1.71
Gemini, default 32k	same images; default setting, 32k cap	Gem	50	$1.83
Multi-class readout	DOTA 30 windows, DocLayNet 50 pages; count-first and list-first, lowest setting; Gemini list-first at default, 16k cap; 14 smoke calls	G6.1, S5.5, Gem	574	$6.25
Zoom on second click	the 260 click items of GUI; two-stage zoom around each model's second click, the same request as the first-click zoom	G6.1, S5.5, Gem	772	$1.60
Click scores	the 260 click items of GUI; each model's zoomed point scored by each model (probability of a hit)	G6.1, S5.5, Gem	2,325	$3.91

Experiment	Data and arms	Models	Calls	Cost
Click choice	the 260 click items; default-setting zooms of GPT-6.1 and Sonnet 5.5, re-grounding and second-opinion arms, scores of default-setting points, pairwise comparisons of cluster representatives, crop-first baselines	same	3,639	$13.56
Click confirmation	600 new ScreenSpot-Pro items; all arms of Table 12; 13,120 requests and 6 retries	same	13,126	$44.93
Fourth click source	the 600 confirmation items; judge scores of the crop-first point and the comparisons the four-source tier needs that were not already asked	Gem	717	$1.44
Total			67,364	$757.99

G5.6, G6.1: GPT-5.6, GPT-6.1. S5.5: Sonnet 5.5. O5.5: Opus 5.5. Gem: Gemini 3.8 Flash. Q: Qwen3-VL-235B (instruct). The open-model calls have no logged per-token price, so the total cost excludes them.

Settings audit. Table 22 records what each setting label meant in practice. The lowest setting used an output cap of 16,384 tokens and the default setting 32,768, except where a 16k default was run to separate the two (Table 9, a 90-call control on the detection sources, and a control on controlled scenes).

Table 22: Reasoning settings as sent and as observed (share of calls with reasoning tokens).

Model	Lowest setting as sent	Observed
GPT-6.1	`reasoning_effort=low`	reasoning in most calls (SCP 100%, PixMo 61%, clicks 39%); default: 100%
GPT-5.6	effort none	no reasoning
Sonnet 5.5	`thinking: between_tools`	lowest: 0%; default (adaptive thinking): 0% to 23% on the dense, FSC and screen runs, 4 of 48 calls on controlled scenes, 0% to 90% on E3 by cell; pilots: not logged
Opus 5.5	effort low (adaptive thinking cannot be disabled)	under 1% of logged calls at effort low; pilots: not logged
Gemini 3.8	`thinking_budget=0`	little on list and box tasks (FSC list 6%, regions 12%); still reasons on clicks and points (GUI clicks about 99%, PixMo 23%, FSC points 32%); default: 51% to 100%
Qwen3-VL	instruct model, no reasoning parameter	no reasoning

D Protocol Details

List-first prompt. The readout confirmation runs used the following system prompt for GPT-6.1 and Gemini 3.8 (native 0 to 1000 frame); the user message was "List every geometric shape in this image with its center, then counts by type. Respond with JSON only."

```
You are a precise visual counting and localization assistant.

Given an image containing geometric shapes, identify every shape.
Report coordinates on a 0-1000 grid: x = 0 is the left edge and x = 1000 the
right edge of the image, y = 0 is the top edge and y = 1000 the bottom edge.
Use integers.
List the shapes in reading order: scan the image row by row from top to bottom,
and within a row from left to right.
The "type" value must be exactly one of: circle, triangle, square, star,
pentagon. A square rotated by any angle is still a "square"; do not use any
other type name.
```

```
Respond with ONLY a JSON object (no markdown, no commentary) in this format:
{"shapes": [{"type": "circle", "x": 402, "y": 311}, ...], "counts_by_type":
{"circle": 5, "triangle": 3, "square": 0, "star": 2, "pentagon": 4}}

Shape types to recognize: circle, triangle, square, star, pentagon.
Locate every shape carefully. Count systematically; do not miss any and do not
invent shapes that are not there.
Write the full shapes list first; after it, give counts_by_type for the shapes
you listed.
```

For Sonnet 5.5 the coordinate paragraph reads "The image is 1024x1024 pixels. Report coordinates as integer pixel positions in this image, where (0,0) is the top-left corner." The count-first prompts of the readout experiment ask for `counts_by_type` before the list; all prompts are released with the responses.

Decoder. The decoder strips a code fence and parses the whole response as JSON (L2). If that fails, it recovers each balanced `{...}` object that has the record fields (declared salvage) and reads `counts_by_type` if present, with its class names mapped by the same alias map (the pure-readout comparisons of Table 7 compare stated and list counts with neither mapped). Coordinates are mapped from the model's frame to image pixels. Classes are lower-cased and mapped by the declared alias map; for SCP, {diamond, rhombus, rotated square} $\mapsto$ square. Records whose class is outside the vocabulary after mapping stay in the audit collection Z^{all}, are excluded from the delivered collection Z that set scores evaluate, and are counted by the first degeneracy flag; Table 9 adds a sensitivity that keeps them and scores them as unmatched. On the detection sources, records with a label outside the vocabulary or a box outside the image are likewise kept in the audit collection, excluded from delivery and counted as invalid (Table 2).

Degeneracy flags. A list is flagged if any of the following holds: more than 10% of records lack a valid class; the list length is outside $[0.5, 1.5]$ times the stated total; more than 10% of records repeat a rounded coordinate pair already in the list; or the finish reason is "length". The same thresholds are used for every model, including the open model.

Scoring and clustering. Scoring uses the compatibility relation of each task in Table 2: for example, points of equal class within 30 pixels on SCP and 20 pixels on SCP-dense, detection boxes of equal class at IoU ≥ 0.75, screen elements at IoU ≥ 0.5 without class, and clicks inside the target box. Maximum matchings are computed with a linear assignment on the 0/1 compatibility matrix (Kuhn, 1955). Clusters across calls use the association rule of Table 2 (boxes: equal class and IoU ≥ 0.5, also on the detection sources where scoring uses 0.75), with a maximum-cardinality, then maximum-overlap assignment against cluster representatives, and the fused record is the coordinate mean of members.

Constructions. Same-model: `inter2` (two calls, intersection), `union2`, `unan3` and `maj3` (three calls, unanimity or majority), `union3`; each with fused (`_fus`) or base-record (`_del`) coordinates. Cross-model: `x2_inter` and `x2_union` with a named partner, `x3_unan` and `x3_maj` over the three models (one call each), `x3_union`. The leave-one-source-out default for an evaluated source is the construction with the best mean relative change over the other five sources, computed separately for each loss and pool.

Alarms. For clicks, the same-model alarm compares two unzoomed calls of one model; the cross-model alarm compares one unzoomed call of the user's model with one of the partner; after zoom, the same-model alarm compares the model's zoomed first click with its zoomed second click, and the cross-model alarm compares the zoomed first clicks of the two models. An item is deferred when the two points are farther apart than δ times the longer image side. The partner is set by a fixed rule, the cheapest other model per click call (GPT-6.1 and Sonnet 5.5: Gemini 3.8; Gemini 3.8: Sonnet 5.5).

Clicks from several models. Candidates are the two-stage zoom points of Step 4, mapped to the original screenshot. A candidate's identifier is the first 16 hexadecimal digits of the SHA-256 of the item, model, setting, parent call and coordinates. The judge's score request shows a native crop of one fifth of the

screenshot's width and height around the candidate, with a marker at the point, the instruction and the platform, and asks for a verdict and the probability that a click at the marker hits the intended element; it never names the model that produced the candidate. A missing or unreadable score counts as -1 for ranking. The comparison request shows the downscaled screenshot with both representatives marked by their identifiers and a native crop around each, asks which candidate satisfies the instruction (element, text, state, relation and whether the exact point lies inside the element), and allows "undecided"; it forbids proposing a new point. Each pair is sent in both orders, and the code asserts that the points drawn equal the representatives' coordinates. Ranking keys, ties and failures follow Algorithm 2: a failed first click leaves that model without a candidate, a failed zoom keeps the first click, a failed comparison gives no win.

Confirmation items and baselines. The 600 confirmation items were sampled from the ScreenSpot-Pro annotations not among the 200 design items, in proportion to each application file and, within it, to each pair of group and element type, with a fixed seed; the sampler asserts that no item and no screenshot is shared with the design items. The release records the SHA-256 of the protocol, the item list and every piece of code used to build requests and to evaluate; the hashes were checked before and after the run. A first batch of 100 seeded items was run through every arm to check costs and transport only. The three-call baseline runs three independent default-setting zooms of GPT-6.1, clusters the three points with the same δ, and returns the earliest point of the largest cluster (ties: the cluster with the earliest point). The crop-first baseline uses GPT-6.1 at its default setting: the first call must return a region of the downscaled screenshot, the second sees that region at native resolution with 8% padding and may return a smaller region or a click, and a third call, if needed, must click; an unreadable reply falls back to the centre of the last region. It follows the coarse-to-fine cropping of GUI-Lens (Fu et al., 2026) without its OCR and detector anchors or verification step, with our own prompts. Paired intervals resample screenshots (4,000 draws, seed 20261014); McNemar tests are exact.

Configuration transfer on the detection sources. The menu contains the four single-call configurations run on every image of the four detection sources (first repetition): list or regions interface, each at the lowest and the default setting. In the regions interface the model returns one row per region of a fixed coordinate registry, with a status of empty, found or unknown, and reports each object in the region that contains its centre; in the list interface it returns one list for the whole image in registry scan order. For each model and evaluated source, F1 picks the configuration with the highest mean of per-source mean F1 on the other three sources, and set error the one with the lowest mean relative change against the list interface at the lowest setting; ties keep the earlier configuration in the order list-lowest, regions-lowest, list-default, regions-default. Intervals are paired bootstrap intervals over the images of the evaluated source.

Reproducing the tables. The code and all requests and responses are at `https://github.com/shijunzhe/vlm-set-reporting`, together with the labels whose licences allow redistribution (DocLayNet, ScreenSpot-Pro, ScreenSpot-v2, GroundCUA, ScreenParse, PixMo-Count). The labels of DOTA v1.0, SKU-110K and FSC-147 are not redistributed: the repository stores a placeholder for each value, naming the official file and the window, scale and rounding applied, and `thirdparty/tp.py` rebuilds the files from the official downloads, writing a file only when its SHA-256 digest matches the file we scored against. `bash reproduce.sh --sources <dir>` performs this rebuild, sets the paths, runs `stageB/analysis/run_all.sh`, which recomputes the data of every table and figure in Section 7, and writes the LaTeX tables and figures; without `--sources` it recomputes the analyses that do not score against those three label sets and keeps the shipped outputs of the others. No step calls a model. The synthetic dense images are regenerated with `python3 -I stageB/code/scp_generate.py --num_images 50 --min_shapes 120 --max_shapes 200 --min_radius 9 --max_radius 16 --seed 2026`, which reproduces all 50 images byte for byte.

E Headline Claims and Their Evidence

Table 23 traces every number used in the abstract and introduction to the operation, comparator, metric and table that support it.

Table 23: Headline claims.

#	Claim	Operation	Comparator	Metric	Evidence
H1	On SCP-dense image 28, GPT-6.1 lists 173 records that match all 173 shapes and states counts that sum to 188. Exact lists: GPT-6.1 20 of 50 scenes, with wrong stated counts in 19; Sonnet 5.5 39 of 50, wrong in all 39. List length misses the true total by 0.04 and 0.02; stated totals by 7.78 and 10.48	none (same response)	stated counts, same response	exact list (every record matched, every shape covered); total-count error; SCP-dense 50, lowest setting	Fig. 1, Table 7
H2	Reading the list cuts per-type count error by 71% to 84%	list count	stated per-type counts, same response	per-type error; GPT -75.6% / -84.2%, Sonnet -77.7% / -70.7% (two protocols)	Table 7
H3	Complete reporting and decoding protocol -85.2% (GPT-6.1), -98.8% (Sonnet 5.5); Gemini 3.8 (lowest) $+15.6\%$ $[-5.3, +39.4]$, not significant	list-first request, scan, allowed names and alias map, readout of Step 6	count-first prompt, stated count	per-type error	Table 7
H4	Alias map takes Sonnet from 5.42 to 0.42	alias map on the same list	same list, unmapped	per-type error	Table 7
H5	Degeneracy check flags 45 of 50 Gemini (lowest) lists; 40 fall back to the stated count, 5 keep the list count	four L3 flags	none	flag rate, readout branch	Table 9
H6	Default setting at the same 16k cap: -54% $[-72, -30]$; 32k cap: further -37% $[-62, +8]$, not significant	setting, then cap	lowest 16k, then default 16k	count readout error	Table 9
H7	Same-model second click catches 17% of GPT-6.1 errors; cheaper second model 61%; slightly lower miss rate at matched coverage; after zoom 40% to 50% against 68% to 78%	agreement alarm, $\delta = 1.5\%$	same-model second call	errors caught; miss rate at equal coverage (point estimates)	Table 11, Fig. 6
H8	Perfect selection 45%; label-fitted selectors 16%	constrained oracle; cross-fitted logistic gate	single call	reduction of E, median of 12 cells	Table 13
H9	Unanimity ($K = 3$) lowers set error significantly in 9 of 15 over-reporting cells, never raises it significantly	leave-one-source-out default	single call	E	Table 14
H10	The evaluated same-model transfer rule did not consistently improve F1	leave-one-source-out default	single call	F1 (2 better, 4 worse, 12 n.s.)	Table 14
H11	Cross-model defaults are better in 20 of 24 cells where the user's model is not strongest; all significantly worse cells are where it is strongest	leave-one-source-out cross-model default	single call	E (12/2/4), F1 (10/3/5)	Table 14
H12	Coordinate averaging is a separate term of either sign	selection and geometry decomposition	deletion-only intersection	E; exact identity on every image	Table 10
H13	Overlays do not correct coordinates of current models (localization changes ≤ 0.23 px); for Gemini 3.8 at its lowest setting a grid lowers set error by 3.5 through fewer omitted class fields, a scan by 11.5	grid, ticks, cell labels	no overlay	localization error; set error	Section 7.5

#	Claim	Operation	Comparator	Metric	Evidence
H14	Two-stage zoom: click error -65% (Gemini 3.8), $+23$ points	zoom	single click	hit rate, ScreenSpot-Pro 200	Section 7.5
H15	Self-consistency over three calls gains at most 23% on dense counts; list-first readout 85% and 99% at about one call's cost	median of 3 stated counts; readout of Step 6	one count-first call	per-type error, cost per image	Table 6
H16	Two-stage zoom cuts click error by up to 65% for less than a three-click vote; vote and gated zoom gain at most 13%	zoom on every item	one click; 3-click vote; gated zoom	click error, cost per item	Table 6
H17	Unanimity beats box fusion (majority clusters, coordinates averaged with equal weights) in 11 of 18 cells, not significantly worse in any	leave-one-source-out default	majority with fused coordinates	E	Table 6
H18	One entry point reproduces the component results from raw responses (counts 1.60, 0.42, 24.52; clicks 90.0%, 87.0%, 87.5%; set error in 18 cells within 0.07, recovery off) and runs the synthetic boundary cases	Algorithm 1 with a replay client	component analyses	as in each table	replay, Section 5
H19	Confirming the whole-image report with four crop reports lowers set error in 4 of 6 cells (up to -45.5%); in the other 2, $p_1 > 1/2$	cross-view confirmation, shipped rule	one whole-image call; second whole-image call; three-call unanimity	E at IoU ≥ 0.5, 30 images per cell	Section 7.5
H20	p_1 on 5 labelled images predicts whether deletion helps on the other images in 83% of draws; the Step 7 rule from the family default in 87%	deletion of unconfirmed records	family default; plain threshold	sign of the selection gain on the other images, 54 pairs $\times$ 200 draws	Table 10, Section 7.4
H21	On aerial and document images with several classes the readout gap is small and the protocol changes count error by -20% to $+0.4\%$ (GPT-6.1, Sonnet 5.5, not significant) and -25.8%, -28.2% (Gemini 3.8 default; DOTA interval reaches zero)	list-first protocol	count-first stated count	per-type count error, 80 images	Table 8
H22	Three-model tier misses 9.33% against 13.67% (-4.33 points $[-6.33, -2.33]$, 34/8, $p = 7 \times 10^{-5}$) on 600 new items; against three GPT-6.1 zooms 12.67% (Holm $p = 0.002$)	Algorithm 2	GPT-6.1 default two-stage zoom; three such zooms with a cluster vote	miss rate, all 600 items in the denominator	Table 12
H23	Against a crop-first baseline in GUI-Lens style, -2.17 points $[-4.17, 0.00]$, Holm $p = 0.060$, not significant; the comparison step adds -0.83 $[-2.00, 0.17]$, not significant	Algorithm 2; comparison step	crop-first baseline (ours); judge score alone	miss rate	Table 12
H24	Misses split into 43 with no correct candidate, 3 lost representatives, 10 wrong choices; a repeated GPT-6.1 call hits 18% of the items GPT-6.1 missed, Sonnet 5.5 29%, Gemini 3.8 37%	three-term split; coverage rate ρ	one and three GPT-6.1 zooms	items with no correct candidate	Section 7.7
H25	A single-call configuration chosen on three detection sources raises F1 on the fourth in 5 of 12 cells and lowers it in 1; macro F1 42.1 to 49.1	leave-one-source-out configuration	list interface, lowest setting	F1, primary scoring	Table 15

#	Claim	Operation	Comparator	Metric	Evidence
H26	Adding the crop-first point as a fourth source lowers items with no correct candidate from 43 to 36 but raises lost representatives from 3 to 7 and wrong choices from 10 to 11; 54 against 56 misses, $-0.33\ [-1.17, 0.34]$, not significant	four-source tier, same rules	three-source tier	miss rate and its three terms, same 600 items	Section 7.7